

TopoDroid USER MANUAL

Version 6.5.4 - 27 Aug, 2026
marco.corvi@gmail.com



TopoDroid is an Android app for cave surveying. TopoDroid takes the raw data, helps you to organize them, and produces the line map of the survey on the spot. You can further draw sketches around the line map while you are surveying, thereby reducing the probability of error entering in your sketch, and helping you to detect errors in the data while you are still in the cave. TopoDroid has many functions to help you get the most out of your work in the cave. In this way a working map is almost ready as you are ending the survey trip.

TopoDroid makes the time you spend surveying in the cave more effective. The purpose of TopoDroid is to help a surveyor to collect the data with more accuracy as well as to make the surveying task more pleasant, not to produce the final map of the cave. For that purpose good PC programs already exist.

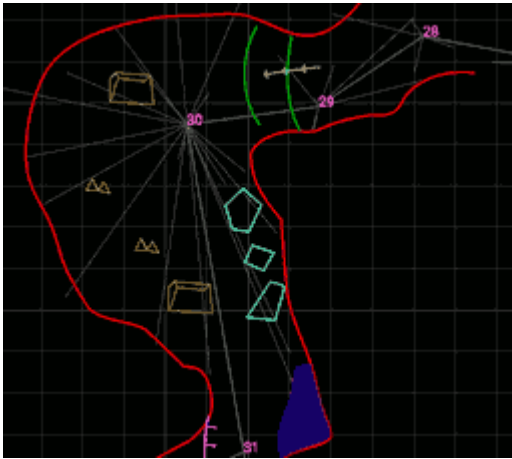
TopoDroid has five different "usage profiles". The lowest levels already have enough functionalities to download data, manage station names, and draw simple sketches. The higher levels can be used to prepare data and sketches for an easier work on the PC.

TopoDroid is designed to handle a survey at a time, not to manage complex cave survey projects that include several different surveys. Nevertheless it includes a *Cave Project Manager*, that can be used to compose surveys into a cave/karst project, and a *3D Viewer*, with which you can see the 3D of surveys and cave projects.

TopoDroid works with [DistoX](#) (v. 1, 2 and BLE), [SAP](#) 5 and 6, [BRIC](#) 4 and 5, [Cavway](#) X1, [DiscoX](#) and [JedEye](#). Survey data can also be entered manually. Beware that Android devices (and active stylus) can influence the measurements taken with an electronic device, if they are too close.

The Android robot is reproduced or modified from work created and shared by Google and used according to terms described in the Creative Commons 3.0 Attribution Licence.

[TopoDroid quick guide](#)



DISCLAIMER

TopoDroid is an open-source free app and it is ad-free. It comes with no warranty: by using it you agree that the authors are not responsible for any loss of data you might incur or damage whatsoever, in particular to any surveying device used with TopoDroid. While the authors do their best to provide a bug-free app, this is not guaranteed.

TopoDroid private folder

The TopoDroid private folder, "Android/data/com.topodroid.TDX/files", contains the folders of the drawing tools, "point", "line", and "area", and the device database "device10.sqlite" with the calibration data, the exported calibration files (folder "ccsv"), the DistoX2 and DistoXBLE firmware files (folder "bin"), and the DistoX/Cavway memory dumps (folder "dump"). The private folder contains also a subfolder for the man-pages translation.

This folder, and its content, is deleted when the app is uninstalled, unless you check the box to keep the app data.

TopoDroid survey folders

The TopoDroid base directory is "Documents/TDX", in the *primary external storage*. It contains the project folder "TopoDroid". The project folder contains the database "distox14.sqlite", the "zip" folder, a folder "tmp" for temporary files, and is populated with a folder for each survey that is created.

This folder and its subfolders are not deleted when the app is uninstalled.



Survey data

Survey data and sketches are saved in the "TopoDroid" subfolder of TDX. Each survey has a subfolder under "TopoDroid".

The "TopoDroid" folder contains the database, "distox14.sqlite", the survey folders, and the folder of the cave projects, "thconfig",

Hardware features

TopoDroid uses the following hardware features: "bluetooth", "camera", "microphone", "GNSS", "multitouch". Functions of TopoDroid that require a feature not present in your Android device, are automatically turned off. Internet connection is required for the AI help-assistant.

It is also used to install translations of the user manual, and to check if a new version is available.

[Android permissions](#)

Granting access permissions

On Android-11 and later, TopoDroid may fail to create the base folder "TDX", or some subfolders or files in it, because it does not have enough permission to access the *primary external storage*.

In this case you can grant permissions using "adb" (Android Debug Bridge).

1. Download "adb" on your PC
2. On Android:
 - enable "Developer options" (go to settings "About Phone", tap seven times the "Build number")
 - enable "USB debugging" in the "Developer options"
3. On the PC enter the command "adb start-server"
4. Connect Android to the PC with a USB cabler:
 - on the Android, if asked, allow adb-connection from the PC
5. On the PC enter the three commands:
 - adb shell appops set com.topodroid.TDX READ_EXTERNAL_STORAGE allow
 - adb shell appops set com.topodroid.TDX WRITE_EXTERNAL_STORAGE allow

- adb shell appops set com.topodroid.TDX
MANAGE_EXTERNAL_STORAGE allow

Native libraries

TopoDroid contains a few libraries compiled native only for selected architectures (arm64_v8a, armeabi-v7a, and x86_64). These libraries are: *station-naming*, *powercrust*, and a *tiff* wrapper for *libtiff*. The last one contains *libjpeg* and *libpng* statically linked.

Errors and crashes

Due to limited resources, the app cannot be fully tested. Make a backup of the apk and the survey database before updating the app.

- Crashes after a selection choice can be due to incomplete translations. Run the app with English language; if the crash disappears contact the translation maintainer.
- If you find an error or the app does not behave as you expect, send me an email. **Include the app version and the Android version.**
- For general questions and feature requests, send an email to TopoDroid mailing list: <topodroid@googlegroups.com>.

App translations:

- Czech: R. Sevcik
- German: M. Keller
- Hungarian: B. Holl
- Italian: F. Toso
- Portuguese: R. Severo
- Russian: A. Kozhenkov
- Spanish: M. Guerrero

Website: <https://sites.google.com/site/speleoapps/>

Sources: <https://github.com/marcocorvi/topodroid>

User manual translations

- A. Kozhenkov - Russian: https://github.com/akozhenkov/TD_manual_RU
- F. Toso - Italian: <https://github.com/fato63/TopoItMan>
- D. Ros - French: <http://souterweb.free.fr/>
- B. Holl - Hungarian: https://github.com/BalazsHoll/TopoDroid_manual_hu

- R. Severo - Portuguese: https://github.com/rsevero/Topodroid_Manual_PT
- M. Guerrero - Spanish: https://github.com/xguerrero/Topodroid_manual_es
- S. Tian - Chinese: https://github.com/tswcmpass/topodroid_man_cn

User manual translations can be quite old if they are not maintained. They are installed by choosing the translation language in the appropriate setting of the Main window. (internet connection required).

If you have installed a manual translation, and there is an update you are prompted to install the translation update when you update the app.

In case of installation failure the message contains a numerical error code:

- 1: failure to create the folder
- 2: failure to retrieve the manual zip (bad URL)
- 3: failure to unpack the zip
- other: HTTP error

TopoDroid websites

Website: sites.google.com/site/speleoapps.

Download: marcocorvi.altervista.org/caving/speleoapps/speleoapks/TopoDroidApksl

Source code: github.com/marcocorvi/topodroid

TopoDroid QUICK GUIDE

This page covers the basics of TopoDroid. Refer to the user manual for more details.

[1] Pair the device with Android

Classic bluetooth devices must be paired with the Android.

Do that with the "Settings" app. Go to "network | bluetooth", turn the device on, and scan for devices. Your device should be found. Select it and enter the PIN "0000" (four zeros) to pair it with Android if asked to.

If your device repeatedly asks the pin, set the socket to "insecure" in TopoDroid settings.

BLE devices can be discovered by TopoDroid through the "BT scan" menu. It may be that the device need to be first paired with Android, to be found. Afterwards, unpair ("forget") it from Android to use it.

[2] Select the device in TopoDroid

Start TopoDroid. To tell it to work with your device, tap the *Device* button. The name and address of your device should appear in the list. To select it tap its address, so that it appears in the top row.

[3] Calibrate the device

If your device is not calibrated, you need to calibrate it. For *DistoX* devices this can be done with TopoDroid: refer to the user manual.

For *SAP*, *BRIC*, *DiscoX* and *JedEye* this is done by the device, without the app.

The *Cavway* has on-board assisted calibration, although TopoDroid supports its calibration.

[4] Create a new survey

Tap on the *plus* button in the Main window, and enter the survey name and the surveyors in the dialog. Close the dialog tapping "Save".

[5] Open the survey

Tapping the survey name in the list of the Main window, takes you to the Survey Data window.

This step can be skipped if you press "Open" in step [4].

[6] Take shots and download them

Take the survey shots: go to the first station, take the splay shots, then measure the leg at least three times. Move to the next station and repeat this procedure.

Tap the *download* button, the leftmost, in the Survey Data window. The download should start and the data appear in a list. Stations are automatically assigned. Make sure TopoDroid did not make any mistake. In case correct them by editing the shot.

[7] Create a sketch

Each TopoDroid sketch consists of a plan view and an extended profile view. Tap on the *sketch* button (the fourth), and tap on *new sketch*: enter the name of a station that serves as sketch origin in the second field. Tap *OK* to open the Drawing window for the new sketch.

[8] Draw the sketch

TopoDroid displays a canvas with a sketch grid, the midline, the splays, and the stations. This is the plan view. Select the "draw" mode (leftmost button) and start drawing (lines). Switch off the "draw" mode tapping it again, and go to the profile view (sixth button). Draw the profile. Close the window and return to the shot list.

[9] Complete the survey

Repeat from point [6], Take a couple more legs. At step [7] tap the sketch name to re-open it in the Drawing window.

[10] Make a ZIP backup

When you are done go to the Survey Info window, choose the *Export* menu, and select the *ZIP* format: this creates a zip archive of your survey.

[11] Export the data

Then export the survey data in the format of your PC cave program. Open again the menu and tap *Export*. Pick your file-format. For some formats you can also choose some export options.

[12] Export the sketch

Back to the Drawing window open the menu (rightmost button) and select *Export*. Pick you favourite format, and tap *Save*. For *cSurvey* you need to export only the plan view. For other formats you must also export the profile view. You might need to export also the x-sections.

[13] Transfer exports

The exported files are in the *out* subfolder of the survey folder. Transfer them to the PC and finish your cave map.

[1] INTRODUCTION

Usage profile

The TopoDroid interface has five usage profiles, differing by the number of actions that are enabled.

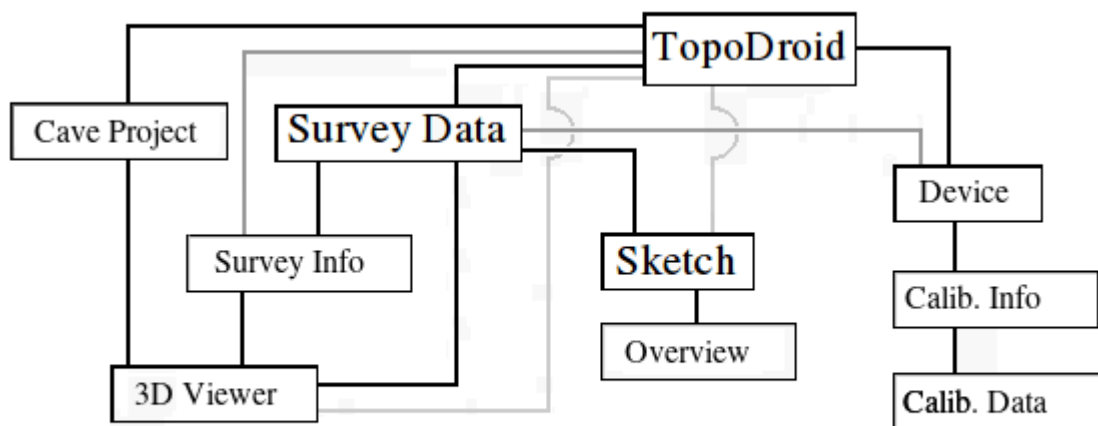
- At the **basic** level [B] drawing is limited to lines, with no editing actions.
- The **normal** level [N] contains all the functions for a detailed cave survey.
- The **advanced** level [A] has also advanced functions.
- The **expert** level [E] has all the functions.
- The **tester** level [T] is only for testing. It has untested features that are partly debugged and may crash the app.
- Features at **developer** level [D] are enabled only in debug build of the app.

By default TopoDroid has only the **normal** functions enabled, but you can switch level, according to your needs, without exiting from the program. This manual describes the complete TopoDroid. Features at a level above "basic" are (usually) marked with the level initial, [N], [A], [E], or [T].

Windows

TopoDroid has a Main window from which you can move to other app activities. The main activities are:

- **Survey** data management (Survey Info and Survey Data windows).
- **Sketch** drafting, Drawing window, is a sub-window of the Survey window.
- **3D viewer**.
- **Device** functions, Device window
- **Calibration** of DistoX/Cavway (Calibration Info and Calibration Data windows) is a sub-window of the Device window.
- **Cave projects**.



TopoDroid windows have a scrollable button bar at the top, for the most frequent actions, and a menu button at the top-right. Dialogs usually have a confirmation button (*OK* or *Save*). If there are both, *OK* saves the input data and closes the dialog, while *Save* saves the input data and keep the dialog open for a new input.

Dialogs can have also a *back* (or *cancel*) button. If not present, the Android *BACK* button saves the changes and closes the dialog. Otherwise it just closes the dialog.

Help and user manual

TopoDroid windows have a menu button (three vertical dots) at the top-right. The last menu opens a quick help about the actions of the window buttons and menus. The quick-help dialog has a "book" button at the top-right which opens the user manual at the relevant page. A long tap on the icon opens the manual at the first page.

If your Android has a *Menu* key, you can use also that to open the man page.

For the dialogs the hardware *Volume Up* key opens the *User Manual* at the relevant page. If your Android has a *Menu* key, you can use that as well. For the dialogs with the row of the action buttons of a slightly darker color, the man page can be opened by a long tap on that row.

Settings

TopoDroid has a large number of settings. These are divided in categories and subcategories:

- General
- Import/Export
- Survey data
- Sketching
- Devices and calibration
- 3D viewer

The settings displayed in the Preferences window depend on the usage profile.

Numerical settings can have minimum and maximum values. If the entered value is outside the bounds, the value of the bound is assigned to the setting.

The app settings can be exported to "settings.txt", in plain text format. The file is stored in the app private Files folder. The settings can be later reset importing the values from this file.

Esoteric settings are marked with [G]. They are collected under the [Secondary settings](#) section.

Text and Buttons size

The "Text Size" setting specifies the dimension of the list text entries, in particular the list of surveys, that of shots, and that of calibration data. There are five choices for the size of buttons: "small", "normal", "medium", "large", and "huge". The "normal" size is suitable for cell phones (5" screen), and is the default. The "large" size is for tablets (7" screen).

Drawing tools (palette)

TopoDroid comes with a considerable number of drawing tools bundled in. By default only the most basic tools are enabled. You can enable more tools as the need arises, with the *palette* button on the Main window. By this means you can get used to the TopoDroid "iconography" a bit at a time.

Additional drawing tool sets can be installed in the palette [G]. Custom tools can be also created and added to the palette, and tools with no use can be removed.

[TopoDroid subdirectories](#)

[TopoDroid databases](#)

[Auxiliary apps](#)

[2] MAIN WINDOW

The Main window has a top bar with (four to six) action buttons and the three vertical dots menu-button. The list of the surveys fills the screen from the bottom up. This list is empty until you create a survey.

The first time you start the app a welcome message is shown. You can go back to it any time later with the *About* menu.

To exit TopoDroid Main window double tap on the hardware *BACK* button.

ACTIONS

- tap on a survey: Survey Data window
- long tap on a survey: Survey Info window

BUTTONS

-  displays the device currently in-use, or a switched-off DistoX if no device is in-use.
-  creates a [new survey](#)
-  [imports](#) a survey (zip archive, or external supported formats)
-  opens drawing tools [palette](#).
-  opens the 3D viewer [T].
-  opens the Cave Project Manager [T].

-  Sketch editing [T]

MENUS

- Close: exit TopoDroid
- work folder [default "TopoDroid"]
- Palette [A/G] opens a dialog to load additional drawing tool sets, or to reload the drawing tools.
- Bulk ops: operations on set of surveys, (only if enabled) [T/G].
- About: the program welcome message. With internet access, this dialog checks if a new version is available.
- Settings: all the program settings
- Help shows a brief descriptions of buttons and menus. At the top-right corner there is a button that opens this user manual. Each TopoDroid window has a similar Help menu

SETTINGS

- *Text size* of the lists items [default 14, min 1]
- *Buttons size*: small, normal (for 5" screen), medium, large (for 7" tablets), huge. [default small]
- *Usage profile*: basic, normal, advanced, expert, tester. [default normal]
- *Keyboard* enables or disables the TopoDroid [keypads](#).
- *No cursor* disables or enables the cursor for TopoDroid [keypads](#) [T]
- *Translated Manual* enables or disables Translated User Manual
- [Language](#) selects the app language

Warning Due to caching the Android might not use the selected language for all texts. A work-around is a cold restart (a force-stop followed by open).

From the settings dialog of the Main window you can go to the other settings dialogs

- Import/Export
- Survey data
- Sketching
- Devices
- 3D Viewer [N]
- *Secondary settings* [A]
- Settings import/export [E]

At "expert" level, or more, you can also export the current settings to file (saved in TopoDroid private folder).

If **bluetooth** is not on, the app asks whether to turn it on. Without bluetooth it cannot communicate with the device and the survey data can only be entered manually. In this case the *Download* and the *Bluetooth* buttons in the Survey Data window are inhibited. In the Drawing window *Bluetooth* is inhibited, and *Download* is replaced by the *Add* button.

[3] DEVICE WINDOW

Tap the *Device* button in the Main window to open the *Device Window*. The list of known devices is shown at the bottom.



TopoDroid works with one device at a time. To select it tap on its entry in the list. The name of the selected device is shown at the top, just below the buttons. If no device is selected, there is a red message saying "Device not selected".

Supported devices

TopoDroid uses only devices connected and paired with the Android system. The devices are selected from the list of devices paired with Android based on their bluetooth name.

The following devices are supported or with support in progress:

Device	Bluetooth model	Support
DistoX v. 1	DistoX	full
DistoX v. 2	DistoX2	full
DistoX BLE	DistoXBLE	full
BRIC 4	BRIC4	full
BRIC 5	BRIC5	untested
SAP 5	SAP5	full
SAP 6	SAP6	partially tested
Cavway X1	Cavway	full
DiscoX	DiscoX	under debug [T]
JedEye	JedEye	under debug [T]

TopoDroid does not recognize devices with a name that is not one of these.

ACTIONS

- tap on a device entry makes it the active device
- long tap on a device entry opens a [dialog](#) to edit the device properties

Device properties

By default, TopoDroid uses the bluetooth MAC address as name for a device. The devices can be given **nicknames** to distinguish among several devices more easily.

Although TopoDroid works with one device at a time, it allows to set a second working device. This options makes it faster to switch between the two devices during the surveying.

Finally, it is possible to "forget" a device: this effectively removes the device from TopoDroid database and from the Android connected devices (if was bonded).

BUTTONS

-  reset bluetooth connection.
-  device information [N,*]
-  [DistoX](#), [BRIC4](#), or [Cavway](#) memory functions [*]
-  toggle the DistoX calibration mode on and off [*] (only for DistoX)
-  list of [calibrations](#)
-  read and display the calibration coefficients from the DistoX/ Cavway [A,*]

[*] These action need communication with the device.

For the *DistoX*, the *Cavway*, and the *JedEye* all the buttons are active.

For the *BRIC*, the first three buttons are active.

For the *SAP* and the *DiscoX* only the first button is active.

If toggling the *DistoX* calibration mode fails at the first try, repeat it a second time.

Reading the device info fails if TopoDroid is connected and downloading data. This can happen when te Device window is opened from the Survey Data window.

MENUS

- Close the window.
- Scan for new BLE devices [N,*].
- Disconnect clears the active device [A]
- Dump and upload [firmwares](#) to the DistoX (v. 2 and BLE) [E,*]
- Display log of Bluetooth packets [T]

- [Settings](#)
- [Help](#)

Bluetooth scan

If a BLE device does not show up with its MAC address, it can be discovered with the "BT scan" menu. To add BLE devices it is enough to do a BT scan with the device on. The "Location" permission is necessary for the scan. For up to Android-11 the device is not found unless "Location" is active. The scan lasts a few seconds and the discovered devices appear in the list. During a scan the "Bluetooth" button turns orange, and a press on it ends the scan.

Bluetooth classic devices (DistoX v. 1 and 2) must be connected and paired to Android using the system Setting app.

The *DistoX* PIN is "0000", four zeros.

The *SAP* PIN is "000000", six zeros.

DEVICE SETTINGS

- *Bluetooth*: whether to enable, disable, or check on start-up [default "check"]
- *Number of new data*: get the number of new data before downloading them (on-demand mode) [default "no"]

The following settings apply only to the DistoX (which uses classical bluetooth).

- *Data connection mode*. "On-demand", "continuous", or "multi" [default "on-demand"]
- *DistoX Socket* type. Normal or "insecure" [default "insecure"]

Data connection mode

There are three connections modes for the DistoX/Cavway.

- in *on-demand* mode data are downloaded in burst, and when there is no more data to download the connection is closed.
- in *continuous* mode TopoDroid remains connected to the device and the shots are downloaded as soon as they are taken. If the device and Android disconnect, TopoDroid repeatedly tries to reconnect. Otherwise, the bluetooth connection is closed.
- the *multi* mode is like the "on-demand" mode, but a long-tap on the "download" button, in the [Survey Data window](#) or in the [Drawing window](#), changes the device. If you have set a "second" DistoX/Cavway (in the [Device window](#)) it is swapped with the current device. Otherwise, you get a dialog to pick one among the known devices.

With the "multi" mode, the current device is shown in the title of the [Survey](#)

Data and Drawing windows.

The connection mode applies only to the data download. Other device functions use a one-shot query mode.

SAP, BRIC, DiscoX, and JedEye are always connected in continuous mode.

Secondary settings: *Connection delay, Double-Device survey, Data pause, Data ready wait delay, Laser pause, Shot pause.*

Devices

DistoX troubleshooting

DistoX functions

BRIC functions

SAP functions

Cavway functions

JedEye functions

[4] CALIBRATION WINDOW

Only for DistoX, DistoX2, DistoXBLE, and Cavway

Tapping the *Calibration* button in the Device window shows the lists of the calibrations of the active device. You can open a calibration or create a new one. In both cases you are led to the Calibration Info window.

The Calibration Info window has text fields for

- the name. Each calibration must have a unique name. For new calibrations the suggested name is the current date
- the date of the calibration
- a description
- the magnetic dip [degrees], if computed by the calibration algorithm

and radio buttons for the calibration algorithm. By default this is set to "auto" (ie, let TopoDroid decide which algorithm to use).

BUTTONS

-  save the calibration information to the database
-  go to the Calibration Data window
-  display the calibration coefficients stored in the database

MENUS

- [Close](#) the window
- [Export](#) the calibration [A]
- [Delete](#) the calibration and its data from the database (non-recoverable action)
- [Settings](#)
- [Help](#)

CALIBRATION SETTINGS

- *Group policy*: how to split the calibration data into [groups](#) [default "Four at a time"]
- *Group tolerance*: threshold to highlight calibration data with large angle distance from the group-mean [default 40 min 0]
- *Algorithm error*: the computation of the coefficients ends if the variation of every coefficient during an iteration step is smaller than this value [default 10^{-6} , min 0].
- *Algorithm iterations*: the maximum number of iteration of the calibration algorithm [default 200, min 50].
- *Download remote data* immediate download.
- *Raw data*: by default TopoDroid displays the azimuth, inclination, and roll. Select if you want to see also the raw calibration data (the components of G and M), either decimal or hex. [default "no"]

[Calibration howto](#)

[5] CALIBRATION DATA

Only for DistoX, DistoX2, DistoXBLE and Cavway

The [Calibration Data window](#) displays the list of calibration data. If the calibration has no data, the window is empty but for the top buttons. Each calibration data line contains the following data information:

- index: incremental index that identifies the data
- group: number of the group the data belongs to
- error: the difference between the calibration-corrected direction and the group average direction, in degrees
- azimuth, inclination, and roll, computed from the raw values without any calibration correction
- optionally the raw calibration data values, ie, the components of G and M in the device frame, either as decimal or hex values

Colors

The calibration data are displayed with yellow and blue text-color, alternating between even and odd group numbers. Data with no group are grey.

The data values background color are:

- black: normal values
- violet: incomplete data (not used in the calibration, but used for the groups)
- brown: saturated data
- gray: deleted data (only if deleted data are displayed)
- red: data with error larger than 1°
- green: data differing from the group average value by more than the *group tolerance*

ACTIONS

- tap on a data: calibration data edit dialog

BUTTONS

-  toggle the DistoX calibration mode on and off. It may fail at the first try (DistoX and XBLE)
-  reset BT, or control the device remotely (DistoX2 XBLE, BRIC, and Cavway)
-  download calibration data from the DistoX
-  assign [group](#) numbers to the data, or reset the group numbers
-  compute the [calibration coefficients](#)
-  [direction distribution](#) of the calibration data
-  read and display the coefficients stored in the DistoX/Cavway

-  write the calibration coefficients to the DistoX/Cavway. Enabled only after the calibration coefficients have been computed. The coefficients can also be written directly from the dialog showing the result of the computation. A warning is issued in the data direction coverage is below 95% or the average error is above 0.5°
-  opens the search dialog

At the moment only on-demand data download is active for the calibration data. The downloaded data are added to the current calibration. Therefore you can take some calibration measurements, download the data, check if you are satisfied with them, if not take further data and download them, and so on until you are happy with your calibration data. Downloaded data can be "disabled" (by setting the group number to "0" (zero); disabled data are not used in the computation of calibration coefficients), or deleted (usually not shown in the list of data). Deleted data are not taken into account in the group-number assignment and in the computation of the calibration.

Bluetooth

This button resets the TopoDroid bluetooth connection.

It also lets the user control the device remotely [A] (if it supports remote control): turn on or off the laser, and take a calibration shot. The data is immediately downloaded if the device data connection mode is set to "continuous".

In this case it is also possible to take (and download) a group of four calibration shots with a small delay between them (controlled by the settings "Laser pause" and "Shot pause").

The bluetooth button is disabled while TopoDroid is taking a calibration shot.

Calibration computation

The "gear" button starts the computation of the calibration coefficients.

When the computation ends, the coefficient dialog is shown.

If there are not enough data or data groups the computation is not carried out and a warning message is displayed.

The calibration fails if the limit of the iteration is reached before the computation ends. This usually happens if the group numbers have been set wrongly. In this case a message is displayed.

Warnings

- **saturated data:** data with abnormal magnetic field value. A possible cause of this is the presence of spurious magnetic fields. Saturated data have a brown background

- **zero data:** incorrectly received calibration data. The data has zero values, and by default it is not assigned a number, although it is used in computing the groups. Zero data have a violet background

MENUS

- Close the window
 - Show/hide deleted: whether to display all data including the deleted data
 - Validate: [validation](#) of this calibration with another one
 - Settings: same as for the Calibration Info window
 - Help
-

[6] SURVEY DATA WINDOW

The Survey Data window displays the list of the shots of the survey.

The title displays the survey name, and, in multi/double device mode, the active device. The title color depends on the current station assignment policy:

- blue: foreshot policies
- dark blue: backshot policies
- pink: "tripod" policy
- violet: "backsight" policy
- red: "magnetic anomaly" policy
- orange: "toporobot" policy

Shot properties are shown through the foreground and background color of the shot data and station names.

Shot data foreground color:

1. [white] "leg" (the first shot of a group of repeated measurements): this shot has both the "From" and the "To" station set
2. [grey] repeated-leg data (the following shots of a group of leg shots): these shots do not need to have the stations set
3. [blue] splay: it has only one station, usually the "From" station
4. [green] X-splay
5. [dark blue] H/V-splay
6. [light green] scan splay

7. [orange] blank shots, i.e, shots with the stations not set yet
8. [yellow]: back-leg (backsight policy)
9. [violet]: blank leg
10. [red]: tampered shot

TopoDroid distinguishes four kinds of shots (displayed with different colors): "leg", "repeated-leg", "splay", and "blank". A series of consecutive shots is recognized as a leg if they are close in distance, azimuth and clino. By default, there must be at least three shots. The first shot is marked as "leg" (and assigned stations), the others are "repeated-leg". If the shots are not close enough, the first is a splay, and the other are "blank".

Splays can be further classified as "cross" (X), horizontal (H), vertical (V), and "scan" (shots acquired in scan mode). [Splay classes](#) X-H-V are disabled by default [G].

Shot data background color:

- [reddish] [unreliable](#) shots, i.e. shots with "abnormal" field intensities (only DistoX2 and Cavway)
- [orange] short leg
- [gray] commented shot

Station name foreground color:

- [bright green] current station

Station names background color:

- [dark green] last downloaded shots
- [gray] shot in "multi-shot" selection set
- [dark yellow] DistoX back-shot data
- [violet] backsight-check splay, with failed check

Shot data: see [SHOT LIST](#) for more information about each entry.

	<i>blank</i>	3.67	163.2	9.4[>]
a6	<i>2 splays</i>	2.74	120.6	11.5[]
a6	<i>unreliable</i>	1.50	152.5	-92.8[]
a6	a7 <i>leg</i>	3.87	184.1	17.4[>]
	<i>repeated leg</i>	3.88	184.1	17.5[]

Station names

Station names are properties of the shot, like the azimuth or the distance. Unlike these, however, the names of the stations are used to "join" together the shots to form the midline. This affects both the midline and the sketches. When the name of a leg station is changed, the midline can change. If the station has attached a x-section, this is unlinked from the midline (and can be later linked to another station). A sketch (plan or profile) can be unlinked from the midline if the name of its reference station (the origin) is changed.

ACTIONS

- tap on the shot data: opens the shot [edit dialog](#)
- long tap on the shot data: start [multi-shot](#) select
- tap on a station: opens the [Saved stations](#) dialog (here you can enable the display of the splays belonging to the station)
- long tap on a station: select/deselect the ["active" station](#)

BUTTONS

-  Data [download](#)
-  [Bluetooth](#)
-  Data [display mode](#)
-  [Sketches](#)
-  Survey [notes](#)
-  Insert [manual](#) shot data
-  [Saved stations](#)
-  [Extend reference](#) azimuth
-  [Station/Leg search](#)
-  Refresh data list

Long taps

- "Download": device switch (only for multi device modes)
- "Sketch": immediately open the most recent sketch
- "Manual data": instruments calibration dialog
- "Search": move to the next result

Multi-shot selection buttons:

-  clear the shots extend
-  set the shots extend to *left*

-  flip the shots extend
-  set the shots extend to *right*
-  special actions dialog [T]
-  delete the shots and put them in the [shot-buffer](#). Hidden secondary-leg shots are deleted and added to the [shot-buffer](#) when the corresponding leg shot is deleted
-  copy the shots to the [shot-buffer](#)
-  close the multi-shot selection

Warning Deleting shots from the survey may leave dangling sketches (ie, sketches without origin station), and media (audio/photo) with broken reference to a shot.

Sketch dialog

Through the "sketch" dialog you create a new sketch or open an existing one. There is a button to create a new sketch, and a two-column list of the current survey sketches. Items in the left column open a sketch in plan view, those in the right column open it in profile view. You can also switch between sketches from within the [Drawing window](#).

MENUS

- Close: closes the current survey
- Survey info: opens the [Survey Info window](#)
- Recover | Paste: [recover](#) a deleted item (shot or sketch), or [shot-buffer paste](#) (long press) [N]/[E]
- Photo: pictures taken for this survey [A]
- Audio: audios recorded in this survey [T]
- Sensor: sensor data taken for this survey [E]
- 3D: [3D viewer](#) [N]
- Device: opens the [Device window](#)
- Settings
- Help [N]

SURVEY SETTINGS

- *Team*: the default surveying team. This value is used to set the "Team" field when you create a new survey. Write your name and that of the cavers you usually survey with. [default unset]

- *Stations policy*: select the automatic [station naming policy](#). [default "splays+foresight"]
- *Station names* specifies the type of the station names, either alphanumeric or numeric [default "alphanumeric"]
- *Initial station name*: default name of the first station for a new surveys. Can be overridden in the *New survey* dialog. [default 0]
- *Thumbnail size*: size of the photo thumbnails [default 200, min 80, max 400] [A]
- *Editable stations*: defines if station names are editable in the shot list [default no] [E]
- *Data default-export*: default data export format [default none]
- *Sketch origin*: whether to suggest always the same origin or not [default no]
- *UNITS*
 - *Length units*: either meters or feet (0.3048 m) [default meters]
 - *Angle units*: either degrees or grads [default degrees]
 - *Sketch units*: sketch grid cell size either in meters, yards, 2-ft, or 0.1 meter [default meters]
 - *Ruler units*: either the same as sketch units (grid-cell), meters or feet [default grid-cell]
- *SHOT DATA*
 - *Leg tolerance*: the angle tolerance (percent) between consecutive shots of a leg [default 0.05, min 0]. The length tolerance is the product of the angle tolerance and the length.
 - *Maximum shot length*: longer shots are "overshoot" and marked as "deleted" [default 50 m, min 20 m]
 - *Minimum leg length*: shorter legs are highlighted with an orange background [default 0 m]
 - *Leg shots number*: minimum number of close consecutive shots for a leg [default 3, min 2, max 4] [E]
 - *Shot vertical-extend threshold*: the azimuth span around the perpendicular to the reference azimuth for the [shot "extend"](#) to be vertical [default 10, min 0, max 90] (used both for legs and splays) [N]
 - *Leg WENS threshold* (*Therion* "vthreshold"): the LRUD for manually entered shots are considered West-East-North-South when the shot inclination exceeds the WENS threshold [default 80, min 0, max 90] [N]
 - *Manual extend reference*: either manual L/R or relative to the [extend reference](#) azimuth [default relative] [A]
 - whether to show *prev/next buttons* in the shot edit dialog [default yes] [A]
 - *Backsight* fields in the shot manual input dialog [N] [default no] [A]
 - *Leg-shot feedback*: feedback on leg-shot recognition: "none", "sound" or "haptic" [default "none"] [N]
 - *Additional settings*
 - *Backshot DistoX*: DistoX in "backshot" mode [default no] [N]
 - *Loop closure* error compensation. It is advisable not to do the compensation, as this allows to see graphically the mis-closures, and spot possible survey errors [default "none"] [N]

- *Timer*: number of seconds of delay before measuring directions with the Android sensors [default 10, min 0] [E]
- *Beep volume*: beep volume, for manual shot timer and leg feedback [default 50, min 10, max 100%] [E]
- *Beep length*: beep duration, for leg feedback [default 500 ms] [E]
- *LOCATION* [settings](#)
- *ACCURACY* [settings](#)

[Secondary settings](#): *Recent shots and their timeout, Fractional extend, Splay classes and coloring, Diving mode, Sensors.*

To exit the Survey Data window double tap on the *BACK* button of your device.

If the automatic backup is enabled and the default data file format is set, the data are exported whenever the Survey Data window is closed.

Warning The first shot entry on the list may not get updated after an editing or a tap, due to a problem of Android. The "Refresh" button can be used to request a reload of the list, if this happen.

Loop error compensation

It is advisable not to do the compensation, as this allows to see graphically the mis-closures, and spot possible survey errors. This is the default behaviour.

Nevertheless TopoDroid implements a few "loop error compensation" policies.

The *normal* policy [E] distributes the loop closure errors on the network branches that make the loops.

In the *weighted* policy [E] the mis-closures of the loops are weighted according to the "branch relevance".

The *selective* [T] compensation applies only the corrections of loops with mis-closure smaller than a certain percent (the "loop threshold" setting) of the loop length. The loops above threshold are displayed violet.

The [trilateration](#) adjust the azimuth angles of shots that make triangles, by forcing the sum of internal angles of the triangle projected on the horizontal plane to be 180 degrees.

[7] SHOT LIST

The shots are shown in a list. The list has no header to save space. The shot information are

- Index (optional: it is shown only if the relative choice in the *display mode* dialog is checked)
- From station

- To station
- Distance
- Azimuth
- Inclination
- Extend direction: either left (<), right (>), vertical (|), or ignore (a blank space). The "extend" of splays is present only if explicitly set by the user in one of the shot edit dialogs.

There can be additional characters for the splay flags, the shot flag, and the presence of a photo (#). Finally the beginning of the comment if any. The shot flag can be: duplicate shot (²), surface shot (÷), only-profile (\$), only-plan (_), and backshot (b).

The splay marks are distinguished by the closing bracket: no mark (plain square bracket) feature (a upper star), ridge (a double bracket) backsight (a down arrow), and generic (a upper circle).

1	O	_	3.14	348.2	16.9	[[*]]	FEATURE
2	O	_	4.99	7.8	22.0	[]	RIDGE
3	O	_	2.17	315.0	-18.0	[^v]	BACKSIGHT
4	O	_	3.44	287.0	14.0	[^o]	GENERIC
5	O	_	3.60	274.3	25.9	[]	NO MARK

By default the *index* is the index of the shot in the survey. There is a setting to use the *BRIC* shot-index instead. The shot survey-index is guaranteed to be unique. The *BRIC* index is not unique because the *BRIC* memory can be reset.

Blank shots, and repeated-legs can be hidden. Splay shots can also be hidden to unclutter the list of shots. However, even if splays are not shown, you can see those at a station by tapping on the station name in a leg shot. To hide them tap again on the station name (even a tap on the name in a splay will do).

TopoDroid can set the station names automatically according to the [station assignment policy](#).

Shot item taps

- a tap on a station name puts it in *editing-mode*, and the station is underlined. If inline station editing is enabled the station name can be edited. The station name edit must be finished with the ENTER key to take effect.
- when the station is underlined, a couple of tap toggles the display of splays at that station (only if splays are globally hidden).
- a tap on the shot data opens the [shot edit](#) dialog, in which you can set the stations as well as other attributes of the shot.

- a long tap on a station name makes it the [active](#) station (it is highlighted green)
- a long tap on a shot starts the multi-shot selection

If you use TopoDroid automatic [station naming](#), you are likely to need the shot dialog only to enter the comment, and to change the "extend" direction. These two actions can be done also from the [Drawing window](#), by selecting the shot in "edit" mode [A] and picking the *Note* button. The extend can be changed also graphically on the sketch. These actions are described in the section on the [Drawing window](#).

Multi-shot selection

The selected shots have a gray background (possibly only in the back of the stations). Tapping a shot adds/removes it from the selection. Subsequent long-taps either select or de-select all entries from the previously selected or de-selected shot to the current one.

In multishot selection, the last tapped shot stations are highlighted in light yellow.

The number of shots in the selection is displayed in the title bar within angle brackets.

The button bar changes to the buttons for the multi-shot actions:

-  clear the shots extend
 -  set the shots extend to *left*
 -  flip the shots extend
 -  set the shots extend to *right*
 -  [special actions](#) dialog [T]
 -  delete the shots and put them in the [shot-buffer](#). Warning: hidden secondary-leg shots are deleted when the corresponding leg shot is deleted
 -  copy the shots to the [shot-buffer](#) (can be copied to any survey using a long-click on the "recover" menu item)
 -  close the multi-shot selection
-

[8] SURVEY INFO WINDOW

The *survey info* menu of the Survey Data window opens the Survey Info window. Alternatively you can open it with a long tap on the survey entry in the Main window.

The displayed survey info are:

- name (not editable)
- date
- team
- magnetic declination (if set)
- survey description
- whether the data-mode is *normal* or *diving* [T]

The survey "team" should not be empty, otherwise the window gives a warning when it is closed. It can be forced empty by trying to close a second time.

The magnetic declination must be entered in decimal degrees. The allowed range of values is [-360°, 360°]. Values outside this range can be used to mark it "unset".

BUTTONS

-  survey annotations
-  survey data [info](#)
-  3D display (Cave3D) [A]
-  [geo-points](#) [A]
-  photo list. Tap on an entry to edit the description or delete the photo [A]
-  sensor measurements. Tap on an entry to edit the comment or delete the measurement [T]

MENUS

- Close
- Export
- Rename [T]
- Delete [A]

- Clear color: clear the custom splay colors [T]
- Instruments offsets [E]
- Calibration-check shots [E]
- Station names remap [T]
- Settings (same as the Survey Data window settings)
- Help

Export dialog

There are a number of export settings, that control the way TopoDroid writes the survey data in the various export formats.

Delete dialog [A]

The delete dialog asks for confirmation to delete the survey. Deleting the survey is unrecoverable: the survey and all associated data (photos, notes, sketch files, etc.), except the zip archive, are deleted. Make sure to export a backup *ZIP* archive before you delete the survey.

Rename dialog

With this dialog you can rename the survey. Renaming fails if there is already a survey with the specified name.

The info in the database, the file of the notes, the binary sketch file, and the image and audio directory are renamed. However reference to section "scrap" in the sketch files are not updated until the files are opened again. Therefore it is advisable to go over and "open and close" all the sketches.

[9] SKETCH WINDOW

The *sketch* button of the Survey Data window opens the dialog with the list of sketches, besides the *New Sketch* button. Tap on this button to create a new survey sketch.

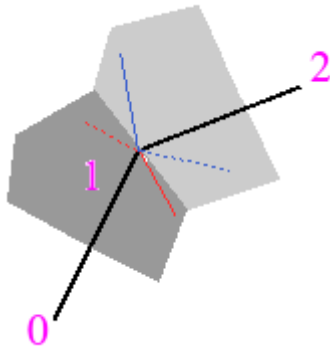
Each sketch is made of two sketch files: one for the plan view, the other for the profile view (either extended or projected). X-sections have only one file each. A new sketch is opened in the plan view. Afterwards the sketch dialog shows two columns, one for the plan, the other for the profile view, and you can choose which one to open from.

Each sketch can contain multiple scraps.

The extended profile view displays the midline with the shots drawn left or right, or vertical, according to their "extend". If a leg has the "extend" not set, the midline is not displayed from that leg on. Splays are projected on the vertical plane of the shot that is closest in azimuth, unless the user has

set a given "extend" through the shot edit dialog. For example, for the plan view in figure, the splays in the light region are projected on the vertical plane of shot 1-2, those in the dark region on the vertical plane of 0-1.

The projected profile view displays the midline and the splays projected in the vertical plane perpendicular to the projection azimuth, defined when the projected profile sketch is created. The projection direction may differ from the projection azimuth by an angle of obliqueness.



Display mode

The *display mode* button (below) allows to select the reference objects to display:

- legs
- splays
- station names or elevations
- grid
- scale bar

Legs

The leg colors are

- normal: white
- anomalous field values: red
- inconsistent with siblings: orange. These are also drawn with a zig-zag line as shown below.



Splays

Splays can be displayed as segments or as endpoint dots [G]. When the window is in selection mode, the splays are always displayed as segments. The colors of splays with mark flags are [only for Cavway]

- feature: yellow
- border: purple
- backsight: blue
- generic: green

The splay class-colors are [E/G]

- commented: dark green
- x-splay (LRUD): green
- h-splay: deep blue (plan), light pink (profile)
- v-splay: light pink (plan), deep blue (profile)
- others: light grey

It is possible to set the splay transparency (opacity level) [G].

Splays can be assigned different colors, individually or in groups (with multi-editing). The colors for splays can be chosen among a palette of colors or with a color picker [G].

A splay is not shown if the shot is marked for no-display in plan or profile view.

If splay dashing [G] is not disabled, they can be displayed with a dashed or a dotted line.

If the splay-dashing is set to *azimuth* the splay is dashed if the angle with the "closest" leg in azimuth is above the dash azimuth-threshold, [default 60°]

If the splay-dashing is set to *clino* the splay is dashed if the splay clino is above the dash clino-threshold: dotted (positive clino) and dashed (negative clino).

If the splay-dashing is set *by the view* the splays dashed by the clino in the plan view and by the azimuth in the extended profile view. In the projected profile the splays are dashed according to the angle they form with the direction of projection.

Furthermore, in the plan view the splay is not displayed if the inclination of the splay is over the plan-view threshold [default 80°]

For x-sections:

- x-splays (LRUD) are green
- other splays are blue: dark at TO station, light at FROM station
- finally, if splay-dashing is enabled, they are dotted or dashed if the angle between the splay and the normal to the x-section plane is below the dash angle-threshold [default 60°] (dotted if in the direction of the normal, dashed if opposite).

The colors of dashed and dotted splay line can be changed with appropriate settings [G].

Scan shot

Scan shots can only be produced by certain devices (Cavway) and cannot be created manually.

- Generic scan: yellow green
- H-scan: light pink
- V-scan: deep blue
- X-scan: dark green

Stations

The station colors are

- violet: default
- green: active station
- red: stations beyond which the midline is hidden
- blue: stations before which the midline is hidden
- yellow: saved stations [T/G]

Interaction modes

The Drawing window has four interaction modes:

- **move**: you can shift the canvas, and zoom in/out. If the "fixed references" box in the references dialog is set, shifts affect only the drawings [T]
- **draw**: in this mode you add iconic symbols (points, including labels), and trace lines and areas (closed regions filled with a color or a pattern).
- **erase**: to erase items, and lines/areas, or portions of them, from the sketch.
- **edit** [A]: for point-wise changes and item properties editing, including item deleting.

Shift and zoom are possible even in **draw** and in **erase** mode using two fingers. If "corner-drag" is enabled, the sketch can be shifted by "dragging" it from the corners of the canvas. Corners are displayed as transparent gray. Zoom controls can be enabled (either temporarily or permanently). If temporary, they show up when you tap at the bottom of the canvas in the middle.

In "draw mode" there is a bottom toolbar with the most recently used tools, and an "arrow" button to select other tools.

In "edit mode", when the selected item is a point, there is a bottom slider for

point scale. If "corner-drag" is enabled, and the point item is orientable, its orientation can be rotated clockwise by dragging down from the top-right corner, and counterclockwise by dragging down from the top-left corner.

BUTTONS

The modes are represented by the three green buttons on the left: *draw*, *erase*, and *edit*. When one of them is "on" the relative button has a bright green color. If none is on the mode is *move*.

When a green button is "on", a long tap on it, opens the dialog to edit the settings relevant to the mode of the button.



The remaining buttons differ by the mode.

In the **move** mode they are:

-  data *download*. If no device is selected, this button is hidden. By default, splays and legs are refreshed when a leg is downloaded (and identified). Use the refresh button to force a centerline refresh. The last download shots can be highlighted blue to help distinguish them from the others [G].
-  [Bluetooth](#) reset/controls
-  drawing [references](#): legs, splays, stations/elevation, sketch grid, scalebar.
-  *plan/profile* switch.  *Splay-display mode*, for x-sections
-  survey [notes](#)
-  ["extend" reference](#)
-  *refresh* of the centerline

The **draw** mode has, besides the green buttons:

-  *undo* removes the last items you have drawn, one at a time.

-  *redo* puts the removed item back in the sketch, one at a time. The list of removed items is emptied when you draw something new. Point-wise editing actions may also affect the list of redo.
-  Drawing *tools* switcher.
-  *splay* display mode: as segment or a point.
-  Insert a *text*
-  Line *join* (only for line)

The icon of the *tools* button reflects the current type of drawing tools: points, lines, or areas.

The *text* button allows to insert a point of type "label" without the dialog of the "label" point tools.

The line *join* is shown only for lines. When line-joining is active, a new line started and/or ended close to lines of the same group is joined to them.

The "Line-join" has no effect on lines that cannot be joined (e.g. "section" line or closed lines).

The **erase** mode has, besides the green buttons:

-  *undo*
-  *redo*
-  type of erasable items: all, points, lines, areas
-  size of the eraser [S, M, L]

The **edit** [A] mode has, besides the green buttons:

-  type of selectable items: all, points, lines, areas, shots, stations
-  size of the selection tool [S, M, L]
-  *delete* the item from the sketch (it asks for confirmation and can be "undone").
-  item *properties* edit. For "section" point it opens the x-section sketch drawing.

-  point-wise and *special actions*. For section-point items this button opens the x-section sketch.
-  *range* selection: either point-wise or local (only for lines and areas).

When there are selected items the first two buttons become *previous/next* navigation buttons in the set of the selected items.

In the *Sketch Window* scan-sets are not deleted as a whole. Rather, individual shot are removed from the scan-set.

The buttons are somewhat different for x-section sketches. In the **move** mode the buttons "switch", "azimuth", and "refresh" are not shown, but there is a button to select which splay to show. It cycles through four states: "both", "viewed station", "behind station", "none". Accordingly, the drawing references dialog has no box for the splays.

Long tap actions

- A long tap on each *mode* button opens the relevant settings dialog [A].
- The *download* button switches the device, in multi device mode [E].
- The *display mode* button switches between station names and elevations.
- In plan view, a long tap on the *extend reference* button makes the reference azimuth graphically editable by sliding the finger across the display. The button turns green [E].
- In the extended profile view, a long tap on the *plan/profile* button flips the sketch [N]. If the checkbox *Fix refs* of the reference dialog is not set, the "extend" of the shots is also reversed.
- For x-sections it toggles the display of splays that are off the section plane [T].
- A long tap on the *tools* button toggles retracing on/off (only for line and areas) [A]
- A long tap on the *delete* button removes the selected point of a line or an area [N].

MENUS

-  When anomalies have been detected in the survey data-reduction, the "menu" button (three dots in vertical) has a red exclamation mark. Long-tap on the button to display a dialog about the anomalies.
- Close the window or Switch to another sketch.
- Export: the sketch can be [exported](#) as *Therion* th2, *cSurvey*, DXF, SVG, PDF, xvi (*xtherion*), shapefile (*QGIS*), and *Tunnel*. All exports generate

two files, one for plan the other for profile, except *cSurvey*, which has both in one file.

- Sketch [statistics](#). For x-sections this computes the section area.
- Reload the sketch from a [backup copy](#), or restore item symbols.
- Zoom-fit: dialog to fit the sketch into the screen, and to select the portrait or landscape presentation
- Rename/Delete/Split the sketch. Renaming and deleting affects both plan and profile view at the same time. Deleted sketches can be recovered (*recover* menu in Survey Data window). Splitting affects only the current view.
- [Scraps](#) dialog [T]
- [drawing tools](#) palette
- Overview window, showing all the survey sketches together. [Only for plan and extended profiles]
- Settings
- Help

Long taps

The following actions are started with a long tap:

- Close/Switch switches to another sketch.
- Reload restores item symbols.
- Palette, in "draw" mode, opens the drawing tool picker dialog.

SKETCH SETTINGS

- *Corner drag*: enable panning by corner-drag
- *Zoom controls*: specifies how zoom controls are displayed on multitouch devices [B]. The zoom controls can be hidden, displayed only for a short interval when the user tap the lower part of the screen in the middle, or shown all the time
- *H-threshold*: leg clino-threshold for horizontal x-sections (profile view).
- *Midline check*: check if all the legs are attached to the midline [disabled]
- *Extend check*: check if all the legs have the "extend" defined [enabled]
- *Toolbar size*: size of the toolbar of drawing items [8]
- *Toolbar update*: how new symbols are inserted in the toolbar: put at front, replace the oldest, or drop oldest and put at front [replace oldest]
- **POINT TOOLS**
 - *Non-zoomed icons*: point items stay of fixed size when zooming
 - *Drawing units* of point symbols
 - *Labels size*: size of text of label points [24 pt]
- **LINE/AREA TOOLS**
 - *Pen width* of lines (NB wall-lines are twice as wide) [1 px]
 - *Drawing units* of linear symbols [3.0]
 - *Style*: fine, normal, coarse, "splines" (interpolated-curve), simplified [normal]
 - *Close*: whether to close lines of "closed" type. [yes]

- *Line-point spacing*: distance between line points in normal style [10 px]
- *Section line direction tick length*: size of the section line tick [5 px]
- *Section points*: whether to add a "section" point automatically for "section" lines [yes]
- *Area border visibility* [on]
- **SKETCH ITEMS**
 - *Pen width* of survey lines (midline/splay)
 - *Text size* of station names [18 pt] [N]
 - *Green dots size* displayed in **edit** mode [5 px] [A]
 - *Splay transparency*: opacity level [0: transparent, 100: opaque, default 80]

Secondary settings:

- Sketch shift/scale
- Sketch split/merge (split, copy, outlines, merge)
- Stylus only drawing
- Backup number
- Backup interval
- Saved stations coloring
- Leg-only refresh
- Affine transform
- Layers (canvas views)
- Graph-paper scale
- Slant x-sections
- Oblique projected profiles
- Line ends cut
- Leg vertical threshold (over threshold LRUD become WENS)
- Minimum angle of corners when a line is straightened
- Bezier interpolation accuracy, and corner threshold
- Line simplification distance, length, and buffer zone
- Splay dashing (mode, clino threshold, azimuth thresholds)
- Auto x-section in exports (csx, svg, dxf, xvi)

Sketch cache

Once opened, the sketches are kept in memory, in a cache. This speeds up switching between the Survey Data window and the Drawing window, as the sketch does not need to be read from file. The sketch cache is cleared when the survey is closed.

Sketch caching can be disabled with a *sketch* setting.

Sketch file format

Each sketch is saved to file whenever it is modified, or when the Drawing window changes to a different view. For performance TopoDroid saves the sketches in a binary format.

Point selection [A]

To do some editing you must select a point: either a point item, a line/area point, a station, or a shot midpoint.

To this aim you tap near the point you want. However if there are several points close where you tap, they are all selected, but only one of them is being edited. You change the "edit" point among the selected set with the *previous/next* buttons.



Line style

TopoDroid has five line styles. With the normal style [default] the lines are drawn as polylines. With the fine style the polyline segments are shorter, while with the coarse style they are longer. In the fourth style lines are interpolated replacing the segments with cubic Bezier curves (splines). These segments pass through the two endpoints and have two control points used define the curvature. With the fifth line style the traced lines are simplified using a point-weeding algorithm. If you export the drawing in dxf with DXF version 6 do not use splines because they get exported as polylines.

Retracing

If the path-tracing style is not complex (Bezier or weeding), line and areas can be drawn in "retracing" mode.

A long tap on the *tools* button turns retracing on/off.

When retrace drawing is active the line/area in the respective *tool* button is white.

Line retracing and line joining are mutually exclusive.

Lines of type "section" and of closed line types cannot be retraced.

To retrace a portion of a line, trace a line that starts and or ends close to the line.

The retraced portion of a line/area is not transformed (Bezier spline, weeding, decimation, etc.).

Areas can be "retraced" by tracing a line that starts and ends close to the border of the area. The shorter between the two pieces of this border is replaced by the newly traced line.

The area can be enlarged or shrunk, and a concavity can be filled (figure below).

Retracing can be undone and redone.



Portrait/Landscape presentation

In portrait presentation the magnetic North and the Vertical are upward in plan and profile view, respectively. The vertical direction of the plan view is always the magnetic North.

If the survey declination is set the true North direction is shown as an angle to the magnetic North arrow.

In landscape presentation they are leftward so that the East is upward in plan view and the Right is upward in profile view.

In landscape presentation the iconic symbols are rotated so that they appear natural holding the device in portrait mode.

The sketches are always saved and exported with the magnetic North and the vertical upward.

Warning. If the sketch disappear from the display, switch between plan and profile to force TopoDroid to refresh it.

File editor mode [T]

The *Drawing window* is opened in file editing mode from the "sketch" button of the *Main window*. TopoDroid sketch ".tdr" files and (simple) *Therion* ".th2" files can be loaded and edited. The changes are not saved and the sketch must be exported to save them.

TopoDroid ".tdr" files are not related to a survey. The stations are loaded from the file as "user station" points.

The syntax of *Therion* ".th2" files is not fully supported. In particular areas with multiline border are not supported, because TopoDroid has only single-line border areas.

When the *Drawing window* is opened as editor, the buttons and the menus change. The menus are:

- *Close* the window
- *Save* the th2 file with the modifications
- *Open* a th2 file
- *Zoom fit*
- *Scraps*
- *Palette*
- *Settings*
- *Help*

In *move* mode there are only the three green buttons for the other modes. The *draw* mode has the buttons: *undo*, *redo*, *tools*, and *line/area continuation*.

The *erase* and *edit* modes have the usual buttons.

The th2 file can contain more than one scrap. As the sketch is not tied to a survey, there is no midline reference, ie, no legs and no splays. Stations are

present only if the th2 file contains "station" points. In this case they are displayed as dots with the station name nearby.

The sketch grid is displayed only when a graph-paper size is set (through the *zoom-fit* dialog).

[Drawing tools palette](#)

[Therion station points](#)

[Undo stack](#)

[Sketch shift and flip](#)

[10] SKETCH DRAWING

The canvas shows the midline with the splay shots. It can be moved around dragging it with one (or two) finger(s), zoom in/out with two fingers. The plan-view magnetic North and the profile-view vertical are upward. It is not possible to rotate the canvas.

Tap the *draw* button: the app goes in **draw** mode and the button is highlighted with a bright green. Now you can draw with your finger (two fingers still pan and zoom). With the *undo* and *redo* button you cancel the last drawn or deleted objects and put them back again.

There are three types of drawing tools: point (icons), line, area (shaded region). Labels are a "point" tool.

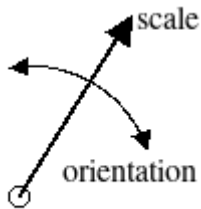
The list of the six most recently used drawing tools is shown in the toolbar at the bottom of the screen. To select a tool tap on its icon in the bottom toolbar. A long-tap on this button reverts the order of rotation of the bottom toolbar

While in draw mode, the sketch can also be moved by dragging it from the corners of the canvas and zoomed with zoom controls if these function have been enabled through the settings. This lets you work with a stylus without the need for two-finger actions.

Point tools

Point tools are iconic symbols. To add a symbol to the sketch touch the canvas where you want to put the point.

If the point is orientable, it is drawn with the set orientation (selectable with the slider in the icon picker dialog). This orientation can be overridden with a touch of the display slightly dragged in the desired orientation. With a further drag the user can set the icon scale besides the orientation. With a longer drag, the dot symbol appears and its direction and size can be adjusted.



Labels

"Label" points are somewhat special. When you add a new "label" point, the keyboard is shown to enter the label text. Text entering is ended by tapping on the canvas or closing the window.

The text can be later changed by editing the "label" point.

Warning: the down-arrow of the navigation bar hides the keyboard, without ending entering the text.

Line tools

Line tools are line-drawing styles. Lines have an orientation: when a line is selected, in edit mode, its orientation is shown by a yellow tick pointing to the left side.

The line thickness is one of the *settings*. There are separate options for the centerline/splays and the drawing line thickness.

Section lines

The "section" line is special as it is always made into a straight segment with a short tick showing the direction of view. A "section" point is automatically added connected to the end of the section line. This point has the "scrap" option preset to the line scrap and is deleted when you delete the section line.

"Section" lines cannot be erased with the *eraser* tool. To remove a "section" line, select it with the *edit* tool and delete it with the "delete" button.

Its points cannot be moved in the canvas, like for points of other lines, neither they can be removed. The point-wise actions do not apply to the endpoints of "section" lines.

Area tools

Area tools are region shaders. To draw an area trace its contour. You do not need to be very precise as TopoDroid will close it.

The water area is special because it can be "closed horizontally" in the profile sketch and in the x-sections, because the surface of a pond is usually horizontal. To draw a water area (in the profile sketch) trace only the lower bottom line of the pond.

Undo, redo, erase and edit

Drawing mistakes can be corrected. The *undo* button removes the last item. To erase a line or a symbol drawn long ago, without losing all the intermediate work, there are two choices: the *erase* mode and the *edit* mode.

The *eraser* is very intuitive: it cancels everything you pass your finger over, except stations, shots, and section lines. Just like a real eraser on the paper: if you pass it in the middle of a line it erases only the central portion (thus leaving two lines). However erasing can be filtered to erase only items of one type: points, lines, or areas.

The *edit* [A] button is not so intuitive, but it is very powerful, as it allows point-wise editing operations. In this mode all the drawn items are shown as little green dots. To select the item to edit tap on one of its green dots (icons have only one dot). If there are several dots close together, they will all be selected (and shown with a pink circle), but only one will be chosen for editing (big green circle). If it is not what you wanted, cycle through the selected points with the *previous/next* buttons.

Except for the endpoints of "section" lines, the points of the the drawn items can be moved on the canvas by dragging them.

The edit tool can be filtered to select only dots of one type: points, lines, areas, shots, or stations.

The size of the green dots and the selection radius are two *settings*.

Segments of Bezier lines have two control points. These are stored with the segment end-point and when this is the selected editing point the two control points are shown as pink circles. Each of them can be moved around by dragging it with the finger.

To clear the selection tap somewhere else in the canvas. The item can be deleted (removed) or its [properties](#) changed ("tool" button). Editing applies also to shots and stations.

Editing an orientable point, its icon can be rotated also with a swipe from the top corners of the canvas (if corner-drag is enabled).

A survey shot (leg or splay) has a green dot in the middle to select it. When a leg is selected in the profile view, a horizontal yellow segment is also displayed on the side opposite to the leg "extend", or on both sides if the leg is extended vertically. The leg extend can be set by tapping the left or right end of this segment, or the middle.

Point-wise edit actions [A]

The point-wise edit actions (fourth to the right button) apply only to drawing items. The button has a drop-down menu of editing actions, for either points [P], lines (except "section" lines) [L], and areas [A]:

- *Snap to nearby point* [P,L,A]: move the point to coincide with the nearest point of the current scrap. No action if there is no nearby point
- *Snap to nearby splays/line* [L,A:T]. Lines are deformed to pass through the splay endpoints that are close enough. If an area point is close to a line point, the pieces of the area border before and after it are replaced with pieces that follow the line path. This feature requires the proper setting enabled
- *Insert point(s)* [L:A]. When a single point is selected, the new point is inserted "after" the selected point, and it is added to the selection set, and you can select it with the Prev-Next buttons. When a range is selected, new points are inserted between every two points of the range. A long-tap moves the selected point to the nearest point and inserts another point, "after" the selected point [T/G]
- *Cut* [L] the line at the point, thus making two lines
- *Straighten* [L:T] the line segment removing the control points
- *Smoothen* [L:T] the line segment inserting control points at 1/3 and 2/3 the segment length. Further editing can move them and give any wanted curvature to the segment
- *Remove* [L:A] the line/area point. The selected line/area point can be removed also with a long-tap on the "delete" button.
- *Append* [L:T, except "section"] to another line of same type. Only for line end-points and the other line end-point must be close to the selected point.
- enter *multiselection* mode [P,L,A:T]

Multiselection mode [T]

Drawing items multiselection is disabled by default. It can be enabled in a "Sketch | Line" secondary setting.

The multiselection mode allows to select several objects of the same type (point/line/area) and act on them all:

- *Delete* the selected items [P,L,A]
- *Reduce* the number of points of the selected items [L,A]
- *Join* the selected item [L]
- *Store* the items in order to move them to another scrap
- *Exit* multiselection mode

The multiselection items are implicitly stored if you switch scrap in multiselection mode. To move the items to another scrap, you must select the target scrap and press the "Restore" menu before you select any point.

Range selection

Lines and areas points can be selected individually (point selection mode) or as a range.

-  single point selection
-  "soft" range selection
-  "hard" range selection
-  whole line/area selection

To specify a range select first one endpoint of the range, then the other endpoint. The range is highlighted orange. Its midpoint gets the big green dot, and it controls how the line range is deformed. The range can have hard or soft bounds. With hard bounds the range is moved as a whole. With soft bounds the points close to the ends of the range are moved by an amount proportional to their distance from the end.

The selection can also be "item-wise", ie, the whole item. In this case you select a point on the line/area, and act (eg, shift) on the whole item.

Stylus only drawing [T]

This feature is usefull when drawing with a fine-point stylus. It should be used **on devices that provide the size of the touch**.

On other devices stylus-only drawing is tricky, because it requires a "slow" tracing of the line, starting with a kink. The "Line-ends cut" setting may be useful.

Orientable points should be first "traced" slightly in the opposite direction, then traced in the required direction.

Stylus only drawing allows you to hold a finger on the screen while drawing with a stylus.

To enable this feature you must set a positive "stylus size". The size, in pixels, should be a value intermediate between the size of the display area touched with a stylus tip and that touched with a finger tip.

Stylus-only tracing does not work if the line-point spacing is too small.

With stylus-only drawing, the multi-finger actions are inhibited. The zoom-controls are enabled and cannot be disabled.

[Sketch station editing](#)

[Drawing tools](#)

[Drawing tools](#) picker

[11] CROSS-SECTIONS

X-sections are kind of special because they do not belong to the sketch but are independent sketches associated to either the plan or the profile. TopoDroid has two types of x-sections: leg x-section and station x-section. The former crosses a leg, the latter is defined at a station.

A plan-view x-section is always in a vertical plane and the upward direction is the vertical.

Profile-view x-sections can be either vertical or horizontal, or slanted if the appropriate setting is enabled [T].

You can assign a comment to an x-section.

Leg x-section

To tell TopoDroid that you want to draw a x-section you have to draw a "section" line, either in the plan view or in the profile view. The section line defines the position of the x-section and its orientation.

A x-section is the shape of the cave in a plane that cuts it. The orientation of the x-section is the orientation of this plane, i.e., the direction of the line perpendicular to the plane, and pointing in the direction of sight. You define the orientation of the x-section by the way you trace the section line. As an indicator the section line has a tick pointing from the line to the direction of view of the x-section.

The size of this "tick" can be adjusted in the *settings* dialog.

You can choose between a photo x-section, and a sketch x-section. In the first case just take a photo that will remind you of the shape of the gallery when you will be drafting the cave map. Make sure to have a fellow caver in the picture who will serve as a size reference.

In the second case you sketch the x-section on a canvas with the usual drawing tools of TopoDroid. As a reference TopoDroid displays the leg shot traversed by the x-section and the splay shots at its endpoints. The station in front (the one that the x-section is looking at) is centered in the sketch grid. The splay shots in front are light grey, those behind are thinner darker grey. All the shots are displayed projected on the plane of the x-section. A white dot marks where the section plane intersects the leg [T].

Leg x-sections can be re-opened and modified by selecting, in "edit" mode, the section-line or the section-point, and tapping the *edit* button.

X-sections in plan view

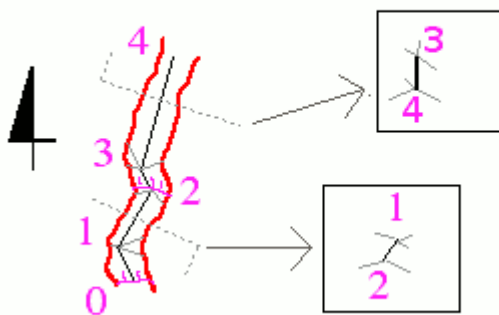
X-sections defined in the plan view are always in a vertical plane (zero

inclination). Their orientation (ie, the azimuth of the x-section plane) is determined by the angle of the section line. Very often the section plane is perpendicular to the gallery, but can be at any needed angle. For example, a section line drawn from left to right (e.g. horizontal, angle 90° from the vertical) has a direction of view to the magnetic North (up, azimuth 0). If drawn slightly downwards at an angle of 120° , it will have a direction of view to an azimuth of 30° .

The crossed leg is displayed vertical when the x-section has the same azimuth as the leg, i.e., when the section line is perpendicular to the leg. Otherwise it is tilted right or left.

In plan view it is possible to make a x-section that crosses multiple legs. All the crossed legs and the splays at their stations are shown as x-section sketch reference. The crossing point is marked on every leg.

Plan view cross sections



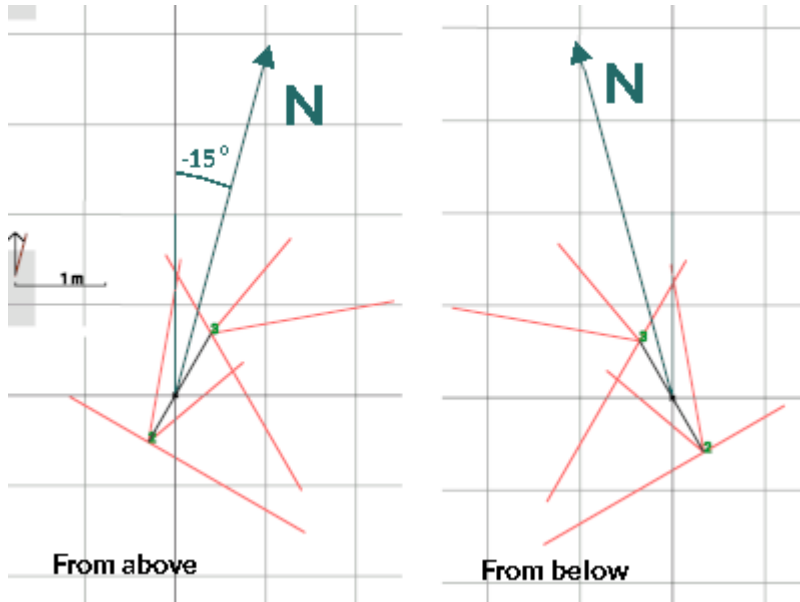
X-sections in profile view

X-sections in the profile view can be either "horizontal" (on vertical shots in pits) or "vertical" (on more or less horizontal galleries). They are considered "horizontal" if the inclination of the leg exceeds the value of the H-threshold setting.

"Vertical" x-sections must be defined tracing an almost vertical section line. They always have the azimuth of the crossed leg (or the opposite one if the direction of the section line is the opposite of that of the leg). The crossed leg is displayed vertical as the x-section has the same azimuth as the leg.

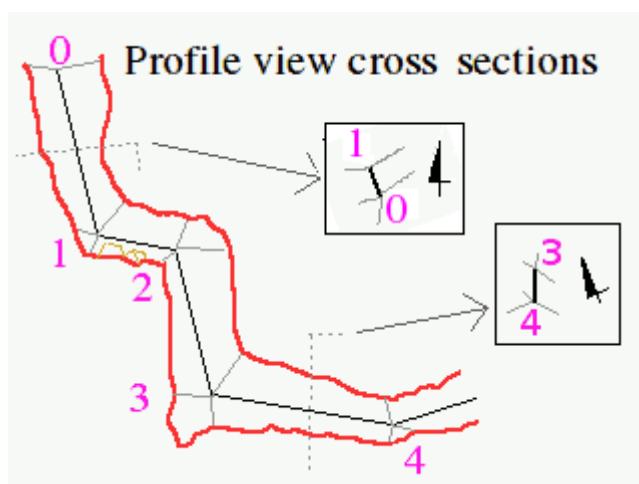
"Horizontal" x-sections are defined tracing an almost horizontal section line. If the section line is traced from the right to left, the x-section is oriented downwards and has -90° inclination. This is a "horizontal" x-section as seen from above. Most of the times you will want to draw "horizontal" x-sections this way, as it is more natural to lay a map on a surface and look down at it. If you trace the section line from left to right the x-section is oriented upwards (inclination $+90^\circ$).

Horizontal x-sections are displayed always with the magnetic North up. They are seen from above if the section line is traced from right-left, and seen from below if it is traced left-right. A reference line 4 m long, oriented to the North, is added to them. If the section is seen from above the North line is to the right of the vertical direction when the declination is negative, and to the left when it is positive. If the section is seen from below the North line is to the left when the declination is negative and to the right when it is positive. The leg is shown at its azimuth



If the "Slant x-sections" setting is enabled it is possible to define profile-view x-sections in planes at variable inclination (a multiple of 10° close to the slope of the section line).

In extended profile view the leg x-section must cross a single leg. Multileg x-sections are allowed in projected profile views. For vertical x-sections the azimuth of the x-section plane is orthogonal to that of the profile view for projected profiles and orthogonal to the crossed leg for extended profiles.



Leg x-section dialog

The leg x-section dialog shows

- the leg stations
- the x-section id
- a text field for a x-section comment

For multiple leg x-sections, instead of the leg stations the dialog lists the id's of the legs crossed by the x-section.

Station x-sections [E]

This is another way to insert x-sections. The "Station" dialog, that you get by tapping on a station point in *Edit* mode, displays the station coordinates and has some buttons:

- *Save* saves the entered comment.
- *Ok* continues the survey from the station. Set the station as the "active station", ie, that to which the next shot will be attached.
- *Splays* toggles the display of splays at the station.
- *X-Section* opens the x-section at the station.
- Three buttons to create a x-section at the station.
- *Saved station* opens the *Saved stations* dialog setted to the current station so it can be saved.

If the mean clino of the legs at the station is smaller than the shot "vertical threshold" setting, the x-section at a station is drawn in a vertical plane passing through the station, otherwise it is horizontal. The station x-section is useful when you take several splay shots around the station to describe the shape of the gallery. The x-section plane can be oriented with the *compass* button. If the station has only two legs the plane is oriented along the bisecant of the angles formed by the (horizontal projections of the) legs. If there is only one leg it is oriented orthogonal to that. If there are more than two legs you have to use the *compass* button.

Station x-sections can be re-opened and modified by selecting, in "edit" mode, the station or the section-point, and tapping the *X-Section* button.

Section-points

When you insert a x-section, either by drawing a section-line or inserting it at a station, a related section-point is added to the sketch. This point marks the place of the x-section on the sketch. Section-points are displayed as a white small cross and an orange line connecting them to the relative section-line or station.

If the x-section is deleted, the corresponding section-point is also deleted.

The outline of the x-section is displayed at the section-point.

The outline can be moved by selecting the section-point (in edit mode) and dragging it around.

Tapping the "properties" button, when a section-point is selected, opens the x-section sketch for drawing.

Section-point items cannot be delete directly. They are removed when the x-section is deleted. The x-section is deleted only deleting the *section-line*, for leg x-sections, or through the station dialog, for station x-sections.

X-section files

Each x-section is saved in a separate file in the "tdr" subfolder. The files get consecutive names "MySurvey-xx0.tdr", "MySurvey-xx1.tdr", etc. The "xx0" numbering part of the name is also displayed in the Section dialog that pops up when you create or edit a section.

The file is deleted when a x-section is deleted.

X-sections can be exported in other formats just like the other sketches, except in *cSurvey* format. X-sections are included in the sketch exports in the DXF, SVG, *cSurvey*, at the associated section-point.

See also

[Sketch station edit dialog](#),

[Sketch line edit dialog](#).

[12] OVERVIEW WINDOW

You get into the Overview window from the *overview* menu of the Drawing window.

The Overview window displays all the sketches together, either in plan or in extended profile view depending on the view of the Drawing window. The Overview window can be either in normal mode (canvas shift and zoom), or in measure-mode.

BUTTONS

-  puts the window in "measure mode"
-  reference. Dialog to switch on/off the display of midline, splay shots, station names, sketch grid, and scalebar.
-  measurement. If "angle", distance and angle between two points. If "polyline", total length of a polyline traced by tapping the positions of the nodes



undo. It removes the last point of the polyline

When the window is in *measure mode* the first button has a bright green color. In this mode the drawing can be translated with the corner-drag (if enabled) and zoomed with the zoom buttons (if enabled).

Segment measure

When the Overview window is in measure-mode you can measure the distance between two points on the canvas. To do this you tap first on the first point: this is displayed as a pink circled cross. Next you tap on the second point: this is shown with a blue cross and the 2D distance is shown in the title bar, together with the differences along the X and Y axes, and the bearing (plan) or inclination (extended profile). The distance units is the the sketch grid units (meter, yard, 2-ft, or dm) by default, but it can be set to m or ft with the proper setting. The angle is in the current angle units (degrees or grad).

Any further tap changes the second point (and the measure values).

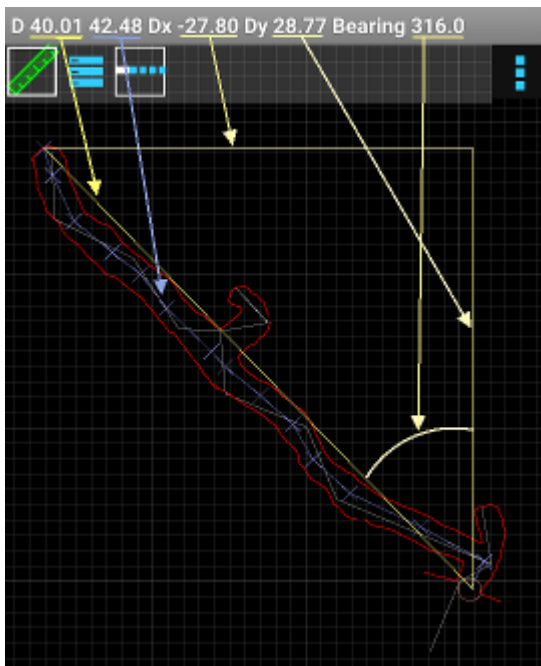
To end the measure tap on the "measure-mode" button.

The distance is computed in the plane of the canvas. In plan view, this is the horizontal distance. In extended profile view the Y-axis difference is the difference in elevation.

Polyline measure

The "polyline" measure is useful to get an estimate of the "length" of the cave, more sensible than the length of the midline. You tap on the start of the polyline and the additional taps mark the nodes of the polyline. If you make a mistake tap the *undo* to remove the last points one at a time.

The title displays the distance between the two end points, the polyline length, the X-Y displacement between the endpoints, and the angle.



MENUS

- Close the window
- Export [T]
- Settings
- Help

Export

The overview can be exported in the following formats:

- *Therion*
- SVG
- DXF (*LibreCAD*)
- shapefile (*QGIS*)
- xvi (*xtherion*)
- PDF

Each sketch is mapped to a "scrap" in the *Therion* export. For the other export formats, the sketches are put in a single drawing.

Station points are not included automatically in the *Therion* export; only user defined station are added.

The stations are added automatically to SVG, DXF, shapefile, PDF and xvi exports.

If there are "section" points, the x-sections are included.

[13] IMPORT EXPORT

When you are done, export the survey data and the sketches in your favourite format and transfer the exported files to your PC, to finish up drafting the cave maps.

Survey data, and zip archive, are exported from the Survey Info window. Survey sketches are exported from the Drawing window (single sketch) and from the Overview window (all sketches of a view in one file).

TopoDroid zip archives are stored in the *zip* subfolder of the "TopoDroid" folder.

Exported data and sketches files are stored in the "out" subfolder of the survey folder.

The destination file of exports of 3D models are chosen with a system filemanager.

ZIP ARCHIVES ".zip"

For safety make a TopoDroid zip archive and transfer it to your PC. The zip archive is a complete backup of your survey, and it contains the command to restore the survey data and other information in the database. You can use it to transfer the survey to a TopoDroid on another Android.

Zip archives are exported from the Survey Info window, from the "Export" menu.

When exporting a zip archive you can check to immediately share it afterward. If you have a custom app to handle exported zip archives you can override the default system sharing apps and have it invoked specifying its "category".

Zip archives are stored in the *zip* subfolder of the current TopoDroid work folder.

The name of the zip archive is the survey name.

By default, the zip file is overwritten when the archive is exported again. If the user unchecks the *Overwrite* box, the zip files are not overwritten and the zip filename is the survey name followed by a number

Zip archives are imported from the Main window with the "Import" button.

Zip import fails if the zip has a database version that is too old or is newer than the one in TopoDroid. The error codes are

- 1: IO error
- 2: zip TopoDroid version too old
- 3: zip database version too old
- 4: zip database version too new
- 5: survey name differs from filename
- 6: survey already present
- 7: failed to open database
- 8: zip SQL error
- 9: number format error
- 10: manifest format error
- 11: manifest not found
- 12: manifest IO error
- 13: unexpected error

Note, splay custom colors are not imported unless the appropriate setting is enabled.

[Survey data import](#)

SURVEY DATA EXPORT

TopoDroid exports survey data to the following cave-program formats:

- [Compass](#) ".dat"
- [cSurvey](#) ".csx"
- [Survex](#) ".svx"
- [Therion](#) ".th"
- [TopoRobot](#) ".text"
- [VisualTopo](#) ".tro", ".trox"
- [Walls](#) ".srv"
- [WinKarst](#) ".sur"
- [Polygon](#) ".cave"
- [Topo](#) ".cav"

Furthermore it exports survey data as

- [CSV](#) ".csv"
- [DXF](#) ".dxf"
- [KML](#) ".kml"
- [GPX track](#) ".gpx"
- [shapefile](#) ".shz"

If the survey is not georeferenced the survey data cannot be exported in the last three export formats.

Georeference points are exported with coordinates latitude-longitude [decimal degrees] and altitude above mean sea level [m].
H/V-splays are treated as normal splays in the exports.

The dialog to export the survey data presents only the format that are enabled in "ENABLE" subset of the "Import/Export" settings.

The settings affecting the exported data/file are listed for each export format.

The exported file can be directly shared by checking the *Share* checkbox.

The following export formats are no longer supported:

- [Grottolf](#)
- [GHTopo](#)
- [PocketTopo](#)
- [PLT file \(OziExplorer\)](#)
- [GeoJSON](#)

Compass ".dat"

Compass data file are supported both in import and in export.

Supported data format are "normal" and "diving".

Data units are 'feet' and 'degrees' (converted to meters and degrees inside TopoDroid).

Settings:

- Station prefix: if given it is prepended to the station names
- Splays
- L/R swap

cSurvey ".csx"

The cSurvey export files require cSurvey v. 1.20 or later.

The cSurvey export from the Survey Info window includes only the survey data, and does not have any sketch.

To make a cSurvey export with a sketch (both plan and profile), you should export from the Drawing window.

The cSurvey export includes the survey notes.

It also includes the calibration-check shots and can include media files (audio records and photo).

Settings:

- station prefix: whether to prefix the station names with the survey name
- media export: whether to include media in the export

Survex ".svx"

Survey data can be exported as Survex files.

Data units follow the user options.

The TopoDroid team string is listed commented, by default.

If the "Team names" setting is set to something other than "Text plain", the TopoDroid team is parsed and split into the Survox *team* commands.

Settings:

- line termination can be either Linux or Windows
- named TO station for splay shots (the TO name is the FROM station with a letter appended)
- add LRUD to export
- Survox "cs out" EPSG number

Therion ".th"

Survey data are exported in *Therion* format, using the data "normal" form.

The TopoDroid survey name is used for the *Therion* cave name, with underscore '_' replaced by space.

Data units follow the user options.

The TopoDroid team string is listed commented, by default.

If the "Team names" setting is set to something other than "Text plain", the TopoDroid team is parsed and split into the *Therion team* commands.

Station comments and marks (fixed, painted) precede the centerline data.

X-splays are exported with station "-" (dash), the others with station "." (dot).

Commented shots are exported under comment (#).

For the extend the syntax "extend <flag>" is used before any leg, with flag equals to "left", "right", or "vert". For splays that do not have extend, TopoDroid writes "# extend auto", as a comment because it is not a valid *Therion* command.

Fractional extend is supported in *Therion* since v. 5.2.2 and it is exported if it is set.

Fixed points coordinates (longitude latitude and altitude a.s.l.) are exported commented. If the point has custom CRS coordinates, these are included as well, preceded by the CRS name.

Furthermore "input" of the th2 files and definitions of maps for each scrap are added at the end, but are commented. They must be uncommented to get the drawings in the *PDF* output of *Therion*.

The *Therion* export includes the calibration-check shots, as comment.

Settings:

- Therion thconfig: if enabled, a simple thconfig file with *Therion* configuration commands is created.
- Therion maps: if enabled the "map" commands are placed before the centerline.
- add LRUD to export

thconfig

With this option TopoDroid exports a *Therion* configuration file. This file

contains the configuration commands to create the pdf of the plan and extended profile maps.

Running *Therion* on the exported file generates a simple *PDF* map.

The TopoDroid parsing of *Therion* data syntax is very limited. Only data in "normal" form are understood. The *Therion* data commands "units", "flag", "extend", "mark" and "station" are supported.

Fixed points ("fix") are parsed but not taken into account as "cs" is not supported.

TopoRobot ".text"

TopoRobot data file are supported both in import and in export.

Survey data can be exported as *TopoRobot* files. The *TopoRobot* station naming policy must have been selected. If the station names follow *TopoRobot* convention "series.number", they are used.

For *TopoRobot* export the filename, including the extension, can be specified in the export dialog. The filename can include only alphabetic characters, a-z and A-Z, and digits, 0-9. If it is longer than 8 characters it is truncated to the first eight characters. The name of the survey in the file is the filename without extension.

The export can also be limited to the last portion of data, by specifying the index of the first shot to export.

VisualTopo ".tro", ".trox"

VisualTopo data file are supported both in import and in export.

VisualTopo splays are included if enabled.

By default LRUD are at the TO station, but they can be set at the FROM station.

Data units are 'meters' and 'degrees'.

Survey in *diving* data-mode are exported in "trox" format.

Settings:

- Tro old format. By default data are saved in "trox" format
- Splays
- LRUD at From
- Station series: if specified the station names are qualified with the series
- Faverjon whether the series is a prefix instead of a suffix.

Walls ".srv"

Walls data file are supported both in import and in export.

Survey data can be exported as *Walls* files (partially tested).

Data units follow the user options.

If the "Walls splays" setting is set the splay shots are exported as "named" splays (as for *Compass*). Otherwise they are exported as "anonymous point shots", (wall-shots).

Settings:

- Walls splay: whether to export splays as splay-shots, "12 12-1", or as wall-shots, "12 -".

WinKarst ".sur"

Survey data can be exported as *WinKarst* files (untested).

Settings:

- Station prefix: if given it is prepended to the station names

PocketTopo ".top"

TopoDroid supports *PocketTopo* only in import.

The "outline" and "profile" sketches are imported. *PocketTopo* uses only a few colors to differentiate line types. Therefore a [mapping](#) from colors to TopoDroid types is used. The mapping can be changed through a setting.

It may happen that the imported sketch is not aligned with the survey data. In this case you can shift the sketch in the Drawing window, to fit the centerline.

Polygon ".cave"

TopoDroid can export survey data in *Polygon* format. Fixed points are exported as lat-long.

When exporting, you can choose to convert the splays to LRUD data or number them in S_xx format or add a - sign to the name of the TO point.

Topo ".cav"

Survey data can be exported as *Topo* files.

Settings (under "Survex" settings):

- line termination can be either Linux or Windows

CSV ".csv"

Survey data can be exported as a list of records with comma-separated fields:

- "from" and "to" stations
- distance, azimuth, clino
- flag (L duplicate, S surface, C commented)

Shot comments are not exported. A brief header is prepended to the data list.

The survey full data can be exported in *CSV* format. In this case each record (line) has

- shot id
- "from" and "to" stations
- distance, azimuth, clino, roll
- acceleration, magnetic field, dip
- timestamp, type (device or manual), device-address
- extend, flag, leg_type, status, comment
- raw values

In the full data export the leg shots are not averaged, but exported individually.

Settings:

- full data: whether to export the full data
- separator: fields separator (comma, pipe or tab)
- EOL: record separator (Linux or Windows)

DXF ".dxf"

Survey data are exported as 3D *DXF* with layers

- stations
- legs
- splays

Data are corrected with the magnetic declination.

Settings:

- Splay endpoints: if enabled they are marked with crosses

KML ".kml"

Survey data can be exported as *KML* file (*Keyhole Markup Language*) if you have georeferenced at least a station. If the survey contains disjoint pieces, the *KML* export includes all the pieces with a georeferenced station. Data are corrected with the magnetic declination.

Settings:

- stations
- splays

GPX track ".gpx"

The *GPX* export includes the stations (as waypoints) and the midline (as a

track).

The data coordinate system is WGS-84 geographic.

Shapefile ".shz"

Survey data can be exported as shapefile. There is a shapefile for the stations and one for the shots (both legs and splays). If the survey contains disjoint pieces, a shapefile is exported for each piece with a georeferenced station. Data are corrected with the magnetic declination. Each shapefile consists of three files: the data file, with extension ".shp", the index file, ".shx", and the attributes file, ".dbf". The files are zip-compressed and the zip is saved with extension ".shz".

If you have georeferenced at least a station and checked the appropriate checkbox in the export dialog, the data are exported in the WGS84 reference system, and corrected with the magnetic declination. Otherwise the data are exported in local coordinates and the magnetic declination correction is not applied.

The "stations" DBF contains only the field "name".

The "shots" DBF contains the fields

- *type*: either "leg" or "splay"
- *from* station
- *to* station ("- " for splays)
- shot *flag* as hex: 1-surface, 2-duplicate, 4-commented
- shot *comment*

Settings:

- stations: whether to export the stations
- splays: whether to include the splays

SKETCH EXPORT

TopoDroid exports survey sketches to the following formats:

- [cSurvey](#) ".csx"
- [Therion](#) ".th2"
- [xtherion](#) ".xvi"
- [DXF](#) ".dxf"
- [SVG](#) ".svg"
- [PDF file](#) ".pdf"
- [shapefile](#) ".shz"

The following sketch export formats are non longer supported:

- [PNG image](#)
- [Cave3D](#)

Screenshots of the sketch window and the overview window can be taken with the volume-up button. Screenshots are saved in the "out" folder of the survey.

cSurvey ".csx"

The *cSurvey* export files require *cSurvey* v. 1.20 or later.

These exports contain both the plan and profile sketches, as well as the survey data. When "section" points are defined the section drawings are added at the point.

The *cSurvey* "cave" attribute is set with the survey name, and the "branch" attribute with the sketch name (without the suffix 'p' or 's'). The midline legs and splays and the sketch items share these attributes.

If your survey has only one sketch this is a complete export of your work (except for the x-sections). If the survey has more than one sketch, you must export each sketch individually and merge the exports with *cSurvey*.

The sketch items are distributed on suitable *cSurvey* layers.

Settings:

- station prefix whether to add a prefix to the station names
- line-point spacing maximum distance between interpolating points on smooth lines

Therion ".th2"

Sketches are exported in *Therion* format as one sketch per file, if exported from the Drawing window, or as a file containing all the sketches with the same view, plan or extended, if exported from the Overview window. In the latter there is a scrap for each sketch, and x-section scraps are included if the automatic "section" points are enabled.

The *Therion scrap* names are composed of the survey name, the sketch name and a suffix, either 'p', for plan, or 's', for profile.

For multi-scrap sketches the *Therion scrap* names of the scraps, beyond the first, are appended also a scrap number, beginning with 1.

X-section *Therion scrap* names are composed of the survey name, a two-letter code, 'xx', and a number.

When a sketch with section-points is exported through the export dialog, the x-sections of the section-points are also exported.

If the sketch contains symbols that are not included in the *Therion* set, you must prepend the prefix "u:" to the unsupported symbol names, before using the th2 file.

Settings:

- automatic station-points. [default: yes].
- add splays to the scrap (as lines of type "splay") [default: no].

- add a xtherion command to read the **XVI** image of the sketch [default: no].
- export scale. Default 1:100
- point spacing minimum distance between points on polylines. Intermediate points are not exported. Default 20 cm

XVI images are not automatically exported with the *Therion* "th2" file, even if a xtherion command to include it is added to the *Therion* file, because it would not contain more information than the *Therion* file. The command can be used to load in xtherion an XVI image created by another program.

xtherion ".xvi"

The "xvi" export can be inserted as background image in xtherion. Lines of type "wall" are red, "slope" orange, "section" gray, all others are brown. Area borders are black. Point icons are rendered in a simplified way. Label rendering supports only latin characters and digits and a few characters ('+', '-', '/', '_', '>', '<', '?'). Other characters are replaced by a "diamond".

The sketch grid cell are 1 m.

The XVI settings are among *Therion* settings:

- grid lines added to the export (cell size in the current sketch grid units). Default no
- export scale. Default 1:100

DXF ".dxf"

Sketches can be exported as 2D DXF files. You can assign layers to individual symbol types (point, line, area) or levels. Use the Z coordinate to separate scraps or levels. The DXF file contains only a few basic line types.

Sketches can be exported as DXF 9 AC1009, 12 (R13) AC1012, and 14 (R14) AC1014. For DXF-9 areas are exported with only the border and are not colored inside.

It is possible to include in the DXF file, the x-sections (for which the sketch contains a "section" point), and the X-Y axis.

Settings:

- DXF version: 9 (R12), 12 (R13), or 14 (R14). Default "9"
- section-point: any section-line has its section-point, the drawing of the section is inserted in the export, at the "section" point
- XY reference: whether to insert the XY axes reference
- Scraps layer: DXF layers by scraps or type. Default "scraps"

PDF ".pdf"

Sketch can be exported as *PDF* files. Sketch *PDF* files are meant to be a quick way to share the survey with non-surveyors in a format that anyone can open.

The *PDF* file has a single page with a white background, and it includes only the reference items (stations, splays, sketch grid, etc.) that are displayed on

the screen.

The text of labels, scalebar value and stations are reduced by 1/4. The 1 m sketch grid is not drawn at scales smaller than 1:100. The color of certain items (eg, labels) is inverted, for visibility.

Shapefile ".shp"

Sketches can be exported as *shapefile*, with stations, shots, and lines. Each item has three files: the shape data file, the shape index file, and the database file. The files comprising the *shapefile* export are zip-compressed and the zip file is saved in the folder "shp" with extension ".shz".

If you have georeferenced at least a station and checked the appropriate checkbox in the export dialog, the data are exported in the *WGS84* reference system, and corrected with the magnetic declination.

Otherwise, the data are exported in local coordinates and the magnetic declination correction is not applied.

The "point" *DBF* contains the fields

- point *name*
- *orientation*
- *level*
- *scrap*
- optional *text* (truncated to 128 characters)

The "extra" *DBF* contains the fields

- point *name*
- *orientation*
- *level*
- *scrap*
- optional *text* (truncated to 128 characters)
- *file*

The "line" and "area" *DBF* contain the fields

- *type* that distinguishes between line and area
- line or area *name*
- *level*
- *scrap*

The "station" *DBF* contains the fields

- *name* of the station

The "shot" *DBF* contains the fields

- *type*: either "leg" or "splay"
- *from* station

- *to station* ("-") for splays)
- shot *flag* as hex: 1-surface, 2-duplicate, 4-commented

SVG ".svg"

Sketches can be exported as *SVG* files.

Sketch exports do not share a common frame of reference, but each sketch export has its own frame.

The magnetic declination is not taken into account in the export.

Items are organized in layers: *sketch grids*, *stations*, *legs*, *splays*, *points*, *lines*, and *areas*.

These layers are under a global *canvas* layer.

Each "scrap" has its *points*, *lines*, and *areas* layers.

The default export scale is 1:100.

If splay classes are enabled, the splays are colored by the class: normal splay are gray, horizontal are light sea-green, vertical light steel-blue, and x-section light sky blue. Splay-classes are not supported in "round-trip" export.

Line effects, such as ticks in pit lines, are not exported, but the direction can be shown enabling the "Line Direction" setting.

Settings:

- SVG program: either *Inkscape* or *Illustrator*. This affects the export scale factor. [Illustrator]
- automatic station-points: if enabled all stations are included in the export, otherwise only the user-chosen stations are included
- grouping items by the type (ie, *Therion* name)
- point spacing maximum distance between interpolating points on smooth lines
- round-trip: sketches are exported ready for *Walls/Compass* round-trip [no]
- grid lines added to the export (cell size in the current sketch grid units) [no]
- line direction tick [no]
- lines width: labels, icons, lines (and area borders), sketch grid lines, shots, orientation stroke
- station size: station names text-size. Default 20
- origin: whether to highlight the origin station with a orange cross [only in the export dialog]

IMPORT/EXPORT SETTINGS

- *Data default-export*: default data export format to present in the export dialog [default none]

- *Sketch default-export*: default sketch export format to present in the export dialog [default none]
- *Sketch auto-export*: if set sketches are exported in the chosen format when they are closed. The exported files are stored in the "out" subfolder of the survey folder. Sketch auto-export puts an extra load on the app, and is not set by default.
- *ENABLE*: settings to enable data export formats.
- **Compass**:
 - *swap LR*: swap L and R on *Compass* export
- **Compass/VisualTopo**:
 - *Estimate LR extend*: on import of *Compass/VisualTopo* files [yes]
- **Compass/cSurvey/WinKarst**:
 - *survey prefix* to the stations [no]
- **Compass/TopoRobot/WinKarst/Polygon/VisualTopo**:
 - minimum *splay-leg angle* for LRUD computation [0.0] This is also used in *Survex/Therion* if LRUD are required.
 - minimum clino of splays used for up/down LRUD [0]
 - maximum clino of splays used for left/right LRUD [90]
- **Therion**:
 - *map commands* before or after the centerline block [after]
 - *auto-stations*: automatically add station points to the *Therion* sketch file(s) [yes]. If you want to use the sketch files in *Therion* they must have station points to define their scale. You can choose manually which point to add (by editing the sketch drawing), or let TopoDroid add the stations. The program will add all the stations that are "inside" the drawing (inside the drawing convex hull).
 - *splays* lines in *Therion* scraps [yes]
- **Therion/Survex**:
 - export with *LRUD* lines [no]
- **Survex**:
 - export with *named splay* TO stations [no]
- **Survex/Topo**:
 - *line termination*: either Linux or Windows [Linux]
- **PocketTopo**:
 - *import color-map*: map of *PocketTopo* colors to sketch item types.
- **SVG**:
 - target *program* [Inkscape]
 - export with *sketch grid* [no]
 - export line orientation ticks [no]
 - stroke widths (labels, icons, lines, sketch grids, shots, orientation ticks)
- **DXF**:
 - export with *splay endpoints* [no]
 - *DXF version* for the DXF export: either 6, 12, or 16. In version 6 DXF export bezier curves are replaced by polylines and areas do not have hatches. Version 12 and 16 DXF exports use splines and hatches, but they are still buggy.
 - export with *stations* [yes]
 - export with *splays* [no]

[The final map](#)
[3D model export](#)
[Cave project export](#)
[Calibrations](#) import/export

[14] CAVE PROJECTS MANAGER

This window is accessed from the *Projects* button of the *Main Window*.

A cave project aggregates surveys that are in the TopoDroid database. The "cave projects" are stored in *Therion*-like files in the "thconfig" folder of TopoDroid, with extension ".tdconfig". The project name must coincide with the survey name in the project file.

A project file is saved whenever the project is closed.

The Project Manager displays the list of cave projects in the "thconfig" folder. The project list is empty if you have not created any project yet.

BUTTONS

- *New project*: opens a dialog to create a new cave project

MENU

- *Close the Project Manager window*
- *Help*

Tapping on a project name opens the [Project Window](#).

[15] 3D VIEWER

TopoDroid 3D viewer is designed for use in cave and it does not require internet connection.

It can load also other file formats: *Therion* (th, thconfig), *Loch* (lox), *Compass* (dat, mak), *Survex* (3d), and *VisualTopo* (tro, trox).

The 3D model can be opened either from a menu in the *Data Window*, or a button in the *Survey Info Window*.

The 3D model of a cave project can be opened from a button in the *Project Window*.

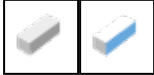
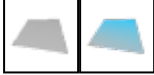
The model is shown with a reference cartesian grid (blue north, green east) or triplet (pink upward). The North is geographic if the magnetic declination is set.

If the data (survey or cave project) has fixed points these are used to compute the coordinates of the points in the model. If the fixed points have cartographic (projected) coordinates the cartesian grid is in the projected coordinate system. The coordinates in cartographic reference systems take into account the meridian convergence.

If the fixed points have only geographic (WGS84) coordinates, the coordinates are referred to the equator and the prime meridian and computed using the differences from the first fixed point: the latitude times the meridian radius, and the longitude times the parallel radius, respectively.

There is no coordinate conversion. Therefore all the surveys in a cave project must be geo-referenced in the same coordinate system.

BUTTONS

-  *Mode*: either rotate the model or move the model or the light (if there is a DEM). Two-finger always scale and translate the model.
-  *View*: either orthographic or perspective.
-  *Stations*: hide or show stations, as green points or names.
-  *Splays*: hide or show splays, as segments or endpoints.
-  *Walls*: hide or show walls (if created).
-  *Surface*: hide or show surface DEM (if present).
-  *Color*: midline color-coding: white, by-survey, denivelation, or depth below the surface.
-  *Frame*: XY grid with vertical line, or XYZ axes, or none.

The button icons are adjusted to the state of the program.

A few buttons have long-tap actions:

- *View*: Adjust the focal parameter of the perspective.
- *Stations*: Toggle station selection on/off.
- *Color*: Change the survey colors and select the surveys to show or hide. Up to Android-9 the station names are shown even when the survey is hidden.
- *Frame*: Open the *leg dialog* (display of surface/duplicate/commented legs).

MENU

- *Open file*: open the 3D model of a survey file.
- *Export*: Export dialog.
- *Info*: Survey 3D info.
- *3D-Rose diagram*: Distribution of 3D directions.
- *Rose diagram*: Distribution of azimuth directions.
- *Reset*: Reset to orthogonal top view, at zoom one.
- *Viewpoint*: Select a point of view among top, north, east, south, and west. For example "south" means looking at the cave from the South.
- *Surface*: Transparency and DEM file. Centreline projection on surface.
- *Wall models*: Create or clear a wall 3D reconstruction. Toggle plan/profile projections of walls
- *Search* stations that starts with a given prefix and highlights them when station dots are shown.
- *Options*
- *Help*

File input

The formats of input files are: *Therion*, *Loch*, *Compass* (dat), *Survex* (3d), and *VisualTopo* (tro, trox)

In any case there is a limited support of the syntax of these formats.

Therion data format must be "normal". The "input" command is supported, therefore it is possible to load whole *Therion* projects starting from the top file.

Loch files are self-contained 3D files created by *Therion*. They can contain also the DEM and a surface texture.

Compass and *VisualTopo* files are survey data files.

OPTIONS

- 3D Viewer:
 - *Below horizon view*: whether to allow viewpoints below the horizon [n].
 - *Station points*: whether to always display station points (green) [n].
 - *Station point size*: radius of station points [8 pxl].
 - *Station text size*: size of station names [20 pt].
 - *Selection radius*: Radius of selection for stations [1.0]
 - *Distance message*: show distance message instead of dialog [n]
 - *Top grid*: Show the cartesian grid on the top [n].
 - *Grid extent*: amount of cartesian grid extent around the survey [10 cells]. This takes effect the next time the 3D viewer is opened.
 - DEM:
 - *DEM buffer*: Extent of surface model around the survey on loading the surface model from a DEM file.
 - *DEM max size*: maximum size of surface DEM for each direction in lox files [400 cells].
 - *DEM reduction*: how to reduce the DEM if too large (subsample or clip) [subsample].
 - *texture root*: filesystem pathname of the storage folder ["/sdcard/"].
 - Wall Model (some settings apply to specific wall models):
 - *Splay usage*: Which splay to use (none, normal, x-sections) [normal]
 - *All splay*: Whether to use all the splays for the walls
 - *Projected splays*: Whether to use splays projected on a normal plane [n]
 - *Splay threshold*: splays outside the shot to keep [0.5]
 - *Split triangles*: Whether to split intersecting wall triangles
 - *Randomize points*: Add a small random vector to splays
 - *Stretch walls*: Shift wall points by a small quantity along the leg
 - *Powercrust*: Algorithm "delta" parameter

TITLE LINE

The title bar shows the viewing parameters:

- *C* and *A* clino and azimuth of the viewpoint.
- *S* is the zoom factor.
- *T* are the X-Y translations of the model.
- *L* are the clino and azimuth of the light.

EXPORTS

Model can be exported as DXF, KML, GPX, STL or STL-binary, CGAL, LAS-

binary, and shapefile.

The exported file is selected with the Android file manager.

The export dialog has options to include splays, walls, stations and surface. Not all items are supported in all export formats, in particular walls and surface:

- STL: Walls
- GPX: Legs, Splays, Stations
- KML: Legs, Splays, Stations, Walls
- SHP: Legs, Splays, Stations, Walls
- DXF: Legs, Splays, Walls (Stations export is to be implemented)

The model can be also serialized in a text file.

The GPX export of projects can have all the project surveys in a single track, or include a track for each survey. The behavior is determined by the GXP export setting "single-track".

If the model is georeferenced the KML, GPX and shapefile exports are in WGS84 geographic coordinates.

GRID/FRAME

It is possible to display a grid/frame of the cartesian axes:

- no grid/frame
- horizontal (East-North) grid with the vertical axis
- horizontal (East-North) grid with the vertical axis and level lines
- East and North axes with the vertical axis and level lines
-

The grid/frame lines are green (west-east), blue (south-north), and red (vertical).

In perspective the grid lines look greenish to the north/south because the green lines stack together closer. And it looks blueish to the west/east. To help distinguish the for cardinal directions, the grid lines get slightly reddish at the west-south corner.

By default the cartesian grid is below the survey, but it can be set above it (optionally). The line spacing (grid cell size) can be checked in the info dialog.

SURVEY

The survey splays can be hidden, displayed as lines or endpoints.

The survey legs can be uncolored or colored by survey, by denivelation (red-to-blue) or by the depth beneath the surface (blue-to-red) if the model has a DEM surface.

When uncolored the survey legs are white, the surface legs green, the duplicate legs orange, and the commented legs light blue.

The "surface", "duplicate", and "commented" legs can be turned on/off with the *leg dialog* which is opened by a long-tap on the *Frame* button (last).

When colored by survey the flagged splay are colored according to their flag, as in the plan/profile views.

WALLS

The wall model is not computed automatically when the file is opened, but it must be requested by the user through the "wall" menu. There are two general models, *simple hull*, and *pseudo convex-hull*, and the "debug-only" *powercrust* model. The first two are suitable when the splay are not dense. The last one is for dense splays.

The wall model depends on the number and the quality of the splays. Poor splays will result in a poor model.

The computation is done in background. When done the wall can be displayed with the "wall" button.

The [debug-only] *powercrust* model includes also the projection of the walls on the horizontal plane (plan view), and on a vertical curtain along the survey (profile view). Their display is enabled through the "walls" menu.

SURFACE

For Topodroid models, a surface (Digital Elevation Model) can be added from a DEM file. If externally loaded *Therion* or Loch models contain surface data (Digital Elevation Model and optionally a texture), the surface can be displayed either gray or with the texture.

The surface dialog has a button to load a DEM from a file. The supported formats are ascii (.asc or .ascii) and *Therion* grid (.grid). Furthermore, there is experimental support for HGT files.

Only the relevant portion of the DEM file is loaded. The DEM extent must cover the area of the model.

Files in **Esri ASCII Format** must have a header like in the following example:

```
ncols 374
nrows 225
xllcorner 7.444608
yllcorner 46.947922
cellsize 0.000213
NODATA_value -99.00
567.1 567.2 ...
...
```

If the number of rows or columns is negative, the data is in reverse order. This option is not included in the original Esri format!

Therion grid files must use a header like in the following example and may contain comments on lines starting with "#":

```
grid 7.444608 46.947922 0.000213 0.000213 374 225
567.1 567.2 ...
...
```

Coordinate Reference Systems (CRS):

The fixed stations of the survey and the DEM data file must all be in the same (cartographic) Coordinate Reference System.

By default, WGS84 is used, so the coordinates and cell size must be entered in degrees (with six or more decimals for good accuracy).

To use a different CRS, all fixed stations must be transformed into that other CRS using the "Convert" button in the respective "Geo-Points" dialog.

Notice: the separately available app "Proj4" must be installed as well (do not use the version from Play Store but the more current one from the Topodroid Website). Check the INFO dialog of the 3D Viewer to verify that the required CRS is effectively in use.

The DEM data must now be in that reference system instead of WGS-84 (depending on the CRS used, cellsize may now be in meters).

The transparency of the surface is adjustable. If fully opaque the portions of surface behind others are hidden.

The projection of the survey centerline on the surface can be displayed (blue).

When the model has a surface, it can be colored by the depth beneath the surface.

If the model has a surface DEM, the surface texture can be loaded from a file. The supported file formats are GeoTiff and OSM.

The GeoTiff file must be in the same Cartographic Coordinate Reference System of the survey. [Tech note: if the file is 24-bit non-compressed or uses a colormap Cave3D tries to read only the portion covering the DEM area, otherwise the whole image is read in and portion area is extracted.]

The OSM texture (WGS84) can be loaded only for surveys and cave projects opened from TopoDroid. The texture displays the OSM lines.

Note. Texture GeoTiff files are read by native code, thus the storage pathname (usually ending with ':') needs to be replaced by the filesystem pathname (usually "/sdcard/"): if this is not properly specified, the GeoTiff texture fails to load. The GeoTiff filename with "storage" prefix is shown as a toast message.

It is possible to visualize a point (yellow dot) on the surface by entering the East/North coordinates in the CRS of the model.

Models loaded from TopoDroid can get also GNSS points. If the app has permission to use the "Location" service, and this service is enabled, it is possible to get surface point with the Android GNSS.

given permissions to use it.

The app displays the last 10 points.

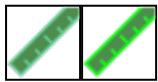
STATION ACTIONS

By default, when the stations are displayed, they can be selected tapping on the station point or name. A long tap opens a *Station dialog*. A short tap opens the *station bar* at the bottom of the screen.

The selected station is highlighted red, and its coordinates are shown in the *station dialog* or the blue *station bar* at the bottom. If the model has a DEM, the depth of the station below the surface is included.

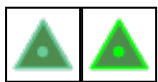
From the *station bar* it is possible

- set the station as "center" of rotation of the model
- "measure" the distance to another station
- "close" the *station bar* and deselct the station



To measure distances between the selected station and another station, check the "measure" button in the *station bar*, and tap on the other station. The 3D distance between the two stations, its East, North, and vertical components, and the plane 2D distance are displayed in a dialog. If the two stations are connected along the centerline the sums of the positive and of the negative "cave" denivelations between the two stations, and the "cave" distance is also shown and the cave-path between them is highlighted green.

Optionally, the values can be displayed in a brief message box instead of a dialog.



When the "station" button in the *station bar* is checked, the station position on the screen is fixed under rotations and the selected station is highlighted orange.

If the station is a connection between surveys ("equate" command) the *Station Dialog* lists all the station names in the different surveys.

The *Station Dialog* is closed clicking the "close" button or tapping outside of it on the screen. In the latter case the red dot on the station remains until the screen is tapped again.

CONTENTS

[Preface](#)

[Quick guide](#)

[Settings](#)

General

[Introduction](#)

[Work folder](#)

[Languages](#)

[Subdirectories](#)

[Databases](#)

[Ecosystem](#)

[Color coding](#)

[Permissions](#)

Main window

[Main window](#)

[Help dialogs](#)

[AI help dialogs](#)

[Setup](#)

[Settings](#)

[Secondary settings](#)

[Setting search](#)

[Keypads](#)

Device

[Device window](#)

[Device nickname](#)

[Devices](#)

[Bluetooth aliases](#)

[Bluetooth controls](#)

[Device troubleshooting](#)

[DistoX](#)

[DistoX1 info](#)

[DistoX2 info](#)

[DistoX packets](#)

[DistoX memory](#)

[DistoX firmware](#)

[BRIC](#)

[BRIC calibration](#)

[BRIC info](#)

[BRIC memory](#)

[SAP](#)

[JedEye](#)

DistoX/Cavway calibration

[Calibration window](#)

[Device calibrations list](#)

[Calibration data window](#)

[Calibration data](#)

[Data groups](#)

[Data distribution](#)

[Calibration coefficients](#)
[Calibration validation](#)
[Import/export](#)
[Calibration howto](#)

Survey

[Survey window](#)
[New survey](#)
[Import](#)
[Calibration check](#)
[Geo-points](#)
[Geo-point insert](#)
[GNSS geo-point](#)
[Geo-point import](#)
[Geo-point edit](#)
[Survey notes](#)
[Survey rename](#)
[Survey data info](#)
[Multi-shot actions](#)
[Final map](#)
[Photos](#)
[Photo dialog](#)
[Photo edit dialog](#)
[Photo view](#)
[Sensor measures](#)
[New sensor measure](#)
[Sensor measure edit](#)

Survey data

[Survey data window](#)
[Data list](#)
[Data download](#)
[Manual input](#)
[Instruments calibration](#)
[Accuracy](#)
[Display mode](#)
[Shot edit](#)
[Audio comments](#)
[Audio records listing](#)
[Extend reference](#)
[Station automatic naming](#)
[Station/Leg search](#)
[Active station](#)
[Saved stations](#)
[Geocoding](#)
[Survey item recovery](#)
[Data shot paste](#)
[Dangling data](#)
[Trilateration compensation](#)

Survey sketch

[Sketch window](#)

[New sketch](#)
[Survey sketch list](#)
[Projected profile sketch](#)
[Sketching references](#)
[Scrap outline](#)
[Sketch drawing](#)
[Drawing tools](#)
[Drawing tool picker](#)
[Drawing tool sets](#)
[Sketch backups](#)
[Undo and redo](#)
[Therion station points](#)
[Sketch item properties editing](#)
[Shot item dialog](#)
[Point item dialog](#)
[Label point dialog](#)
[Section point dialog](#)
[Photo point edit dialog](#)
[Line item dialog](#)
[Area item dialog](#)
[Point tools](#)
[Line tools](#)
[Area tools](#)
[Station editing](#)
[Shift and flip](#)
[Sketch statistics](#)
[Sketch export](#)
[PocketTopo import color-map](#)
[Sketch renaming](#)
[Sketch merge and split](#)
[Sketch zoom fit](#)
[Cross-sections](#)
[Overview window](#)
[Overview references](#)

Import Export

[Import export dialogs](#)

Cave Projects

[Cave project window](#)
[Cave Project](#)
[Cave Project dialog](#)
[Cave surveys](#)
[Cave equates](#)
[New cave equate](#)

3D viewer

[3D viewer](#)

TopoDroid SETUP

TopoDroid has a large number of [settings](#).

The setup dialogs lead the user to choose the most important ones:

- **text size:** this affects the size of text in the list displays as well as the size of station names and labels in the sketches
- **button size:** this is the size of buttons in the windows action bars and in the dialogs
- **icon size:** this is the size of icons in the sketches

TopoDroid SUBDIRECTORIES

Here is some background about how the survey data, the calibration data, and the sketches are stored.

The app creates its default base directory in the *primary external storage*, "Documents/TDX". This directory is not removed when you uninstall the app. In this directory the app creates the surveys folder "TopoDroid" with the SQLite database "disto14.sqlite" that stores the survey data, the surveys sketch files, and other surveys files.

The app also creates subdirectories for the different types of files. The following subdirectories are created only in the app private directory, that is "Android/data/com.topodroid.TDX/files" in the *primary external storage*.

- **bin:** firmware files (DistoX v. 2 and BLE)
- **ccsv:** CSV text files of exported calibrations (DistoX, Cavway)
- **dump:** memory dump files (DistoX, Cavway)
- **point:** point symbol drawing tools files
- **line:** line symbol drawing tools files
- **area:** area symbol drawing tools files
- **man:** user manual translation

"bin" is used to store firmware files, both the factory files included in TopoDroid and the firmware dumps read from a DistoX2.

The database "device10.sqlite", with the calibration data, and the device information is in this directory.

For each survey a survey-folder (named after the survey name) is created, with the following folders:

- **audio:** *audio* comments (wav files)
- **note:** survey annotations text file
- **photo:** survey pictures (jpg files)
- **tdr:** binary sketch files
- **out:** exported files

Android PERMISSIONS

Storage

TopoDroid saves the survey data and sketches, and the calibrations, in the *primary external storage*.

It cannot work without read/write access to the storage.

Internet

The network connection is necessary to install a translation of the user manual, and to use the AI help and setting assistants.

TopoDroid does not use the network in any other way, and it is safe to disable this permission if you do not want to use AI assistance nor install user manual translations.

Bluetooth

TopoDroid requires bluetooth access to communicate with the devices.

If the survey data are entered manually, it can be used without bluetooth.

TopoDroid uses also bluetooth pairing, to pair with newly discovered devices. Discovery and pairing of DistoX2 can be done with the "settings" app.

Discovery of BLE devices are done by TopoDroid itself

Camera

TopoDroid can use the camera to take pictures.

TopoDroid camera management is not so feature-rich as that of a "camera" app, but it stores the direction of view (azimuth and inclination) in the pictures.

It should be safe to use TopoDroid without this permission.

Audio

TopoDroid uses the microphone to record audio notes (for shots and audio points).

If this permission is not granted, audio recording is disabled.

It should be safe to use TopoDroid without this permission.

Location

Location is necessary to scan for BLE devices.

TopoDroid uses the device GNSS to get the geographical position of a point, but it can import it from [external apps](#).

GNSS is used also to track a point on the DEM in the 3D viewer.

TopoDroid DATABASES

It is useful to have a rough idea about the organization of the information and the data in the databases. The units of work in TopoDroid are the surveys and the calibrations. A survey is a single survey session of a piece of cave. A calibration is a single DistoXi/Cavway calibration.

Surveys are stored in the "distoX14.sqlite" database. There is one such database in every project folder.

Calibrations are stored in the "device10.sqlite" database, together with the device information. This database is global and is stored in the TopoDroid private "files" folder.

The most important table in the survey database is:

- **surveys:** survey information

The surveys records store the survey name, date, team, magnetic declination, initial station name and possibly a comment.

Related to the surveys are the tables for:

- **shots:** survey data
- **stations:** saved stations
- **plots:** 2D sketches; plans, profiles, x-sections
- **photos:** pictures associated to survey shots
- **audios:** audio comments associated to survey shots
- **sensors:** measurements associated to survey shots
- **fixeds:** (geolocalized stations)

The most important tables in the device database are

- **devices:** devices information
- **calibs:** calibrations information

The calibs records contain the calibration name, date, device, algorithm, and the results of the last computation: coefficients, average error, maximum error, number of iterations.

Related to the calibrations is the data table

- **gms:** calibration data

Finally each database has the table

- **configs:** configuration values

You can learn more about the data organization in the database by looking at it with a SQLite viewer.

Auxiliary apps

When working with TopoDroid, you might find useful some complementary and auxiliary apps:

- *Proj4* is a (very simple) coordinate converter app, used by TopoDroid to convert long-lat values to other coordinate systems
- A *GNSS app*. You can use it to take the point coordinates and import the coordinates in TopoDroid.
- *CrowdMag* is an app by NOAA National Geophysical Data Center, useful to get information about the current state of the Earth magnetic field.
- an app to take pictures (usually there is a *Camera* app)
- a decent File Manager, such as *MiXplorer*, to look at the files and to make backup copies
- *SQLiteViewer* (or any other SQLite manager app) can be helpful to inspect the TopoDroid databases. One can also change the records in the database, at his own risk.

GNSS apps

- *GPSTest* is considered "The GNSS app". TopoDroid reads the coordinates from a *GPSTest* log file. The coordinates can also be stored in the clipboard, and pasted into the TopoDroid geo-point insert dialog.
- *GPX Recorder* and *GPS Position*: TopoDroid reads saved tracks (gpx and csv files, respectively) and uses the coordinates of the last point.
- *GPS Point*: TopoDroid read the coordinates exported to a (plain-text) point file.
- *MobileTopographer*: TopoDroid read the coordinates from the point file in *MobileTopographer* pointlist. Unfortunately this app is no longer free.

In addition, depending how you process your surveys, you may want to install

- a image viewer/editor (usually there is a preinstalled app)
- a DXF viewer/editor
- a spreadsheet viewer/editor
- a SVG viewer

- a ZIP archiver
- *GoogleEarth*
- a track file viewer

Proj4

TopoDroid stores point coordinates in the reference system geographical long-lat WGS-84. Altitudes are stored both referred to the ellipsoid and to the geoid.

The app Proj4 is used by TopoDroid in two ways to transform coordinates between WGS-84 and other coordinate systems.

- to [convert](#) the coordinates of a point from long-lat (WGS-84) to another coordinate reference system.
- to [enter](#) the coordinates of a point in a different coordinate system.

The proj syntax for coordinate reference systems can be found on spatialreference.org.

TopoDroid COLOR CODINGS

Survey Data window

Station naming policy

Survey name at top bar:

- forward leg [default]: blue
- backward leg: dark blue
- foresight + backsight: violet
- tripod: pink
- magnetic anomaly: red
- TopoRobot: orange

Survey data

Foreground color:

- white: leg shot
- grey: leg repeated shot
- blue: splay
- green: x-splay
- dark-blue: h-splay
- deep-blue: v-splay

- orange: blank shot
- violet: failed back-splay
- others: user-assigned color

Background color:

- black: normal
- red: unreliable shot
- grey: commented shot
- orange: short leg

Survey stations

Foreground color:

- same color as survey data: non active station
- green: active station
- light yellow: last tapped shot in multiselection

Background color:

- black: normal
- dark green: latest data
- grey: multiselection
- dark yellow: DistoX backsight

Sketch legs

- white: normal
- red: unreliable
- orange: inconsistency with siblings

Survey splays

- blue: normal
- dark-green: commented
- green, dark-blue, deep-blue: splay classes
- others: user-assigned color

Drawing window

Survey stations

- pink: normal
- green: active station
- red: midline hidden after the station
- violet: midline hidden before the station
- yellow: saved station

Calibration data

Foreground color:

- blue, orange: data groups
- grey: unused data

Background color:

- violet: partially downloaded data (not used)
- red: error above 1 degree
- green: large difference from group average
- brown: saturated data
- grey: "deleted" data

TopoDroid SETTINGS

TopoDroid has a large number of settings, which are arranged in sections and subsections. Seldom used settings are collected under "second" sections.

Settings can be exported and imported. It is possible to restrict the export/import to specific categories of settings.

The "AI" button in the *Settings Window* opens a dialog for an AI-guided change to the settings.

- [Main](#)
- [Import/Export](#)
- [Survey data](#)
 - [Units](#)
 - [Shot data](#)
 - [Location](#)
 - [Accuracy](#)
- [Sketching](#)
 - [Points](#)
 - [Lines](#)
 - [Sketch](#)

- [Devices](#)
 - [Calibrations](#)
 - [3D viewer](#)
 - [DEM](#)
 - [Wall model](#)
 - [Main - secondary](#)
 - [Survey Data - secondary](#)
 - [Splay data - secondary](#)
 - [Sketching - secondary](#)
 - [Lines - secondary](#)
 - [Devices - secondary](#)
 - [Import/export - secondary](#)
 - [Settings import/export](#)
-

Main

- *Usage profile* {basic, normal, advanced, expert, tester}. [normal]
 - *Text size*: size of the text in lists. ≥ 1 . [14]
 - *Buttons size* {small, normal, medium, large, huge}. [normal]
 - *Symbols size* [18 pt]
 - *Hidden Navigation bar* [no]
 - *Screen orientation* (unspecified, portrait, landscape) [unspecified]
 - App [language](#) {default, Bulgarian, Chinese, English, German, Hungarian, Italian, Portuguese, Russian, Spanish.}. Default "default" (i.e. system locale)
 - *User manual*: available translations. To change this setting from english to something else TopoDroid needs internet access {English, Spanish, Italian, Russian, Hungarian, French, Portuguese, Chinese}.
 - *Analytics* of functions usage [enabled]
-

Import/Export

- General
 - *Data export format*: default data export type {cave,cav,csv,csx,dat,dxf,grt,gtx,srv,sur,svg,svx,th,top,tro} [none]
 - *Sketch export format*: default sketch export type {none,th,csx,dxf,svg,xvi} [none]
 - *Sketch auto-export format* [none]
 - *Compass LRUD splay angle*. LRUD minimum splay-leg angle.
 - *UD splay clino*: Up/Down minimum splay clino. [0°]
 - *LR splay clino*: Left/Right maximum splay clino. [90°]
 - *Point spacing*: distance between line points
- Enable: sets which export format options are available at the export dialog. Supported formats:
 - *Compass* (DAT)
 - *cSurvey* (CSX)
 - *Polygon* (CAVE)
 - *Survex* (SVX)

- *Therion* (TH)
- *Topo* (CAV)
- *TopoRobot* (TAB)
- *VisualTopo* (TRO, TROX)
- *Walls* (SRV)
- *Winkarst* (SUR)
- *CSV*
- *DXF*
- *KML*
- *GPX*
- Import:
 - [PocketTopo color map](#): map of *PocketTopo* colours to TopoDroid types
 - *Estimate LR extend* on import of *Compass* and *VisualTopo* files. [yes]
- *Survey* export:
 - *Line separator*: end-of-line chracters {Linux,Windows}. [Linux]
 - *Survey splays station*: add splays TO station. [no]
 - *LRUD*: add LRUD to the export. [no]
- *Therion* export:
 - *Config*: create *Therion* thconfig file [no]
 - *Maps first*: put *Therion* map commands before survey data. [no]
 - *Splays*: add splay lines to *Therion* scraps. [yes]
 - *LRUD*: add LRUD to the export. [no]
 - *Therion scale*: export scale. [1:100]
 - *xvi image*: insert reference to xvi image in th2 files
- *cSurvey* export:
 - *Station prefix* [none]
 - *Media export* [no]
- *Compass* export:
 - *Compass station prefix*. [no]
 - *Splays*: add splays to *Compass* files
 - *Swap LR*: swap Left and Right on *Compass* export. [no]
- *Polygon* export:
 - *LRUD*: include LRUD
 - *Splays*: add splays to *Polygon* files
- *VisualTopo* export:
 - *Splays*: add splays to *VisualTopo* files
 - *LRUD at FROM* station
 - *Trox format*: file format, tro or trox
- *Walls* export:
 - *Splays*: include splays in *Walls* files
- *SVG* export:
 - *SVG roundtrip*: round-trip ready *SVG* export [no]
 - *Grid*: add sketch grid to *SVG* export. [no]
 - *Line orientation* [no]
 - *Splays*: include splays. [no]
 - *Point stroke* [0.5]
 - *Label stroke* [1.5]
 - *Line stroke* [2.5]
 - *Grid stroke* [2.5]
 - *Shot stroke* [2.5]

- *Orientation tick* size. [12]
 - *Station size* [32]
 - *Label size* [48]
 - *SVG program*: target SVG program. *Inkscape* or *Illustrator*
 - *Shapefile* export:
 - *Georeferenced sketch*: export georeferenced plan-sketch [no]
 - *DXF* export:
 - *DXF splay endpoints* [yes]
 - *DXF version* {6,12,16}. [6]
 - *Splines*: use splines for Bezier segments
 - *Reference*: X-Y axes are *DXF* sketch export
 - *Scrap layers*: *DXF* layers by scraps
 - *KML* export:
 - *KML stations*: add stations to *KML* export. [yes]
 - *KML splays*: add splays to *KML* export. [no]
 - *GPX* export:
 - *Single track* export data as single *GPX* track [yes]
 - *CSV* export:
 - *Complete CSV*: export whole data to csv. [no]
 - *Separator*: field separator [pipe]
 - *Line separator*: line separator [line-feed]
-

Survey Data

- *Team*: default survey team. By default it is unset. When you create a new survey you must enter the team.
 - *Stations policy*: survey stations name assignment policy {none, splay+fore, splay+back, fore+splay, back+splay, back+splay+fore, tripod, magn.anomaly}. The number of choices varies with the usage profile. Default "splay+fore".
 - *Station names*: {alphanumeric,numeric}. [alphanumeric]
 - *Initial station name*: by default it is unset, and the program uses "0". Special characters are not allowed.
 - *Thumbnails size*: photo thumbnails size [80,400]. [200 px]
 - *Editable stations*: inline-editable stations [no]
 - *Sketch origin*: origin in common to all sketches or not
-

Units

- *Length units*: {m, ft}. [m]
 - *Angle units*: {deg, grad}. [deg]
 - *Sketch units*: sketch grid units {m, yd, 2ft, dm}. [m]
 - *Ruler units*: measure ruler units {cell, m, ft}.
-

Shot Data

- *Leg tolerance*: tolerance among the shots of same leg > 0%. [0.05, ie, 5%]

- *Maximum shot length* [50 m]
 - *Minimum leg length* [30 cm]
 - *Leg minimum shots*: minimum number of shots per leg. [3]
 - *Shot vertical extend*: shot vertical threshold. Region around 90°, where shots are extended "vertical" [0,90]. [10°]
 - *Leg WENS threshold*: leg vertical threshold [0,90]. [80°]
 - *Manual extend reference*: "extend" reference. Settable or manual L/R extend reference. [no]
 - *Prev/Next buttons*: prev/next buttons in the shot edit dialog. [yes]
 - *Backsight*: backsight fields in user entered shot dialog. [no]
 - *Leg-shot feedback*: leg identification feedback {none, sound, haptic} [no]
-

Location

- *Coordinate units*: location units {ddmmss, dec.deg}. [ddmmss]
 - *CRS*: name of CRS, as used in Proj4. [Long-Lat, ie, geographic WGS84]
 - *Neg. altitudes*: allow negative altitudes
 - *Editable altitude*
 - *Fine-position*: fine position time in seconds
 - *GNSS app*: preferred GNSS app {non-specified, Mobile Topographer, GPX Recorder, GPS Position, GPSTest, GPS Logger} [non-specified]
-

Accuracy

- *Acceleration tolerance*: percent tolerance for the shots acceleration > 0. [1%]
 - *Magnetic field tolerance*: percent tolerance for the shots magnetic field > 0. [1%]
 - *Magnetic dip tolerance*: tolerance over the shots magnetic dip > 0. [2.0°]
 - *Sibling threshold*: threshold for sibling shots tolerance > 0. [5%]
-

Sketching

- *Corner drag*: enable corner-drag in Drawing window and Overview window. [no]
 - *Zoom controls* {off,temporary,permanent}. [off]
 - *X-section H-threshold*: x-section horizontal threshold
 - *Midline check*: check if stations are attached to midline. [yes]
 - *Leg extend check*: check if leg extends are set [yes]
 - *Toolbar size*: recent tool number {3, 4, 5, 6}. [4]
 - *Toolbar update*: how new tools are inserted in the toolbar: put at front, replace oldest, or drop oldest and put at the front [replace oldest]
 - *Sketch cache* [yes]
-

Points

- *Non-zoomed point icons*: point icons not zoomed [no]
 - *Point tools scale* [1.2]
 - *Labels size* [24 pxl]
 - *Labels scaling* [no]
 - *Section point offset* [20 pxl]
-

Lines

- *Line width*: drawing line width (N.B. wall line are twice this). [1 pxl]
 - *Line style scale*: scale factor of line drawing styles
 - *Line style* {Bezier, fine, normal, coarse, simplified}. [normal]
 - *Line-point spacing*: minimum distance between line points. [10 pxl]
 - *Close closed lines*: whether to close lines of "closed" type. [yes]
 - *Slope line l-size*: section lines "direction tick" size [1,20]. Default 8 [x unit]
 - *Section line tick size*: size of the section line tick
 - *Area border visibility* area border default visibility [yes]
-

Sketch

- *Survey line width* [0.5,10]. [1 pxl]
 - *Station name size* [24 pxl]
 - *Green dots size*: green dots radius [0.5,100]. [5 pxl]
 - *Splay transparency* [80% opacity]
-

Devices

- *Bluetooth*: check if BT is enabled on start. [yes]
 - *Data connection mode*: {on-demand, continuous, multi}. [on-demand]
 - *Number of new data*: get the number of new data before batch download
 - *Socket*: BT socket type {normal,insecure}. [device dependent default]
 - *Auto-pairing* [yes]
 - *Connection feedback*: type of connection status feedback {none, sound haptic} [none]
 - *Unname devices*: manually add unnamed devices
-

Calibrations

- *Group policy*: default group policy {TopoDroid, PocketTopo}. [TopoDroid]
- *Group tolerance*: tolerance among shots in the same group. [40°]
- *Algorithm error*: calibration algorithm termination error. [0.000001]
- *Algorithm iterations*: calibration maximum iterations. [200]

- *Download remote data*: download remote calibration data immediately
 - *Raw data*: show calibration raw data. {None, G-M Decimal, G-M Hex, G Decimal, G Hex, M Decimal, M Hex} [None]
-

3D viewer

- *Below-horizon views*: allow below-horizon views [no].
 - *Station points*: always display stations [no].
 - *Station point size*: size of station points [8 pxl].
 - *Station text size*: size of station text [20 pts].
 - *Selection radius* [50 pxl]
 - *Distance message*: show distance message instead of box [np].
 - *Top grid*: grid over instead of below model [no].
 - *Grid extent* [10 cells]
 - *Visible names*: whether station names are always visible even when the survey is hidden (above Android 9) [no]
-

DEM

- *DEM buffer*: extension of DEM buffer around survey [200 m].
 - *DEM max size* for lox files [400 cells]
 - *DEM reduce*: DEM reduction mode. {subsample, clip} [subsample]
-

Wall model

- *Splay usage*: the parser use of the splays data {none, normal, x-section} [normal]
 - *All splays*: whether to use all splays for Convex Hull model [yes].
 - *Projected walls*: whether to use splays projected on a normal plane [no].
 - *Splay threshold*: splay minimum value of the splay-shot cosine (Hull model) [0.5].
 - *Split triangles*: whether to split intersecting wall triangles (Convex Hull model) [yes].
 - *Randomize points*: small random vector added to splays (Convex Hull model) [0.1 m].
 - *Stretch walls*: shift wall points by a small quantity along the leg (Convex Hull model) [0.1 m]
 - *Powercrust Delta*: exclude splays differing from nearby too much (Powercrust model) [0.1 m].
-

Main - secondary

- *Single back*: whether to close windows with single back [no]
- *Navigation bar*: whether the navigation bar is hidden and is only shown with a slide-up [no]

- *Palettes*: additional palettes menu [no]
 - *Keyboard*. Enable [second keyboards](#) instead of Android default keyboard [no]
 - *No keyboard cursor* [yes]
 - *Bulk surveys operations*
 - *Packets log* [no]
 - *Sketch editing: Therion th2 editing* [no]
 - *AI key* (Gemini API key)
-

Survey Data - secondary

- *Diving mode*
 - *Shot tampering*: allow modifying downloaded data
 - *Backsight-check*: if first splay after a leg should be considered a backsight check of the leg (only for forward policy with leg after splays: failure of backsight check is highlighted violet). [no]
 - *Recent shots*: recent data highlight.
 - *Recent data timeout*
 - *Fractional extend*: whether to allow fractional extend. [no]
 - *Backshot DistoX*: whether to swap stations for DistoX backshots. [no]
 - *Plane fitting*: enable plane fitting of splays
 - *Sensores*: enable Android sensors [no]
 - *Loop closure*: none, normal, weighted, selective, triangles [no]
 - *Loop threshold*: loop selective closure threshold (percent of loop length)
 - *Android compass*: enable Android compass/clino
 - *Shot direction timer*: shot direction timer countdown. [10 s]
 - *Beep volume*: beep volume for shot direction timer and leg feedback [10,100%]. [50%]
 - *Beep length*: beep duration for leg feedback [milliseconds]. [500]
 - *Blunder shots*: skip blunder leg-shots (shots that differ only in length)
 - *Splay station re-assign*: maintain assigned splay stations on renumbering
 - *Splay group-rename*: splay group-station assign only forward
-

Splay data = secondary

- Splay classes [no]
 - Splay coloring: none, color-picker, color-palette [none]
 - Splay plan-view threshold. [80°]
 - Splay dash-mode: none, by azimuth, by clino, by the view. [none]
 - Splay dash clino threshold. [50°]
 - Splay dash azimuth threshold. [60°]
 - Splay dash angle threshold. [60°]
 - Color of dash splay
 - Color of dot splay
 - Color of latest splays
-

Sketching = secondary

- *Sketch shift/scale* [no]
 - *Sketch split/merge* (split, copy, outlines, merge) [no]
 - *Stylus only drawing*: stylus maximum size for stylus X hand touch in pixels. Zero means ignore [0]
 - *Backup number*: number of sketch file backups [5]
 - *Backup interval*: minimum interval between backups [60 s]
 - *Saved station*: enable saved station coloring [no]
 - *Legs-only refresh*: only update midline on new legs [no]
 - *Affine transform*: enable full affine sketch transforms [no]
 - *Layers*: enable canvas layers [no]
 - *Graph paper scale*
 - *Slant x-sections* [no]
 - *Oblique projected*: maximum oblique projected angle [0°]
 - *Line ends-cut*: number of points to drop from the line ends when tracing stylus-only [0]
 - *Zoom-fit bound*: zoom-fit minimum "scale" [0.1]
-

Lines - secondary

- *Straightening angle* [45°]
 - *Interpolation accuracy*: for Bezier lines
 - *Bezier corner threshold*: Bezier corner detection threshold
 - *Weeding point distance*: max point/segment distance for weeding. [0.5 m]
 - *Weeding segment length*: max segment length for weeding in meters. [2.0 m]
 - *Weeding segment buffer*: buffer around a line segment in pixels.
 - *Line/Area snap* [no]
 - *Smooth/straighten segments*
 - *Straighten lines* [no]
 - *Path multiselection*: enable line-area path multiselection.
-

Device - secondary

- *BT aliases*: manually set BT aliases for measurement devices.
- *unnamed devices* [no]
- *Connection delay* [0 sec]
- *Double-Device survey*: enable surveys with two DistoXs [no]
- *Data pause*: pause after data download [250 msec]
- *Data ready wait delay*: waiting delay for data to be ready [500 msec]
- *Laser pause* [2000 msec]
- *Shot pause* [2000 msec]
- *Firmware check*: firmware sanity check [yes]
- *BRIC mode*: BRIC data download mode. {Prim. only, All info} [all]
- *BRIC zero shot*: whether to record BRIC zero-length shots [no]
- *BRIC index*: whether to use BRIC index as shot ID [no]
- *SAP5 bit-16 bug* [yes]

Import/export - secondary

- *Zipped symbols files*: store/load symbol files to/from zip archives
- *Import datamode*
- *Auto X-sections*: insert X-Sections in sketch exports
- *Stations*: include stations in sketch exports
- *LRUD splay count*: number of splays used for LRUD [default false]
- *Zip-share category*: categorized zip-share [default false]

If LRUD count is false all splays (within tolerances) are used to compute LRUD. The transverse dimension is set to the maximum value of the splay projections on the horizontal or vertical axes. An in-tolerance splay can be taken into account for both horizontal and vertical dimensions.

If LRUD count is true only the first four splays that satisfy the tolerances are used, and an in-tolerance splay is used only for the vertical or the horizontal dimensions. Each dimension is set to the value of the first acceptable splay. A dimension can be left with value zero by shooting a splay in a direction already set.

By default the tolerances are very permissive: 0° for vertical directions and 90° for horizontal.

Zip sharing can be categorized to restrict its handling to specific apps. If there is no app that can handle sharing, the action fails.

Settings import/export

Settings can be exported to the file "settings.txt" (in TopoDroid private folder). This is a text file listing the values of all the settings.

Settings can be imported from the same file. It is possible to choose whether to import all the settings or only the functional ones.

If there is a problem parsing the file, the settings are not changed.

Some interface settings (like the buttons size) need to restart the app to become effective.

Setting search

The setting search dialog allows searching for a word in the descriptions of the settings.

The search uses case-insensitive pattern matching.

Wildcards and regular expressions are acceptable.

The result of the search is the list of settings sections that contain a match. Tapping the name of a section in the list shows the setting window with the settings of the section.

The button "cancel" closes the dialog.

PocketTopo COLOR MAP

This dialog is opened selecting the *PocketTopo* color-map setting (in the Import/Export section).

It displays a table with three columns.

- PocketTopo color
- associated TopoDroid line symbol (*Therion* name)
- associated TopoDroid point symbol (*Therion* name)

The symbol names used in the color-map are the "*Therion* names", ie, the filenames of the symbol files.

A few symbols are hardcoded in the program: *user*, *wall* (line), *water* (area), and *section* (point and line).

Also *label* is hardcoded. It cannot be used in the color-map because is rendered with a text.

The color-map is used in the import of ".top" files, to map PocketTopo colors to TopoDroid drawing items.

CUSTOM SETTINGS

Secondary settings are for TopoDroid esoteric features.

They are accessible only at tester level.

However, their effects can be visible at a lower level (denoted in square brackets).

By default, the flags are disabled.

General settings

- *Single BACK*: close windows with single tap of the BACK button
- *Navigation bar*: whether to hide the navigation bar and show it with a slide-up
- Additional [palettes](#): to install tools from other sets than the basic cave-tool set
- *Packet logging*: raw data packets are copied into the packed database
- *Sketch editing*: enables the sketch editing tool in the Main window
- *API key*: [Gemini API key](#) [default non-set]

Survey data settings

- *Diving mode* enables data "diving-mode" - distance, azimuth and depth
- *Shot tampering* enables editable values for downloaded shot [B]
- *Recent shots* can be highlighted in blue, and selectively displayed
- *Recent data timeout* the time after which a shot is no longer "recent"
- *Fractional extend* for legs in the extended profile
- Swap stations for *DistoX backsight* data
- *Plane interpolation*: direction and dip of the plane of a set of splays
- [Sensors](#) measures [A]
- *Loop closure*: error compensation method [E]
- *Loop threshold*: threshold used in selective loop compensation [E]
- Using Android compass/clino [A]
- Remote control shot *timer* [E]
- Remote control timer *volume* [E]
- *Blunder* leg-shot [T]
- *Splay station re-assignment*
- *Splay group-rename*

Splay data settings

- *Splay classes*: classification of splays in classes (H,V,X, and normal)
- *Splay coloring*
- Maximum *clino* of splays in plan view [80°] [A]
- *Splay dash mode*: none, by clino, by azimuth, by the view. [none]
- *Splay dash clino threshold*: splay shots with inclination above this threshold are dashed. This applies when dashing by the clino, or in plan view dashing by the view. It also applies to projected profile in dashing by the view to dash splays that makes an angle below this threshold with the direction of projection [50°]
- *Splay dash azimuth threshold*: splay shots with azimuth off more than this threshold are dashed. This applies when dashing by the azimuth, or in extended profile dashing by the view [60°]
- *Splay dash angle threshold*: splay shots making an angle greater than this threshold are dashed, in x-sections [60°]
- *Splay dash color*
- *Splay dot color*
- *Splay latest color*

Sketching settings

- Sketch *shift and scale*
- Sketch *split/merge* (split, copy, outlines, merge)
- *Stylus size* [pxl, 0: ignore]: filter finger touch when drawing with a stylus [T]
- Number of sketch *backups*, kept in a round list [5] [A]
- *Minimum interval* between two backups [60 s] [A]
- *Saved stations* colored orange in the sketches

- *Leg-only refresh* recompute (and redraw) the midline only when a leg is downloaded (in continuous data-download mode) [not] [T]
- *Affine transform* enable the sketch affine transformation
- *Canvas levels*: none, by item, by type [none]
- [Graph-paper scale](#) adjustment [0]
- [Slant x-sections](#): whether to allow slanted x-section for profile views [no]
- [Oblique projected profiles](#): maximum value of the projection angle to the normal of the projection plane. It cannot be over 80°. If below 10° the projection is normal. [0]
- *Line ends cut*: number of line-points to drop from ends when *Stylus only drawing* is set to something greater than zero.

Leg-only refresh does not affect *on-demand* data-download mode. It is also not used at tester level, because, at this level there is an experimental incremental refresh.

If *slant x-sections* is enabled, the plane of x-sections in profile views can have an inclination. Only inclinations multiple of 10° are used. If *slant x-sections* is not enabled, section lines traced at more than the value of *HThreshold* are vertical x-sections, otherwise they are horizontal x-sections.

Line settings

- Minimal *corner angle* in making lines straight (fourth button of line-item edit dialog) [45°]
- *Interpolation accuracy* for cubic Bezier segments [1.0]
- *Corner threshold* for cubic Bezier segments [20.0]
- Maximal *point/segment distance* for weeding points [0.5 m]
- Maximal *segment length* for weeding [2.0 m]
- *Weeding segment buffer*: buffer around a line segment [10 pxl]
- *Line/Area snap* actions
- *Smooth/straighten segment* actions
- Making whole line "rocky" action [no]
- *Multi-path* actions: path multiselection is initiated with a point-action menu. The multi-path actions are: delete, decimate, join, scrap-move. "Finish" terminates the multiselection without doing anything.

Device settings

- *Connection delay*: delay [1/10 sec] before trying to connect to the device [0, no delay] [B]
- *Second device*: whether to enable working with two devices (DistoX only) [T]
- *Data pause*: pause after the download of each data [250 ms] [A]
- *Ready data*: wait time for the data to get ready [500 ms] [A]
- Pause after the remote *Laser-on command* [1500 ms] [A]
- Pause after the remote *shot command* [2000 ms] [A]
- *Firmware* sanity check [yes] [A]

- *BRIC mode*: either primary data only, all data, or all data except the index. With all data TopoDroid uses as shot index the value from the *BRIC*. [all] [T]
- *BRIC zero data*: whether to record zero-shots [no] [N]
- *BRIC index*: whether to use the shot-index of the *BRIC* as shot ID [no]. It requires that all the info of a shot are downloaded.
- *SAP5 bit-16 bug*: workaround [T]

Note. If the delays after the remote "Laser-on" and "Shot" commands are too short, the commands may fail.

Import-Export settings

- Include *tool files* in zip archive
- Survey *datamode* on import (normal, diving) [normal]
- Export csx in *cSurvey-transfer* format (required cSurvey 1.20)
- *Stations*: add stations automatically to th2 files
- *LRUD splay count*: number of splays used for LURD

TopoDroid Project Folders

In order to manage cave survey projects involving several survey sessions, it is advisable to use a specific work directory (folder) for each cave project. All project folders are stored inside the TopoDroid base folder "Documents/TDX", the default subfolder is "TopoDroid".

The folder name "tmp" is reserved. If you type it, the default folder is used.

The drawing symbol files and the files related to the devices (database, firmware, memory dumps, calibrations) are stored only in the app private *files* folder (in the *primary external storage*: for example "/sdcard/Android/data/com.topodroid.TDX/files").

The drawing symbols files are stored in the *point*, *line*, and *area* folders.

The firmware files are in the *bin* folder.

The exported calibration files are in the *ccsv* folder, and the memory dumps in the *dump* folder.

The project folders contain only the survey database of the project and survey folders.

The drawing symbols in use in the project are configured in the project database. When a project is created, only the default drawing symbols are enabled in the configuration. Other symbols can be enabled from the "Palette" button of the *Main Window*.

The name of the current project folder is one of the general settings of the app in the Main window.

Warning

Future Android versions may force the project folder to be under the app private "files" folder.

TopoDroid KEYPADS

TopoDroid comes with a numerical keypad and a simple "qwerty" keyboard.

The numerical keypad is used for numerical values (lengths, angles, etc.). It has the ten digits, the decimal point, and the plus/minus sign. For the input of coordinates it has also the degree (°) and minute (') keys.

7	8	9	°
4	5	6	'
1	2	3	DEL
+/-	0	.	>>

The qwerty keyboard is used for station names. It has the ten digits, the 26 letters of the latin alphabet (lowercase and uppercase), and the period ('.').

1	2	3	4	5	6	7	8	9	0
Q	W	E	R	T	Y	U	I	O	P
A	S	D	F	G	H	J	K	L	DEL
Aa	Z	X	C	V	B	N	M	.	>>

These keypads are enabled by default. They can be enabled/disabled through a settings of the Main window.

Note. TopoDroid keypads are not Android IME (Input Method Editor). The cursor position is at the right end of the input. If the no-cursor setting is not enabled, it is denoted by an underscore '_'.

LANGUAGES

By default TopoDroid uses the system default language if TopoDroid contains the proper translation. Otherwise English is used. You can override this choice selecting another language among the supported ones:

- English
- Bulgarian (A. Hazday)
- Chinese (H.J. Luo)
- German (M. Keller)
- Hungarian (B. Holl)
- Italian (F. Toso)
- Portuguese (R. Severo)
- Russian (A. Kozhenkov)
- Spanish (M. Guerrero, J. Pardo)

The choice of the language is one of the general settings of the app.

Warning Due to caching the Android might not use the selected language for all texts. A work-around is a cold restart (a force-stop followed by open).

Warning

Setting the language with the "Settings" app changes the language of TopoDroid, however items already loaded using the TopoDroid language setting (menu lists, dialogs, ...) are NOT reloaded with the system language.

DEVICE NICKNAME

This dialog is opened long-tapping on a device entry in the Device window.

It displays some info about the selected device and has a text field to enter the **nickname**

. If the second-device setting is enabled there is a checkbox to set the device as second device (DistoX only).

Buttons

- *OK* save the nickname to the database and closes the dialog. If the checkbox is selected the device is set as second device (DistoX only).
- *Forget* remove the device from TopoDroid database, and from the Android list of connected devices. There is a confirmaton dialog
- *Cancel* closes the dialog without doing anything

DEVICE SELECT

TopoDroid supports the following device:

- *DistoX* (first version)
- *DistoX2* (second version)
- *DistoXBLE* (third version)
- *SAP5* and *SAP6* (Shetland Attack Pony), only models with bluetooth
- *BRIC4* and *BRIC5* (Bluetooth Ruggerized Integrated Cartographer)
- *Cavway X1*
- *DiscoX*
- *JedEye*

All devices must be paired with the Android.

Normally TopoDroid works with only one device. The data can be downloaded continuously (*continuous* mode) or in batch (*on-demand* mode). The difference between the two modes is that the *continuous* mode remains active when the data download is finished, or the connection with the device is lost, while the *on-demand* mode is closed.

SAP, *BRIC*, and *JedEye* download the data only in continuous mode.

With DistoX's TopoDroid can work with more than one device at once, although the data are still downloaded from only one device at a time. The *data download mode* must be set to *multi*. When it is so a long-tap on the *download* button brings up a dialog to choose from which DistoX to download the data.

Shetland Attack Pony (SAP)

TopoDroid supports the model of the Shetland Attack Pony, version 5 (*SAP5*) and 6 (*SAP6*), with bluetooth.

The device must have been previously "paired" with Android, using the *Settings* app. The PIN is "000000" (six zeros).

Afterward the *SAP* is listed among the known devices in the *Device Window*. It appears as "SAP XX MAC-address", where "XX" is the two-letter code of the *SAP*, for example "HG".

To select the *SAP* as current device tap on the its entry in the list.

While connected, the *SAP* will transfer all shot values to the app, whether the shot data are stored or discarded on the device.

The *SAP5* data transmission protocol does not include acknowledgment of the reception of the data. Therefore

- if *SAP5* is not connected, the shot values are not transferred to the app.

- upon reconnection, the *SAP5* does not transfer the shot data stored while not connected.

The *SAP6* data transmission protocol includes the acknowledgment.

The *SAP* keeps in its internal memory the data that have been "stored". These can be retrieved as a Survex file, using the *SAP PonyTrainer* program.

The calibration of the *SAP* is done by the device and does not require an external program.

BRIC

TopoDroid supports the *BRIC* version 4 and 5.

The device must have been previously "paired" with Android, using the *Settings* app.

Afterward the *BRIC* is listed among the known devices in the *Device Window*. It appears as "*BRIC* code MAC-address", ('X' being 4 or 5) where the code is the four-digit code of the *BRIC*, for example "0039".

To select the *BRIC* as current device tap on the its entry in the list.

Once connected to a client, the *BRIC* remembers the connection, even if turned off, therefore it stays "always connected".

When TopoDroid is connected, shot readings are immediately transferred to Android. If TopoDroid is not connected, the data are kept in memory, and transferred when the app is next connected.

The *BRIC* data transmission protocol includes acknowledgment by Android, but it does not include acknowledgment of the reception of the data by the app. The protocol has a few seconds delay between the connection and the first data transfer, and TopoDroid has a one-second delay after the last data, before closing the connection. By this means usually no shot is lost, although this is not guaranteed and shots are lost if the app is not notified by Android before the next shot arrives.

The *BRIC* stores the data in internal memory in CSV files. If connected to a PC via USB, the device appears as external memory and the CSV files can be copied to the PC.

The calibration of the *BRIC* is done by the device and does not require an external program.

The *BRIC* can be controlled from the app. In particular it is possible to turn the laser on/off, and to take shots. The *BRIC* has also a "scan mode" in which readings are taken continuously, about two every second.

TopoDroid can follow three different data protocols (according to a setting in the "secondary" set). The first receives only the primary data (distance, azimuth and clino) which do not include the data ID number assigned by the *BRIC*. The other two receive the complete data. They differ as whether TopoDroid uses the provided ID number or assigns its own numbers. When the complete data are received TopoDroid records the *BRIC* error codes and values in the shot comment.

Cavway

TopoDroid supports the Cavway version X1.

The device must have been previously "paired" with Android, using the *Settings* app.

Afterward the Cavway is listed among the known devices in the *Device Window*. It appears as "CVWYn code MAC-address", ('n' being 1, for "X1") where the code is the four-digit code of the Cavway, for example "0002". To select the Cavway as the current device, tap on its entry in the list.

The calibration of the Cavway is done by the device and does not require an external program. However, the calibration data can also be downloaded in the app, and the calibration can be computed in TopoDroid and uploaded to the device. The calibration computation on the Cavway and in TopoDroid are the same.

Add-device dialog

With this dialog the user can add a device to the TopoDroid device list. The dialog has the fields

- Address
- Model and number
- Nickname (optional)

The *address* is selected from a drop-down list which shows the MAC-addresses of the paired devices that are not in the TopoDroid database. The address can also be checked from the details of the connected device in the *Settings* app.

The *model* is selected from a drop-down list of supported devices.

The *number* must be a four-digit number. It can have leading zeros. The number of the device can be read from the info on the device itself.

The nickname is optional and can be left blank.

BLUETOOTH

The *Bluetooth* button is present in the Survey Data window, in the Drawing window, and in the Calibration Data window.

For *DistoX*, *SAP*, and *DiscoX* the *Bluetooth* button resets the Bluetooth connection.

For *DistoX2*, *DistoXBLE*, *Cavway*, and *JedEye* it open a drop-down menu [T]:

- *Reset* the Bluetooth connection
- turn on/off the laser
- take a shot or a calibration data
- take three shots in a row, ie, a leg (Survey Data window or Drawing window)

In the Survey Data window and Drawing window, the shots are downloaded immediately if the device communication mode is "continuous".

In the Calibration Data window, the data is downloaded immediately depending on a setting.

After turning on the laser there is a short pause before taking a shot. Similarly there is a pause between shots when three are taken. The lengths of these pauses are defined in settings.

For the *BRIC* the *Bluetooth* button open a drop-down menu [T]:

- *Reset* the Bluetooth connection
- turn on/off the laser
- take a shot readings
- starts a scan mode (only if the *BRIC* connection is set to receive the complete data)
- turn off the device
- reset the shot counter of the *BRIC* (only if the survey has no data)

Bluetooth Alias

With this window the user can set an alias for a known bluetooth device.

The device model must be selected among the given options.

It is important to select the correct device, because this determines how TopoDroid exchanges data with the device.

Unless the model is an old DistoX (first version based on Disto A3) the device code must be entered. This can be a four-digit number, or contain characters. TopoDroid does not check the code, but it is advisable to use the code reported by the device.

The alias for the device must be entered as reported in the Android listing of the paired bluetooth devices.

The button "Add" adds a new alias or change an existing alias.
The button "Close" closes the window.

The window contains also a listing of the bluetooth aliases.

To remove an alias tap the entry in the list and confirm to remove.

Device TROUBLESHOOTING

Device pairing

BLE devices can be discovered by TopoDroid, and then the app can connect to them.

Bluetooth classic devices must be paired with Android first.
You can pair Android with the active DistoX in the Android *Settings*.

If Android asks the *DistoX* PIN enter 0000 (four zeros).
For the *SAP* PIN enter 000000 (six zeros).
Android should not ask the Bric4 PIN, nor the DistoXBLE PIN.

Once paired with the DistoX, the Android should keep the pairing, and not request the PIN again. If it keeps requesting the PIN every time you download data from the DistoX, try using the *insecure* socket setting.

CAVWAY REFERENCE GUIDE

Here is a summary of the Cavway functions.
The official user manuals are available at <https://cavedevice.com>

The *Cavway* has five buttons,

- "DIST"
- "Left" (<)

- "Right" (>)
- "Memory/Confirm" (M)
- "Clear" (CLR)

To turn the *Cavway* on press the "DIST" button. To turn it off press and hold the "CLR" button. The *Cavway* goes off by itself after the amount of time on inactivity defined with the *Idle off* option.

The switch off and lock the *Cavway* press and hold the buttons "CLR" and ">". To unlock it, after pressing the "DIST" press "M" within two seconds.

The *Cavway* interface is menu based, with several screen-pages.

Button Functions

- "DIST": turn on the laser; take a measurement
- "<" and ">" are used mostly as navigation button used to move in the menus
- in measurement mode, with laser off, a tap on "<" or ">" turns the device in memory mode.
- "M" enters the menu page and is used mostly to confirm choices.
- "CLR" goes back to the previous screen.
- a long tap on "<" shows the details of the current measurement
- a long tap on ">" clears the untransmitted flag on all shots

Measurement

Press "DIST" to turn on the laser. Press it again to take a measurement. The readings are shown on the screen. If the *Cavway* detects an anomaly (a discrepancy between the readings of the two sensor sets), it shows an error message on the screen.

Memory mode In this mode the *Cavway* shows the values (distance, azimuth and clino) of five shots. The first column contains "E" if an error was detected, or "C" for calibration shots. The shot that have not been transferred yet have a "*" to the right. Leg shots are in bold.

The shot under cursor has a dark background. The buttons "<" and ">" scroll up and down the list. Pressing "M" the details of the shot under cursor and shown: readings, detected errors, and values of G/M/dip.

Menu mode

The first menu-screen has five menus

- "Calibration", to start a calibration
- "Clr. Unsent", to clear the not-transmitted flag from all shots
- "Options", to change the device settings
- "Info", to see device informations (fw, hw, serial number, battery voltage, and so on)
- "Advan. Menu" to move to the second menu-screen

The second menu-screen has

- "Shot Options", settings for shots
- "Calib. Options"
- "Time" to set the date and the time
- "Units"
- "Factory Reset"

The device options are

- "Reference" point: rear, (extension) tail, tripod screw, front, custom
- "Shot delay", from 0 to 9 seconds
- "Screen backlight", from 0 (no backlight) to 10
- "Volume" (beep sound), either ON or OFF
- "Idle timeoff", 60, 120, 180, 240, 300 seconds

The buttons "<" and ">" move among the options. The button "M" cycles through the values of the setting.

The shot options are

- "Error detection", either ON or OFF
- "Angle difference" for leg identification
- "ABS G/M" error limit
- "Dip" error limit
- "Reset to default"
- "Reset statistics"

The calibration options are

- "Error detection", either ON or OFF
- "Group detection" angle limit

Calibration

The "Calibration" menu turns the *Cavway* in calibration mode. In this mode the device registers raw readings in groups of four, as long as the angle difference between shots in the same group is small. If more than four shots are taken the device records only the last four.

When a group of four is complete, a new group is started when there is a shot with a different direction.

The distribution of group directions is shown on the screen with two circles, one for positive inclinations, the other for negative inclinations.

The group shot count can be reset by pressing the ">" button.

The *Cavway* records at most 25 groups.

The button "M" starts the computation of the calibration coefficients.

Afterwards it shows the results of the computation: average angle error, std deviation and maximum angle error.

Another press of the button "M" stores the computed coefficients.

The button "CLR" exits from the calibration mode.

Advanced operation

Shot timer

To take the shot with the timer do a long press on 'DIST'.
The timer can be used also for taking calibration data.

Splay marks

By pressing the "M" button while the laser is on, the user can select among four flags to mark the shot that is being taken.

The mark name appears on the screen. Further press of 'M' cycle through the four marks. The last used mark is shown at the first press of 'M' for a following shot. If you do not want to mark the shot press 'CLR', which also switches the laser off.

The four marks are: *generic*, *feature*, *border*, and *backsight*.

TopoDroid assign special colors to the marked splays: *generic* green, *feature* yellow, *border* purple, *backsight* blue.

Scan mode

To enter scan mode press 'M' and 'DIST' for two seconds while the laser is on.

Alternatively one enter scan mode by pressing ">" for two seconds, while the laser is on. The choice of which button(s) to press is a setting.

Another setting specifies the acquisition speed. If set to "low" the *Cavway* takes a shot only when it is held fixed on the target point. If set to "high" it collects data more frequently. The default is a "medium" speed.

TopoDroid cluster all the shots of a scan and shows only the first shot in the shot listing window.

Online sensor values

If the '<' is pressed when the laser is on, the display shows, in order,

- field values, and dip angle
- readings of the M sensors
- readings of the G sensors
- azimuths, clinos, rolls, absolute values of G and M, and dip for the two sensors

Cavway INFO

This dialog is opened from the *Info* button of the Device window when the current device is a Cavway.

The *Cavway* info dialog shows

- MAC address
- firmware version
- hardware version
- whether the cavway time is synchronized with the Android
- calibration information (date and error statistics)

There is also a button to synchronize the time on the *Cavway* with that of Android.

Cavway Memory

The *Cavway* memory dialog allows retrieving the latest data from the device memory. You must specify the number of data to download and press the button "Dump". The data are shown in the dialog.

For survey data date, distance, azimuth, inclination, roll and the two dip values are shown. Leg data has an 'L' in front. If the shot has an error, there is an 'E'. Calibration data have a 'C', the date, and the raw sensor readings: G1, M1, G2, and M2.

If you specify the name for a dumpfile, the data is also stored in the file.

DistoX REFERENCE GUIDE

Here is a summary of the DistoX functions.

The official user manuals are available at <http://paperless.bheeb.ch>

PIN code:

If the DistoX does not pair automatically use PIN "0000" four zeroes.

Error codes:

- 252: too high temperature
- 253: too low temperature
- 255: too weak signal
- 256: too strong signal
- 257: too much background light
- 260: interrupted laser beam

Main functions:

- DIST: power on / start laser / measure
- CLR: cancel current operation, switch laser off
- CLR (2 secs): switch device off
- MEM: memory entries
- REF: toggle distance reference

DistoX2 /DistoXBLE - X310

- TIMER (laser on): start timer (automatic measurement)
- TIMER (laser off): set timer interval (with PLUS/MINUS)
- SMART: extra measure info
- FUNC: device info

Two seconds (keys pressed together):

- DIST (with laser on): measure
- CLR: power off
- CLR-SMART: toggle calib. mode
- CLR-MEM: clear unsent memory
- CLR-FUNC: toggle bluetooth
- CLR-MINUS: locked power off
- REF: front reference
- REF-FUNC: back reference
- REF-PLUS: toggle display light
- REF-MINUS: toggle beep
- MEM: distance units
- MEM-SMART: angle units (deg/grad)
- MEM-FUNC: toggle silent mode (shot not transferred)
- MEM-MINUS: toggle backsight mode [2.4]
- SMART-MINUS: toggle triple-shot check [2.4]
- DIST-PLUS: continuous measurement, CLEAR off [2.6]

Five seconds (keys pressed together):

- FUNC-SMART: change battery type
- FUNC-CLR-MEM: factory reset
- DIST-PLUS-MINUS (switching on): bootloader mode

Extra measure info (SMART key):

1. azimuth, H distance, V distance, distance
2. roll, dip
3. M, G
4. on-board accel. raw x,y,z (*)
5. separate accel. raw x,y,z (*)
6. magnetic raw x,y,z (*)

1. combined accel. x,y,z
2. magnetic x,y,z
3. on-board accel. x,y,z (*)
4. separate accel. x,y,z (*)

(*) only last data, not for memory data

Device info (FUNC key):

1. battery
2. hw/fw version, serial number
3. display light level (1:10) [2.3]
4. beep tone (0:5) [2.6]
5. endpiece offset (-126:127) [2.3]

DistoX - A3

- AREA (back ref.): continuous display of azimuth+clino (*)
- AREA+REF (front ref.): continuous display of roll+dip (*)
- AREA+AREA: firmware version and serial number
- UNITS: toggle display light
- AREA+AREA+REF+REF+CLR: toggle silent mode
- MEM[10 times]+UNITS: toggle calib. mode

(*) Do not take measurements in this mode

Two seconds:

- UNITS (in back ref.): distance units
- UNITS (in front ref.): cycle angle units / BT

Five seconds (keys pressed together):

- CLR-PLUS: toggle beep

"Angle units / Bluetooth" cycle (UNITS key)

- compass off, BT off
- compass on (deg), BT off
- compass on (grad), BT off
- compass on (deg), BT on
- compass on (grad), BT on

DistoX A3 INFO

This dialog is opened from the *Info* button of the Device window when the current device is a DistoX.

DistoX A3 information:

- device address
- device code
- angle units
- compass and clino (on/off)
- normal/calibration mode
- silent mode (on/off)

The dialog has a button that *clears* the DistoX memory, ie, it clears the hot-bit that marks the data to transmit. After clearing, the DistoX must be switched off and on for the action to take effect.

DistoX X310 INFO

This dialog is opened from the *Info* button of the Device window when the current device is a DistoX2.

DistoX X310 information:

- device address
- device code
- firmware version
- hardware version

DistoXBLE INFO

This dialog is opened from the *Info* button of the Device window when the current device is a DistoXBLE.

DistoX X310 information:

- device address
- device code
- firmware version
- hardware version

DistoX HOW TO CALIBRATE

The DistoX needs to be calibrated before using. The calibration is a transformation of the data that accounts for the non-perfect orthogonality of the axis of the sensors, their misalignment with the laser direction, and effects of the electronics on the magnetic field.

To compute the coefficients of this transformation, you must take a number of "accurate" calibration data recording the raw readings of the sensors. This occurs when the DistoX is in "calibration mode".

The calibration data must be taken in a place free from magnetic influences, eg, in a cave or in a wood. Beware of metal objects and electronic objects that might affect the DistoX.

The data must cover all directions and for each direction you must take a group of four data at different rotation about the laser axis.

The best way is to take 14 groups of four data each, eight groups in the directions of the corners of a cube, and six in those of the centers of the faces. The four data in each group should be taken rotating the DistoX by 90° between one and the next. You can take more than 14 groups of data; the more the better.

At a minimum you must take four groups of four data (with different roll) in four directions at 90° in the horizontal plane, as if these directions were going from the center of a cube towards the center of four side-faces. The other 40 data do not have to be taken very carefully if the *PocketTopo* group policy is used, because they are treated as individual data. You can take more individual calibration data.

The calibration data that will form a group must be taken with care.

Inaccurate group data will result in a bad calibration. Therefore use fixed points (eg, a point in the wall, or on a tree) and far apart (a few meters) so that the accuracy of pointing the laser is better than the wanted calibration accuracy (2 cm at 5 m are about 0.2°).

After you have taken the calibration data, download them with TopoDroid. You do not need to wait to take all the calibration data to download them: when you download some data these are added to those already in the

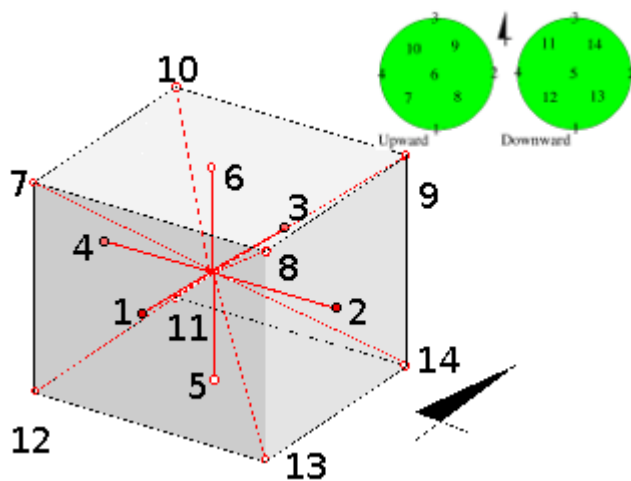
calibration. Therefore if, by mistake you take five data instead of four, download the data and delete one of them in TopoDroid.

Before computing the calibration you must tell TopoDroid how the calibration data are divided in groups, i.e., how groups are assigned to data. The groups are denoted with positive numbers. A data with group number zero or no group, is not used to compute the calibration. Deleted data have group -1 and are usually not shown in the list.

TopoDroid implements two policies to assign groups automatically:

- *TopoDroid*: all data divided in groups of four data. This is the default policy
- *PocketTopo*: four groups of four data, followed by individual data

The group numbers can be assigned or changed by editing the calibration data. Tap on a data entry in the list to open the data editing dialog. When the groups are defined, you can compute the calibration. Besides the calibration coefficients, the dialog shows the average error (and the "delta" of the original algorithm), the standard deviation of the error, and the maximal error. Furthermore there is a histogram with the distribution of errors for the calibration data. This plot has two vertical lines: a yellow mark at 0.5° and a red one at 1.0°



To summarize, to calibrate the DistoX you need to:

- turn the DistoX in calibration mode
- take the data of the calibration data
- download the data to the app
- assign the groups to the data (automatically or manually)
- compute the calibration coefficients
- upload the coefficients to the DistoX
- revert the DistoX to normal mode

DistoX/CAVWAY CALIBRATIONS

This dialog is opened from the *Calibration* button of the Device window.

Buttons:

- *new calibration*
- *import* a calibration from the CSV export

Underneath there is the list of the calibrations of the active device.

Open a calibration

Tap the calibration entry in the list to open it in the Calibration Info window.

New calibration opens the Calibration Info window, where you can enter the data for the new calibration, and create it. After it is saved to the database you can open it (ie, the Calibration Data window), download the calibration data, compute the calibration coefficients and upload them to the DistoX/Cavway.

CALIBRATION DATA

This dialog is opened tapping a data entry in the Calibration Data window.

- *Shot ID (index)*
- *azimuth, inclination, roll*
- *calibration error*
- *group* number entry field
- *Save* the changes

The buttons are

-  mark the data "deleted"
-  reassign the groups from this data onward

The item group can be set or changed

- *positive*: the data takes part to the computation of the calibration coefficients
- *zero*: the data does not enter the computation
- *negative*: the data does not take part in the group assignment, nor in the computation of the calibration coefficients

Data with the same positive group number form a group.

Deleted data are not removed from the database.

They are not taken into account in the automatic group assignment, and do not enter to the computation of the calibration coefficients.

They are not usually shown in the calibration data list. If displayed (see [Calibration Data window](#)) they have a gray background.

The group number assignment can follow the TopoDroid scheme or PocketTopo scheme [T].

Calib Data SEARCH

The dialog is opened by the *Search* button of the [Calibration Data window](#).

It allows o search for calibration data with error exceeding a given minimum error, which can by typed in the text field.

The calibration data with error value larger than the minimum error are highlighted with an orange background to the error value.

There are three buttons:

- *Close* the dialog without doing anything.
- *Clear* the display of search results.
- *Search* perform the search and display the result.

CALIBRATION DATA GROUPS

This dialog is opened from the *Group* button of the [Calibration Data window](#).

Before computing the calibration coefficients the calibration data must be subdivided in groups of data having the same azimuth and inclination (but different roll).

Beware that a wrong group assignment can result in a failure to compute the calibration coefficients (usually due to too many iterations in the algorithm).

The groups are identified by an integer number starting with 1.
A data with number 0, or negative is listed, however negative-group data are excluded from the computation of the calibration coefficients.

Data with odd groups are displayed in blue, with even groups in orange.
Data with 0 or negative group are displayed in gray.

Data flagged as "deleted" are normally not listed and even if listed they are excluded both from the assignment of groups and the computation of the coefficients.

You can assign the group number to each calibration data by hand (with the *Calibration data edit* dialog), or let TopoDroid assign the group numbers for you.

There are two automatic group assignment policies:

1. *TopoDroid*: four data at a time for each group;
2. *PocketTopo*: four-at-a-time for the first 16 data (four groups), then each data is a group by itself;

The two policies gives equivalent calibrations, but with *TopoDroid* policy the app can compute the deviation of the calibrated data from the expected values, on more data, and create a more populated histogram of residual errors.

Actions:

- *Reset groups*: all the data groups are reset
- *OK*: groups are assigned (only to non-deleted data without a group)

CALIBRATION DATA DISTRIBUTION

This dialog is opened from the *Distribution* button in the Calibration Data window.

It displays the distribution of the calibration data, in azimuth and inclination, i.e, on a 3D sphere.

The sphere is cut on the horizontal plane, and the upwards and downwards hemispheres are displayed as two circles. The center of the circles are the

vertical direction, while the points on the border are the horizontal directions. On each circle, (magnetic) North is upward, South downward, East to the right and West to the left.

Green areas indicate directions covered by the data. Red zones are directions for which data are missing. The dialog can show either the direction distribution of the raw data, or that of the data corrected with the calibration.

BUTTONS

- **Raw** computes the distribution using the raw direction values.
- **G** displays the distribution of the data acceleration vectors in the frame of the device
- **M** displays the distribution of the data magnetic field vectors in the frame of the device
- **Roll** displays the distribution of the data roll angles in the frame of the device
- **Calibrated** computes the distribution after correcting the data with the calibration coefficients [only after the calibration has been computed].

A proper calibration data set should have no red spot in all the distributions.

CALIBRATION COEFFICIENTS

This dialog displays the 24 coefficients of the calibration linear transformations for the **G** and the **M** vectors (12 coefficients each) and 3 coefficients of the non-linear terms (only DistoX2 v. 2.3 or higher).

It is used in three different situations:

- The *Read* button of the Device window display the coefficients read from the DistoX/Cavway.
- The *Read* button of the Calibration Info window displays the coefficients stored in the database
- The *Compute* button of the Calibration Data window displays the computed coefficients

If the calibration info (errors, iterations, dip) are available the dialog displays also

- a histogram of the residual errors of the calibration data,
- the average error [degrees]
- the "delta" of the original algorithm
- the standard deviation of the error distribution [degrees]
- the maximum error [degrees],
- the number of iterations taken by the calibration computation,

- the average angle between magnetic field and vertical [degrees]
- the average difference between the roll angles computed using G vectors of the data in each group and using M vectors [degrees]
- only for double-sensor devices the statistics (average, std-dev, and max [degrees]) of the differences of the device directions between the two sensors.

In case some calibration info is not known the value 0 is shown.

Assuming that the azimuth-clino error distribution is normal, the standard deviation of the errors should be slightly over half the average value.

The error of a calibration data is the angle between the data direction and the average direction of the group the data belongs to, after having taken the calibration correction into account. The errors are reported in degrees.

The histogram has ticks on the horizontal axis at 0.5° (yellow line), 1.0° (red line), and 1.5° .

On the vertical axis the tick marks are 10 and 20.

The "delta" of the original algorithm is the square-root average of the residual differences between calibrated G-M vectors and the vectors that satisfy the minimization condition. It gives an estimate of the calibration accuracy (precision of the instrument): "delta" is roughly 5/4 of the accuracy in degrees.

The dialog displays also the magnetic dip [degrees] computed by the calibration algorithm.

After a computation of the calibration coefficients, the dialog has a button to **write** the coefficients to the DistoX/Cavway.

A warning is given if the data direction distribution is below 95%, or the calibration average error is above 0.5° .

For the **Cavway** there is a button to switch between the coefficients of the calibration of the first sensor set, and those of the second sensor set. The reported errors are from the calibration of the first sensor set. Because of the board layout the first sensor set has smaller magnetic influences. Nevertheless, the differences of direction between the data of the two sensor sets are very small, for a good calibration.

DistoX CALIBRATION VALIDATION

A calibration can be validated against another calibration of the same DistoX.

Calibration list

With the Validation menu you get the list of the other calibrations of that DistoX, and you choose which one to validate with, by tapping its entry.

Calibration validation result

The validation result dialog displays

- (blue) "accuracy" of the second calibration on the data of this calibration. For each group of data the direction of the average of the group data is computed, as well as those of individual data. The accuracy is reported as average and std deviation of the difference between the directions of individual data and that of the average [degrees]
- (red) "accuracy" of the current calibration on the data of the second calibration.
- (grey) "precision" of the calibrations, measured as average, std deviation and maximum of the angle differences between the direction computed with the two calibrations, on the data of both.

The yellow vertical line marks 0.5° , and the red line 1.0° .

CALIBRATION Import/Export

Calibration import

A calibration can be imported from a CSV calibration file.

The import dialog is opened from the *Import* button in the *Calibration* list dialog.

This dialog displays the list of the calibration CSV files.
Tap the entry you want to import.

The import fails if the file does not contain a calibration, or the calibration is already present.

It can also fail if the device bluetooth address written in the calibration is different from that of the current device.

Calibration CSV export

A calibration can be exported in CSV (text) format.

The export dialog is opened from the *Export* menu of the Calibration Info window.

The calibration CSV export format contains the calibration details (name, date, device bluetooth address, comment, algorithm), followed by the calibration data, and the calibration coefficients.

Each data line contains:

- data index
- G and M values (six integers)
- group
- azimuth, clino and roll
- error
- status

The exported file is saved in the *Android // data // com.topodroid.TDX // files // csv* folder of the Android device. The contents of the *Android // data* folder can only be accessed connecting the Android device to a computer and after turning on the "File transfer" setting for the Android's USB connection.

BLUETOOTH PACKET LOGS

This dialog is opened from the *Packets* menu of the Device window.

It displays the list of logged packet. The most recent packets are at the top of the list.

Items with a black background are packets received from the Device. Those with a lighter background are sent to the Device.

The text color depends on the type of the packet:

- white: data packet (D)
- grey: vector packet (V)
- brown: G packet (G)
- orange: M packet (M)
- green: command packets (C)
- yellow: other packets (X)

Bluetooth packets are logged only if the *Packet* setting is enabled.
The packet database is in the private app files folder.

The dialog has two buttons to clear the log of the packets older than a day or than a week, respectively.

DistoX/Cavway MEMORY

This dialog is opened from the *Memory* button of the Device window.

Memory dump

The *Dump* button reads the content of the DistoX/Cavway memory. Reading the device memory directly is a slow process, and the device must be kept "ON" while doing it. It is advisable to read a small portion of memory, say 20 entries, at a time: enter the initial and final data memory index in the two text fields to the right. Data are read from the initial index up to one before the final index. For the DistoX2 the memory indices range from 0 included to 1064 excluded.

For the DistoX version 1 the memory they range from 0 included to 4096 excluded.

For the Cavway ... (TODO)

If you have entered a filename the memory dump is also saved to a file (in the "dump" folder).

The result of the memory dump is shown in a list at the bottom of the dialog. Each line contains

- the memory location
- a letter-code: foresight shot ('d'), backsight shot ('b'), vector (v), calibration data ('g' or 'm')
- the values of the memory.

Uppercase letter code denotes data that have not been downloaded yet, except "vector" data for which upper case denotes backsight flagged. Shot are shown as length (meters), azimuth and clino (degrees).

The "accuracy" data of the DistoX2 are not shown.

For calibration data, if the "Raw Calibration Data" setting is not zero, memory data is converted to human-readable form, otherwise the raw hex values are shown.

Entries without a recognized type are marked with a question-mark ('?') code.

Only for DistoX A3

- **Read** the memory cursors
- **Store** the memory cursors
- **Reset** a portion of memory, ie, marks it "to-be-downloaded"

DistoX2 FIRMWARE

TopoDroid includes the *DistoX2* firmwares v. 2.1, 2.2, 2.3, 2.4, and 2.5, which are all compatible with hardware 1.0.

Only v. 2.5 is compatible with hardware 1.1.

TopoDroid includes also firmwares v. 2.6.1, 2.6.3 and 2.6.4 which are compatible with hardware 1.2, and firmware v. 2.7.0 compatible with hardware 1.3 (BLE).

WARNING.

Uploading an incompatible firmware makes your DistoX unusable. Carefully read all the messages before uploading a new firmware.

Firmware dialog

Through the firmware dialog you can save a copy of the current firmware of your *DistoX2*, or upload a new firmware file.

The dialog has a pair of radio buttons to specify whether to upload or to download the firmware. In the first case the user must select a firmware file. In the second case the user must enter the name of the output file for the firmware.

Next there is a bar with buttons

- *help*
- *close* the dialog without doing anything
- *OK* confirms either the upload or the download of the firmware

Before uploading a new firmware TopoDroid checks that it is compatible with the hardware version. Unfortunately there is no sure way to read the firmware version from the file content; To prevent uploading a bad file, compares a signature block in the file (the first block after the bootloader blocks) and on the file checksum, against the known values.

The check result codes are (negative for failure):

- 2100: firmware 2.1
- 2200: firmware 2.2
- 2300: firmware 2.3
- 2400: firmware 2.4
- 2412: firmware 2.4c
- 2500: firmware 2.5
- 2501: firmware 2.51
- 2512: firmware 2.5c
- 2610: firmware 2.6.1
- 2630: firmware 2.6.3
- 2640: firmware 2.6.4
- 2700: firmware 2.7.0
- -200: failed either 2.1 or 2.2
- -230, -2300: failed 2.3
- -240, -2400: failed 2.4
- -250: failed either 2.5 or 2.51
- -246, -2412: failed 2.4c
- -256, -2512: failed 2.5c
- -99: generic failure

The DistoX (v. 2 or BLE) must be booted in **bootloader mode**. Refer to the DistoX2 documentation for details.

The device must be turned on in bootloader mode because TopoDroid reads the signature block from the firmware that is already installed on the DistoX, before uploading a new firmware.

If the hardware versions are not compatible there is a warning.

Beware that TopoDroid does not warn against uploading a firmware pre 2.5 on hardware 1.1.

Upon uploading a firmware, TopoDroid displays a message saying how many bytes have been written to the DistoX. If these are less than the firmware file size, the upload has failed.

Firmware update procedure

1. Make sure the DistoX is the selected active DistoX in TopoDroid
2. Open the firmware dialog ("Firmware" menu) and select the firmware file to upload
3. Check version of hardware and firmware of your DistoX: make sure laser is switched off (CLR) and press FUNC twice to show the version. You may also check the DistoX firmware and hardware versions with TopoDroid device info dialog.
4. Switch off the DistoX, and switch it on in bootloader mode: press PLUS, MINUS and DIST together. The display illumination switches on, but the screen remains empty.
5. Upload the new firmware with TopoDroid
6. Switch off the DistoX by pressing CLR
7. Switch the DistoX on and check the new firmware version.

If the DistoX does not turn on after a firmware upload, you need to restart it in bootloader mode and try to upload a firmware again. To restart it in bootloader mode you must open it and jump two pins: see the docs on internet for details.

Note

The firmware updates are always logged in the TopoDroid log file.

BRIC REFERENCE GUIDE

Here is a summary of the BRIC functions.

The official user manuals are available at <http://www.caveexploration.org/gear/bric4>

To turn the *BRIC* on press the external button three times quickly in a row. To turn it off press and hold the external button. The *BRIC* goes off by itself after the amount of time on inactivity defined at the *Idle off* options.

Internal buttons

- *Menu* enters the menu functions
- *Up* scrolls forwards the data on the display
- *Down* scrolls backwards the data on the display
- *Back* ...

Menu functions

The main menu is entered by pressing the *Menu* internal button. A press of the external button exits the menu and return to normal mode. You can also return to normal mode by repeatedly pressing the *Back* button.

In menu mode the internal buttons are

- *Enter* confirms the choice
- *Up* scrolls up among the choices
- *Down* scrolls down among the choices
- *Back* goes back to the previous screen, or to normal mode

Options:

- *Dist*: meters or feet
- *Temp*: Celsius or Fahrenheit
- *Shot delay*: between 0 and 5 seconds
- *Charge current*: 100 mA or 500 mA
- *Error sensitivity*: from 0.2° to 2.8° in steps of 0.2° [1.0°]
- *Idle off*: amount of time the *BRIC* should be inactive before it shuts itself off. In steps from 30s to 1800s.

Error info: reports the errors info of the last ten measurements.

Calibration:

- *Display report*
- *Loop test*
- *CAL quick AZM*: quick azimuth calibration
- *CAL full INC&AZM*
- *CAL range finder*

Set Clock (scroll down beyond *second* to confirm)

- *Year*
- *Month*: from 01 to 12
- *Date*: day of the month, from 01 to 31
- *Hour*: from 00 to 23
- *Minute*: from 00 to 59
- *Second*: from 00 to 59

Bluetooth

- *Name*: eg Bric4_0039
- *MAC address*: eg D52FDE2E98DE
- *Connected to*: client MAC address
- *Reset BLE*
- *Advanced Menu*: opens the BLE Advanced menu

The BLE Advanced menu has three options

- *RST to AT mode*
- *RST to Run Mode*
- *Curr Comm*: CPU

The BLE Advanced menu shows

- the BLE transmit and receive counts
- the data sync status: total measurements and BT messages sent
- the last BLE remote command

Display

- *BL light*: 0 to 5
- *Color*: white, red, blue, green, purple, cyan
- *Contrast*: 0 (invisible) to 30 (completely dark) [16]

Advanced Menu

- *Memory clear*: opens the *Memory clear* screen
- *Sensor Raw Data*
- *Firmware CPU*: firmware and hardware versions - *Bootloader mode* option
- *Firmware BLE*: firmware version - *Update* option
- *Cal Settings*: several settings related to the calibration processes
- *Reprocess full Calibration*
- *Reprocess AZM Calibration*

The *Memory clear* screen shows the total measurements and BT-sent messages. It has an option to resets the data index to 0.

Error codes

The *BRIC* reports up to two errors with the shot data. The error code are recorded in the shot comments together with the error value. Nothing is recorded if there is no error (code 0).

1. no error
2. first accelerometer high value [normal 1]
3. second accelerometer high value [normal 1]
4. first magnetometer high value [normal 1]
5. second magnetometer high value [normal 1]
6. accelerometers disparity in an axis (1:X, 2:Y, 3:Z)
7. magnetometers disparity in an axis (1:X, 2:Y, 3:Z)
8. rangefinder calculation error
9. rangefinder weak signal

10. rangefinder strong signal
11. rangefinder pattern error
12. rangefinder response timeout
13. rangefinder error
14. rangefinder wrong message id
15. inclination disparity
16. azimuth disparity

BRIC CALIBRATION

The *BRIC* calibration is done on-board.

There are two calibration procedures for the orientation: a complete calibration, and a quick calibration that updates the correction of the magnetic sensor values.

Complete calibration

The complete calibration requires to acquire four data at different roll in several directions (at least 14), covering all possible orientations.

Go to *Menu, Calibration, CAL: Full INC&AZM*.

A message with instructions appears. Press any button to continue.

The display shows status and group distribution while performing the calibration.

Take four shots for each point rotating the device about 90 degrees around the laser direction.

Four dots show the number of shots taken for the group.

The markings in two horizontal bars show where the distribution of the completed groups magnetic and accelerometric values. Cursors under the bars show the position for the group that is being taken.

- The device automatically detects when a new group is started
- If fewer than four shot are taken the group is not saved
- If more than four data are taken, only the last four are used
- It is possible to *Reset* the current group and restart it

When 14 groups are taken the *Done* key appears. Pressing it starts the computation of the calibration. More groups can be taken before computing the calibration.

After the computation of the calibration a report is displayed.

Quick azimuth calibration

Go to *Menu, Calibration, CAL: Quick AZM*.

A message with instructions appears. Press any button to continue.

A grid with a cross-hair is displayed.

The device must take samples of the sensor values that cover all directions.

Turn the device slowly around in every direction. When the device need to take a sample the backlight turns off and sampling for that point begins.

When the sample is taken there is a beep.

When the grid is full the calibration computation begins and a report is displayed.

Note. The calibration can be interrupted at any time by pressing *Abort*.

Distance calibration

Go to *Menu, Calibration, CAL: Rangefinder*.

A message with instructions appears. Press any button to continue.

Place a target at a specified distance of 1 m, or 3 ft, from a station point. The distance must be measured with an accurate instrument.

Take four shots and ensure their values are equal.

Press *Done*. A calibration report is displayed.

BRIC INFO

This dialog is opened from the *Info* button of the Device window.

BRIC information:

- Device address
- Device name
- BLE firmware
- Firmware
- Hardware
- Battery level

BRIC MEMORY

This dialog is opened from the *Memory* button of the Device window.

With this dialog you can

- **reset** the *BRIC* last time (year, month, day, hour, minute, second)
- **clear** the *BRIC* memory

BRIC Last Time

After resetting the *BRIC* last time the device will resend all the stored data more recent than the "last time".

TopoDroid will refuse to reset to a time in the future.

The fields of the last-time are shown above the "Reset" button.

The minutes and seconds of the Android time are shown on the right below the last-time fields to help set the *BRIC* time in agreement with the Android time.

Clear Memory

Clearing the memory erases all the data from the memory and resets the shot counter to 1.

The old surveys remains on the *BRIC* internal sd-card.

SAP5

The *SAP5*, Shetland Attack Pony, has a single button. It is turned on and off by tapping two times in a row quickly on the button.

The *SAP5* needs to be paired with the Android before it can be used in TopoDroid. Open your Android' *Settings* app, and go to "Connected Devices". If the *SAP5* is not listed, turn it on and select "Pair a new device". The *SAP5* should appear in the list of available devices. Tap on it and enter the PIN "000000", six zeros.

When the *SAP5* is switched on, the laser is on. To take a shot point is on the target and "press and hold" the button, until the laser light has a glitch, or turns off. The readings are shown on the screen, either one at a time (azimuth, clino, distance, and plane extension) or all three at once. You cycle through the displays by tilting back and for the device. If you press the button when a value is on the display the laser turns on and the device is ready to take the next shot.

After the display of the readings, the screen shows the menus:

- *Store*, to assign stations to the shot and save it in the internal memory
- *Discard* goes back to take another shot
- *Main menu*, with
 - the device *Settings*
 - *Measure* goes back to take a measurement
 - *Calibrate* enters the device calibration procedure
 - *Visualize* a rough survey plan view
 - *Info* displays (among others) info on the shot, the device hw/fw, the battery charge, and sensor readings in real-time
 - *Off* turns off the *SAP5*

The battery level is shown on screen. The *SAP5* has a micro USB socket on the back. When it is connected to the PC it charges and the data can be downloaded using the "PonyTrainer" program.

When it is connected to TopoDroid, each shot is automatically transferred to the app as soon as it is taken. If the connection is interrupted, the shots taken when not connected are *not* transferred when the connection is re-established. Therefore the *SAP5* is suitable for surveys in which it can always remain in connection with the Android.

Refer to the *SAP5* documentation for further details about the device.

JedEye

The *JedEye* is an Arduino-based handheld instrument designed for dry cave surveying. It measures azimuth (compass), clino, roll, and direct distance (lidar). A full user manual is published at <https://sebkister.github.io/JedEye-Documentation/>.

The device is operated through two on-device buttons labelled *Next* (move the menu cursor) and *Select* (activate), and is charged over USB-C.

Bluetooth pairing

The *JedEye* uses Bluetooth Low Energy only - there is no PIN-based pairing. To make it visible to TopoDroid, switch the device to its *Distance Meter* screen via the on-device menu. It then advertises as *JedEye_<serial>*.

In TopoDroid, open the *Device* screen and run *BT Scan* from the menu. The *JedEye* appears in the list; tap it to add it. Pairing only needs to be done once - subsequent connections are automatic when the device is back in the *Distance Meter* screen.

Taking a shot

On the *Distance Meter* screen the OLED shows the current lidar distance and the current heading and clino in real time. A small Bluetooth icon in the bottom-right corner indicates that BLE advertising is active.

To capture a measurement on the device, point it at the target and invoke *Measure* with the *Next* / *Select* buttons. The captured values are streamed to TopoDroid as a single shot. If TopoDroid's shot window is open, the new shot appears immediately in the list and is auto-assigned a station name like any other leg shot.

TopoDroid can also drive the *JedEye* remotely. While connected, tapping the Bluetooth button in the shot window opens a popup with *laser on*, *laser off*, *shot*, and *device off* options. *Shot* tells the *JedEye* to capture and send a measurement without touching the device - convenient when it is mounted on a tripod or otherwise out of reach.

Connection mode

The *JedEye* is always connected in *continuous* mode. Each measurement is transferred to TopoDroid as soon as it is taken. Shots taken while the device

is not connected are *not* buffered or re-transmitted later - the *JedEye* is best used in sessions where it can remain connected to the Android throughout.

Battery

The *JedEye* runs on an internal battery and charges over USB-C. The current battery level is shown in the device's *Info* screen and as a small icon on the main display.

NEW SURVEY

This dialog is opened from the *Add* button in the Main window.

Information for the new survey:

- *name*
- *date*
- *team*
- *initial station*
- *magnetic declination*
- *survey description*
- *diving data-mode* [T]

The *name* is mandatory. The survey names are used to list surveys in the Main window and TopoDroid distinguishes surveys by their name (although each survey has a unique numerical id in the database).

The name must be unique in the database. Names that differ only by the characters case are not different because the filesystem is case insensitive. TopoDroid raises a warning if the new survey name coincide with the name of an existing survey except for the characters case.

WARNING. The name of the sketch cannot contain special characters: '/', ':', '<', '>', '*', '\\', '?', '|', '''.

The name of the survey cannot contain spaces. Spaces are replaced with '_' (underscore).

The *date* is written with format "yyyy.mm.dd". For example January 6, 2015 is "2015.01.06". It is preset to the current date.

The *team* can be a plain text with the surveyors' names, or the names can be separated by commas or semicolon for better *Therion/Survex* export. It is also possible to use a [dialog](#) to enter the separated team names.

The "Team names" setting control how the team field should be entered and how it is handled in the exports.

The *team* should be filled in, otherwise the dialog has a warning when saved.

The team can be forced to be empty by pressing the "save" button a second time.

The *description* is empty. It can safely be left so, although it is a good practice to take the time to write a description of the survey.

The *magnetic declination* is "unset" by default. If you know it you can enter it here, otherwise you can look it up in the web after you have found the coordinates of a fixed point (say, the cave entrance) with the GNSS [A]. The declination must be entered in decimal degrees. The accepted range of values is [-360°, 360°]. Any value outside this interval is considered as "unset".

Buttons

Save: the new survey name is added to the database.

Open: saves and starts the *Shot list* window immediately.

Diving data-mode [T/G]

The diving data-mode is an experimental feature. Data can be entered only manually, and in the following format:

- FROM station
- TO station
- depth at the FROM station
- azimuth of the direction FROM-TO
- distance between the FROM and the TO station
- LRUD at the FROM station (optional)

Backshot data are not supported in "diving" data-mode.

TEAM DIALOG

The team dialog allows entering team names without bothering about separators between names. The program takes care of combining the team names in a string with proper separators.

The characters comma (',') and semicolon(';') are reserved for separators and must not be used inside names.

SURVEY IMPORT

This dialog is opened with a tap on the *Import* button of the Main window.

TopoDroid imports survey data from TopoDroid zip archives or from external files.

To import files or zip archives tap on the "Import" button. Files and zip

archives are selected through a system filemanager.
The name of the file must not contain a colon ':'.

The dialog displays the list of supported import file formats, besides "zip" archives:

- *Compass* ".dat"
- *Survex* ".svx"
- *Therion*. Only data ".th"
- *VisualTopo* ".tro" and ".trox")"
- *Walls* ".srv"
- *TopoRobot* ".text"
- *PocketTopo*, including sketches ".top"
- *CaveSniper* ".csn"
- *BRIC* and *Cavway* csv ".csv"

For some formats it is possible to customize the import through a few flags.
A file import fails if a survey with the same name is already in *TopoDroid*.

The import file is selected with the system content provider.
File import fails if the file extension is not one those listed above

Warning The files to import must be picked from the filesystem, not from a "media container".

On import from file, LRUD values are converted to x-splays (shown in green).

Compass import ".dat"

- *LRUD* if enabled LRUD values are imported and converted to splay shots
- *Leg first* if enabled LRUD splays are placed after the leg
- *Diving datamode* enable for data diving mode in the *Compass* file [T]

TopoDroid converts imported compass survey from *normal* datamode to *diving* datamode [T], if the relative setting is enabled [G].

Survex import ".svx"

N.B. *Survex* syntax is partially supported

Only the data modes "normal", "topofil" and "passage" are handled to some extent. In particular for the "passage" data mode, the "left" and "right" values are not used because they cannot be converted to splays.

TopoDroid has only single survey sessions. Therefore, data of multi-session files are merged in a single session.

Therion import ".th"

N.B. *Therion* syntax is partially supported.

Only *Therion* survey files with data in "normal" datamode are importable.

The "declination" command is taken into account: the read azimuth are corrected with the value of the declination.

If the station short-names have no clash the file can be imported with "short-names". Otherwise, it must be imported with full names, ie station names qualified by the survey path.

TopoDroid does not import *Therion* scraps.

TopoDroid supports the *Therion* "input" command. Therefore, whole *Therion* projects can be imported as a single survey, starting from the top *Therion* file.

It is necessary to use full names for stations.

The survey files should have the "declination" command if the dates span a long time.

VisualTopo import ".tro" and ".trox"

N.B. *VisualTopo* syntax is partially supported.

VisualTopo (tro trox) import options:

- *LRUD* if enabled LRUD values are imported and converted to splay shots
- *Leg first* if enabled LRUD splays are placed after the leg

TopoDroid can import also *VisualTopo* surveys with *Prof* instead of *Clino* in "trox" format. The data are stored in *diving* datamode.

Walls import ".srv"

N.B. *Walls* syntax is partially supported.

Importing data from a *Walls* file can result in disconnected survey.

TopoRobot import ".text"

TopoRobot import is very experimental.

PocketTopo import ".top"

TopoDroid imports both the survey data and the sketches. This import maps PocketTopo colors to TopoDroid drawing tools (points and lines). The mapping can be modified through a setting.

CaveSniper import ".csn"

BRIC and Cavway csv import ".csv"

BRIC and Cavway csv import is experimental.

Survey data imported from BRIC csv files contains only distance, azimuth, and inclination.

The survey data must be edited after importing to add the other information.

Survey data can be imported from Cavway csv files created with the CavwayAssist program

CALIBRATION-CHECK SHOTS

This dialog is opened from the Calib. check menu in the Survey Info window.

The dialog shows the calibration check legs. Tapping on a leg-shot it displays the distribution of the angle differences between the shots of the leg and the average leg direction.

*If there is the backsight leg, or a leg that is opposite (ie, differing less than 0.1 m and 1.0°) the distribution of the angles between any two shots, one for each leg, is displayed on the right in light gray color.
The yellow line is 0.5° and the red line is 1.0°*

Otherwise, if there are two other calibration check shots that make a triangle with the selected one, the distribution of the mis-closure errors is shown on the right in gray color.

*The mis-closure error is computed (approximately) as the ratio between the mis-closure distance and the radius of the circle enclosing the triangle.
The yellow line is 0.5% and the red line is 1.0%.*

GEO-POINTS LIST

This window is opened tapping the GNSS button in the Survey Info window.



The window shows the list of geo-localized points. TopoDroid does not use geo-localized data for the sketches of survey sessions. Geolocalization is used only for exports to PC cave programs and formats that require geographic coordinates.

There are buttons to insert a new geo-localized point.

- [GNSS](#) gets the new point coordinates with the GNSS (if "Location" permission has been granted)
- [Insert](#) the coordinates of the new geo-point manually
- [Import](#) the coordinates from a point file created by a GNSS app.
- Start GPSTest app

Warning. The accuracy of a geopoint obtained with a phone is usually 15-20 m at best.

Tapping a geo-point entry in the list opens the [Geo-point edit](#) dialog.

Internally TopoDroid stores the coordinates in decimal degrees. In the coordinate dialogs you can type them either in decimal degrees or as "dd:mm:ss.ss". The TopoDroid keypad has special keys for "degree" and "minute". If you use a keyboard that does not have colon ':', use the space instead. If the keyboard does not have the decimal point '.' use the slash or the comma instead.

Menus:

- Close the window
- Cycle through the external GNSS apps for importing coordinates. If there is only one GNSS app this menu option is not shown. This menu affects the External GNSS app setting.
- Help

External GNSS apps

TopoDroid can import geopoint coordinates from

- MobileTopographer point files. A dialog is opened to choose the geopoint.
- GPX Recorder track (gpx) file. The last track-point is used.
- GPS Position position (csv) file. TopoDroid computes an average of the positions.
- GPSTest fix from a log file. TopoDroid uses the last fix.
- GPS Logger track file saved in "csv" format. TopoDroid uses the last track-point.

The GNSS app in use is shown in the window title.

The station name of geopoint imported from GPX Recorder, GPX Position, and GPSTest is set by TopoDroid automatically as a number prefixed by '#'.

MobileTopographer point-list files are stored in the folder MobileTopographer/pointlists.

GPS Recorder track files are stored in the folder gpx_recorder.

GPS Position files are stored in the primary external storage.

GPSTest logging must be enabled in the app setting. It is enough to log the fixes. The log file is saved when the app is closed, and is stored in the folder gnss_log.

Furthermore geopoint coordinates saved in the clipboard with the GPSTest app can be transferred to the fields in the geo-point insert dialog. Make sure to set the checkbox to store also the altitude in the clipboard.

GPS Logger track files must be exported in "csv" format. TopoDroid uses the ellipsoid altitude and the accuracy.

GNSS app advices (search internet for details and how to do)

- *Enable "force full GNSS measurement"*
- *If available disable "kinematic Kalman filtering"*
- *If available set location "static mode"*
- *If available disable "A-GPS". Disable "mobile data" and clear A-GPS data*
- *Disable location "wi-fi and bluetooth fix"*

Note. Disabling battery optimization for the app is necessary only if you want to prevent the system closing the app when it is not in foreground.

Location settings

- *Coordinates units: either dd.mm.ss or decimal degrees [A]*
- *C.R.S. coordinate reference system [default is long-lat] (disabled)*
- *Negative altitudes [default no]*
- *Editable altitudes [default no]*
- *Fine position time [default 60 s]*
- *GNSS app [default non-specified]*

GEO_POINT INSERT

This dialog is opened from the Add button in the [Geo-points](#) window.

This dialog has six text fields:

- **name** (station)
- **longitude**
- **latitude**
- **geoid altitude** [m]
- **accuracy** [m]
- **comment**

The name does not have to be a survey station.

To the left of the longitude there is button to specify whether the longitude is east (positive) or west (negative). Tapping the button switches between "E" and "W".

To the left of the latitude there is button to specify whether the latitude is north (positive) or south (negative). Tapping the button switches between "N" and "S".

Longitude and latitude can be entered as degrees-minutes-seconds (dd:mm:ss.ss) or as decimal degrees. Alternative separators for colon are space, single quote, and the degree symbol. Alternative separators for the decimal point are comma and slash.

Longitude and latitude can have a negative sign.

Altitudes are entered in meters. Altitudes are referred to the geoid and can be negative.

Buttons:

- **Clipboard**: the coordinates saved to clipboard as "lat,long[,alt]" are copied from the clipboard to the respective text fields. With this button you can use the app GPSTest to get the coordinates (altitude must be checked when saving data to the clipboard in GPSTest).
- **View** invokes a map app (provided you have one installed) to display the point.
- with **Proj4** you can enter the coordinates in another Coordinate Reference System. The app Proj4 is invoked, and you must select the CRS and enter the coordinates on the left side: longitude/East, latitude/North, and geoid altitude. Units of altitude and cartographic coordinates are meters. Units of geographical coordinates can be either decimal degrees or degree-minute-seconds. Next you tap the conversion arrow and the coordinates are converted to geographical WGS84 and sent back to TopoDroid.
- **Save** saves a record with the fixed station data in the database.

GNSS GEO-POINT

This dialog is opened from the GNSS button of the [Geo-points](#) window.

To use the GNSS the Android "Location" service must be enabled. For security reasons a generic app, like TopoDroid, cannot enable the GNSS. If it is not enabled, TopoDroid shows a dialog asking whether you want to enable it (through the "Settings" app). Remember to disable it when you are done, to save battery life.

Fields:

- **station** name
- **comment**

Buttons:

- **GNSS** starts/stops the geo-point location
- **View** the point in a map app
- **Save** the point

While searching for satellites the "GNSS" button becomes red (or green) displays the number of satellites in sight. When enough satellites are in view the button becomes green and the coordinates are displayed underneath. You should wait until the coordinate values are stable and the location accuracy is shown. Then stop the GNSS location tapping the "GNSS" button again.

The reported accuracy depends on the phone hardware and the environment.

On Android before API-26 only the horizontal accuracy is displayed. On Android API-26 and later both the horizontal and the vertical accuracy are displayed.

Settings

Fine location time: how long to let the app do the fine location of a geo-point [default 60 s].

GEO_POINT IMPORT

This dialog is opened from the Import button of the [Geo-points](#) window.

With this dialog you can import in TopoDroid the coordinates of a point acquired with one of the supported GNSS apps.



MobileTopographer

TopoDroid reads the point coordinates from a MobileTopographer point-list file.

If you do not select a MobileTopographer point-list file the app behaviour is undefined.

1.  Open MobileTopographer and find the point coordinates.
2.  When you have reached a sufficient accuracy in the coordinates, save the point.
3.  You can also enter the point coordinates by hand (and save the point).
4.  Save the point list. The list is saved in "MobileTopographer/pointlist"
5. Import the coordinates in TopoDroid. The saved points are shown in a list; tap on the point to import.



Optionally, you can go to the list of saved points in MobileTopographer, and move the point you just got to the top with the arrows.

GPS Logger

The app displays the point coordinates and other infos. The track files are saved in the folder "gnss_log". They have an header with device info, and the points data in CSV records: constellation, latitude, longitude, altitude, speed, accuracy, and time.

The log can contain also raw GNSS measurements, navigation messages, and NMEA sentences.

1.  Start recording the track points.
2.  Stop the track recording.
3. Finalize the track. The track is added to the "tracklist".

GPSTest

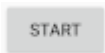
The app displays the point coordinates and satellites data: fix, constellation,

carrier frequencies, carrier/noise ratio, flags, elevation, azimuth. The point can be shown on the map (option in the "settings").

1.  Open the share/export dialog.
2.  The point coordinates can be copied to clipboard
3.  or saved to file. It is important to check the "Include altitude" box.

GPX Recorder

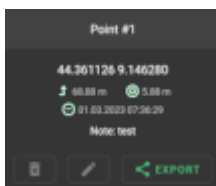
The points coordinates are saved in a track file that can be read by TopoDroid. Track files are in the folder "gpx_recorder". The format is xml and the points data are latitude, longitude, altitude, and time.

1.  Start recording the points coordinates
2.  Stop the recording.

GPS Point

TopoDroid reads the coordinates exported to a point file: latitude, longitude, altitude, and accuracy.

1. The app displays a list of saved points
2.  Open the GNSS interface
3.  Acquire the GNSS position
4. The app displays the point with the circle of uncertainty on the map. The point coordinates, and other infos, are shown in a box.
5. Save the point when done.
6. Select the point from the list, and export to (text) file.



GPS Position

The app displays the point coordinates and the satellite data.

1. 
2.  Display the satellites data: fix, azimuth, elevation, carrier/noise ratio, and flags (almanac, epherals)
3.  The position can be copied to clipboard
4. or it can be saved to a file.

GEO-POINT EDIT

This dialog is opened tapping a point entry in the [Geo-points list](#) window.

The dialog shows

- longitude, latitude and geoid altitude
- point **name** and **comment**
- magnetic **declination** at the point (at the current date)

Buttons

- **View** the point in a Location app (provided you have one installed)
- **Clear** converted coordinates. This action is immediate.
- **Convert** the geo-point coordinates to another reference system (using the app Proj4)
- **Delete** the geo-point record

If the coordinates of the geo-point have been entered manually, they can be edited. Coordinates taken with GNSS or imported from a GNSS app cannot be changed. The geoid altitude is editable, irrespective of the input method, if the appropriate setting is set.

A long-tap on the longitude or the latitude changes the display from DMS to degree and vice versa.

The geoid altitude is computed from the ellipsoidic altitude using the geoid model EGM2008.

The **Convert** button computes the geo-point coordinates in another Coordinate Reference System, using the app Proj4. The geoid altitude is used in the conversion. The converted coordinates (East, North, and altitude) and the estimated meridian convergence [deg] are displayed and stored in the database with the geo-point.

The magnetic declination is computed using the World Magnetic Model most recent model (WMM). The uncertainty of the WMM model declination can be as large as 0.4 degrees in temperate regions.

Check the **save** box to set the magnetic declination into the survey.

The **Save** button commits the changes to the database.

SURVEY NOTES

This dialog is opened from the Notes button of the Survey Data window, the Survey Info window, and the Drawing window (in "moving" mode).

With this dialog you can write and edit a generic text with notes that integrates the data and the drawing of your survey.

Buttons:

- **Save:** save the changes and close the dialog
- **Back** key: close the dialog without saving the changes

The notes are saved in a text file in the "txt" subfolder of the survey folder. You can also edit this file with other text editors.

SURVEY RENAME

This dialog is opened from the Rename menu of the Survey Info window. The menu is present only at "tester" level.

Buttons:

- **Rename** the survey
- Add a **prefix** to all the survey station names.
- **Close** closes the dialog.

Survey rename

The new survey name must be typed in the field to the right of the "Rename" button.

The "section" points have the survey name written in the option "-scrap". The survey name is automatically updated the next time the sketch is opened in TopoDroid.

WARNING. *The name of the sketch cannot contain special characters: '/', ':', '<', '>', '*', '\\', '?', '|', '\"'.*

The name of the survey cannot contain spaces. Spaces are replaced with '_' (underscore).

Station prefix

The prefix for the station names must be typed in the field to the right of the "Prefix" button.

SURVEY SPLIT / MOVE

This dialog is opened from the Split/Move button of the shot secondary edit actions.

You can split the current survey and create a new survey. The the shots for the one being edited included to the end are moved to the new survey. Alternately you can move the shots to an already existing survey. For this there is a selector to choose the survey to which the shots are moved.

Buttons:

- **Split** split the shots into a new survey
- **Move** move the shots to the selected survey
- **Cancel** closes the dialog without doing anything

Warning

Moving shots from a survey to another may leave dangling sketches and media objects (photo and audio) without reference to a shot.

Surveys bulk operations

This dialog is opened from the Main Window menu "BULK OPS."

It allows to do bulk operation on a set of selected surveys.

The available operations are

- **import:** import all zip archives from a folder
- **delete:** delete the surveys. There is a confirmation dialogue
- **export:** export the surveys. The export format can be chosen afterwards

Other buttons

- **toggle:** toggle the surveys selection
- **cancel:** close the dialog

SURVEY DATA INFO

This dialog is opened from the Info button of the Survey Info window.

The survey data info are

- *number of (regular) legs, their total length and projected length (on horizontal plane)*
- *number of duplicate legs*
- *number of surface leg*
- *number of splay shots*
- *number of stations*
- *the survey length, that is the leg length not including flagged or repeated legs*
- *number of cycles in the midline graph*
- *number of disconnected surveys (this is 1 if the legs are all attached together in one survey)*
- *number of bluetooth devices used in the survey, and number of data from each device*
- *the first and last dates of the survey data*

Length is in the units set in the app settings.

If only one device has been used in the survey, there are also

- *std. deviation of the absolute value of the acceleration field (percent)*
- *std. deviation of the absolute value of the magnetic field (percent)*
- *std. deviation of the magnetic dip angle (degrees)*

The last three are also displayed as histograms. These plots are centered about the mean of the values that falls within the bounds.

The yellow lines mark the value of the accuracy threshold setting (percent for G and M, degrees for the dip). These values are shown next to the histograms.

DATA DOWNLOAD

The Download button starts a connection with the device to download data. While the connection is active it stays red. If you are using the on-demand download-mode the connection is closed when there are no more data to retrieve. With the continuous download-mode the connection remains open (and data are continuously downloaded as soon as they are taken) until the

device and the Android lose the Bluetooth connection. Which mode is best suited for you depends on your habits. With the on-demand mode you can concentrate on sketching and other tasks without being bothered by incoming data, and when you download the data you can focus on them (they are highlighted blue). On the other hand on-demand download requires pauses to get the data, and these are especially long if you took several shots in the between.

When the Bluetooth connection with the device is broken (eg. the device gets too far from the Android, or it is switched off) in on-demand mode, the data download is interrupted. In continuous mode TopoDroid retries to connect to the device every few seconds.

When the Survey Data window is closed the data download connection is terminated. The download connection is not interrupted when the window is paused (the display switches off). If you open a sketch while the download connection is active in continuous mode, the connection is maintained and the data are downloaded (and displayed in the sketch).

For SAP, BRIC, and JedEye data are always downloaded in continuous mode.

DistoXBLE

The data download from the DistoXBLE may fail the first time immediately after the **Android Bluetooth** has been enabled (ie, turned on). In this case switch the DistoXBLE off and on, and try again to download data.

BRIC

The data download from the BRIC can be limited to only the essential values (distance, azimuth, clino, and little more), or complete, with all the information transferred by the BRIC protocol.

The complete protocol includes the "index" of the shot. TopoDroid uses the BRIC index as data index in the database table, if it receives it, and the BRIC data mode is set to use also the index. Since the data index must be unique, if TopoDroid receives a shot with an index that is already in the database, instead of the BRIC index it uses a novel value for the data index.

The BRIC index can be reset with the Advanced menu | Memory clear BRIC command.

It is possible to disable the use of the BRIC index [G], and let TopoDroid assign incremental values to the data index.

MANUAL DATA INPUT

If you do not have a bluetooth device you can still enter the shot data manually. The Add button opens the dialog for the shot data input. You must enter the station names and the shot data (length, azimuth, and clino).

If only the backward fields are filled in (the forward fields are left empty) the shot is entered with the backward data.

To add a splay shot enter "." (period) or "-" (dash) in the TO station. LRUD can be added to a splay shot, thus providing additional splay shots. In this case the LRUD refer to the FROM station.

There is a warning if, upon editing a leg shot, it differs from another leg between the same stations more than one fourth in length, or 15°.

Buttons:

- **Save** inserts the shot into the survey, and updates the station names incrementing them so that the dialog is ready to input the next shot
- **OK** inserts the shot in the survey, and closes the dialog.

Station names

If a station name contains the following characters, '/', '', ':', they are replaced with '-', '+', and '-', respectively. Multiple spaces are replaced with ' '. Backslashes are dropped.*

If the TO station name is "." or "-", it is replaced with the empty name.

Using Android for the direction

Azimuth and clino can be set using the Android sensors [G]:



the reference axis is the direction of the long side of the device. A timer (10 s by default) lets you point the device before measuring.



the reference axis is the normal into the display: look at the TO station on the screen and start the timer when the station is in the middle of the cross-hair.

A warning is shown if the sensor accuracy is low or there is a sensor failure.

LRUD

If LRUD are specified the respective splay shots are also inserted.

Additionally you can select whether these refer to the FROM station (default) or to the TO station.

If the shot clino is smaller than the Leg WENS threshold setting, the LRUD are in a vertical plane: U is upwards, D downwards, L horizontal to the left of the shot (i.e., with azimuth that of the shot minus 90°), and R horizontal to the right of the shot.

If the shot clino is greater than the Leg WENS threshold setting, the LRUD are in a horizontal plane: L is west, R is east, U is north, and D is south.

Units

Data must be entered in the current units. For example if the length units are "feet", distances must be entered in "feet".

Diving mode [T/G]

In diving mode the shot data are depth, azimuth, and distance. The azimuth is assumed taken with a normal compass, with graduation increasing clockwise. The azimuth entries are converted as if taken with a caving (geological) compass. For example an entry of 340° is converted to 20°.

Diving mode is enabled through a secondary setting.

INSTRUMENTS OFFSETS

This dialog is opened from the Offsets menu of the Survey Info window.

It is used to define offsets for the instruments of manually entered data. The offset values must be written in the current units. For example if the length units is "feet", the tape calibration must also be in "feet". Internally the offset values are stored in default units, meters and degrees.

- **tape**
- **compass**
- **clino**
- **LRUD**: whether the tape offset affects LRUD or not

Buttons:

- **Save** records the instruments offsets and closes the dialog.

Note.

The instruments offsets are not stored in the database, and are reset when the Survey Data window is opened.

STATION NAMING POLICIES

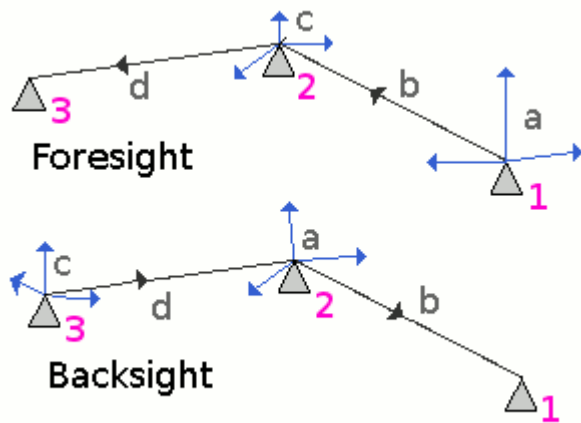
When you download data TopoDroid can automatically assign them station names. A repeated measurement is considered a leg shot, the others are splay shots. If you stick to a convention taking measurements TopoDroid automatic station naming can be very handy. TopoDroid supports a number of conventions to automatically assign stations to shots. You can select the one that fits your habits through the settings.

1. **splays+foreshot**. At the FROM station: you take splays first and then forward leg shots. This is the default convention.
2. **splays+backshot**. At the TO station: you take splays first and then backward leg shots
3. **backsight**. Before taking splays and foreshot, a backsight check leg is taken.
4. **tripod** [A]
5. **magnetic anomaly** compensation [E]
6. **foreshot+splays**. At the FROM station: you take the forward leg shots first, and then the splays
7. **backshot+splays**. At the TO station: you take backward leg shots first and then splays
8. **TopoRobot** [T]

Examples of the conventions (shots taken at the same station are grouped in parenthesis):

1. ... (1-. 1-. 1-. 1-2) (2-. 2-. 2-3) ...
2. ... (2-. 2-. 2-. 2-1) (3-. 3-. 3-2) ...
3. ... (2-1 2-. 2-. 2-3) (3-2 3-. 3-. 3-4) ...
4. ... (2-1 2-. 2-. 2-3) (4-3 4-. 4-. 4-5) ...
5. ... (2-1 2-. 2-. 2-3) (3-2 3-. 3-. 3-4) ... (same as "3")
6. ... (1-2 1-. 1-. 1-.) (2-3 2-. 2-.) ...
7. ... (2-1 2-. 2-. 2-.) (3-2 3-. 3-.) ...
8. ... (1-. 1-. 1-. 1-2) (2-. 2-. 2-3) ... (same as "1")

Sequence a-b-c-d for the first (foresight) and the second (backshot) policies.



Each time the automatic station assignment is carried out, it starts from the last leg (or second to last for the backsight and tripod policies). If the survey has yet no leg, it starts from the first shot. Therefore shots before the last leg are not modified, while stations are assigned to all the shots following the last leg. This may affect the station of splays immediately following the last leg, when the station-assignment policy is changed.

Active station

The active station can be chosen manually, both in the Measurements window (by long-tapping on the name of the chosen base) and in the Drawing screen (by clicking on the Apply button in the Station dialog). The manually selected active station is shown in green in both screens. If the active station is not set, the last station is used.

Backsight policy

At each station you take the backshot to the previous station, next the splay shots, and finally the forward shot to the next station. Backsight shots are checked: if a shot is not close to the opposite of the previous leg, it is not considered a backsight shot. The backsight shots are marked "duplicate".

Tripod policy

At station "1" you take splays and then shots to the station "2". Then you move into station "3", take shots to station "2", then splays, then shots to station "4". Move to station "5" and so on.

Magnetic anomaly compensation policy ***[Tester level]***

The magnetic anomaly compensation is done estimating at each station the "local" magnetic declination by comparing the azimuths of the leg reaching to the station and the backsight leg. The azimuth of all the shots at the station are then corrected with the "local" declination. An overall declination remains because the declination at the first station cannot be determined. Furthermore the survey azimuth angles become relative as in a traverse, therefore the angle errors add up instead of compensating. The magnetic anomaly compensation requires taking backsight shots. However, with magnetic anomaly compensation the check whether a shot is a backsight does not take the azimuth into account. The magnetic anomaly compensation is carried out during the data reduction (for the sketches).

TopoRobot policy

The "TopoRobot" policy is just like the first, but station names follow the TopoRobot convention.

Station naming policy and usage profiles

Certain station naming policy are enabled only at high usage profiles:

- "Tripod" requires "advanced" level*
- "Magnetic anomaly" requires "expert" level*
- "TopoRobot" requires "tester" level*

Reducing the usage profile when one of these is active, automatically set the station policy to the default policy (splays + foreshot). The policy is automatically reinstated again if the usage profile is raised to the required value or above.

Blunder leg-shot [T]

A blunder shot is an intermediate leg shot that happened to be off-target because of the distance reading. Typical leg sequences with blunders 'B' are "A-B-A-A" and "A-A-B-A".

Blunder leg-shot detection is an experimental feature of the default station naming policy. It must be enabled with a secondary setting.

Blunder shots are detected and marked as blank entry that will be hidden when the list is refreshed.

Blunder shots can be recovered.

ACTIVE STATION

The "active" station is the station from which the survey will continue. If you have set the "active" station, TopoDroid will continue numbering stations from it. If the "active" station is not set, the last one is used.

Thus you can start a branch in the middle of the survey by selecting the station you want to attach as "active" station. Alternatively, you can replace the FROM station of the first shot of the branch, afterwards.

The "active" station is highlighted green in the Survey Data window and in the Drawing window. A long tap on another station switches the "active" station to that. A long tap on the "active" station itself turns it off. The "active" station can be set in three ways

- with a long tap on the station name in the Survey Data window.
- from the Saved stations dialog (button Active")
- through the Station edit dialog in the Drawing window.

SAVED STATIONS

This dialog is opened from the  Station button in the Survey Data window, or from the edit station dialog of the Drawing window.

The "saved stations" are station names saved for later reference, e.g., at a crossway, to go back and continue the survey from there, or to mark a possible lead to check in a future trip.

Note The names of stations attached to shots or to "saved stations" are just names and do not identify entities. Therefore, when a station name of a shot is changed, the same name in a "saved station" does not change.

The dialog displays the list of saved stations. You select one by tapping it.

The selected station is shown in the text fields. Opened from the Survey Data window, the text field contains the last station, or the active station if this is set. Opened from the edit-station dialog, the text field contains the station.

Each saved station must have a brief description (comment) or a flag set. It is possible to assign a geocode to a saved station [T].

The dialog displays

- the selected **station**
- the station **flag**, if any ("fixed" or "painted")
- the station **comment**

Buttons:

- **Clear** the station fields: name comment and flag
- **Geocode** sets/changes the station geocode [T]
- **Save** the station into the database
- **Delete** the station from the database (non-recoverable)
- **Close** closes the dialog
- **Splay** toggles the display of splays at the station (only if the dialog is opened from the Survey Data window)
- **Scan** toggles the display of scan sets at the station (only if the dialog is opened from the Survey Data window)
- **Active**: sets the station as the "active" station, and closes the dialog

STATION NAMES MAP

This dialog is opened from the "STATIONS" menu of the survey window. The dialog is used to replace the station names of the survey.

The dialog contains a two-column list of station names. The left column contains the current names. The right column is for the replacement names. Buttons:

- **OK**: to confirm the replacement
- **Cancel**: to abort the replacement
- **Import**: to read the replacement named from a file.

If two stations are mapped to the same name, there is an alert before the name replacement. The user can cancel the replacement and fix the name conflict or decide to continue with the replacement.

Input file

The input file must be a text file with a pair of names (old and new station names) separated by a single space on each line.

If the first name of a pair is not a station of the survey the pair is not taken into account. If the first name is a survey station, the second name is set as its replacement.

Station Photo

This page displays the station photo, if any.

Actions:

- *take/retake the picture*
- *delete the photo*

STATION/SHOT SEARCH

This dialog is opened from the Search button of the Survey Data window.

It has a text field to enter the name of the station to search, and a checkbox to include also the splay shots in the search.

Station search buttons:

- *Leg stations: searches for legs with the specified station. Only one station should be informed.*
- *All stations: searches for legs and splays with the specified station. Only one station should be informed.*
- *Leg search: searches for legs that connect the two informed stations. The names of two bases should be informed separated by space.*

There are also three buttons to search for legs, and one for splays [T]:

- *duplicate legs*
- *surface legs*
- *legs with extend not set*
- *reverse splays*

*The **search** buttons closes the dialog and move the first matching shot to the top of the display.*

The search is carried out only on the displayed data, and the search matches are highlighted with a yellow background on the stations column.

A long tap on the Search button scroll the shot list moving the next match to the top on the display.

The search is cleared by tapping any button.

EXTEND REFERENCE

This dialog is opened from the Azimuth button either in the Survey Data window or in the Drawing window.

The extend reference is used to assign the profile "extend" direction to the midline legs.

The "extend" direction is not automatically set for the splays.

The Azimuth button displays either a long arrow inside a circle, oriented to the current extend reference direction, or a short arrow directed left or right (for fixed "extend").

When the profile "extend" is compute using the "extend reference" it is set to right if it differs from the reference by less than 90°. Otherwise it is set to left.

When the Manual extend reference setting is enabled, the Azimuth button has only two states, either LEFT or RIGHT. Tapping the button switches between them. The new shots extend is set according to the state of the button.

This dialog has

- "extend reference" **azimuth**
- slider to change its value between -180 and 180°
- **text field** to enter the azimuth numerical value, between 0 and 360
- **compass** sets the value using the Android magnetic sensors (using the direction of the long side)
- **left** forces "extend" to LEFT (-1)
- **right** forces "extend" to RIGHT (+1)
- **ok**: use the azimuth for the "extend"

Sketch extend reference mode

The extend reference azimuth can be changed also in the plan view of a sketch [E]. A long tap on the Azimuth button turns the window in "extend reference" mode. In this mode a finger slide across the display rotates the reference azimuth (the canvas is not shifted). To get out of this mode lift the finger from the screen.

Leg extend

When a new shot is downloaded TopoDroid sets its "extend" automatically. This holds also for shots entered manually, if the "extend" in the shot dialog is left "unset". By default the "extend" is either LEFT, RIGHT, or VERT (vertical), depending on the angle the shot makes with the reference azimuth.

If the angle is smaller than 90° minus the extend vertical threshold the leg is

extended RIGHT. If it is larger than 90° plus the extend vertical threshold the leg is extended LEFT. Otherwise it is extended VERT.

The Extend Reference dialog allows to change the reference azimuth, by rotating it clockwise or counterclockwise. Alternatively, the reference azimuth can be set using the Android compass sensors. A warning is shown if the sensor accuracy is low or there is a sensor failure.

It is also possible to fix the "extend" for coming shots to either LEFT or RIGHT, by selecting the appropriate buttons in the dialog.

Splay extend

The "extend" of splay shots is determined according to that of the legs at the splay station and the azimuth between the splay and the legs. The splay is extended as if it were projected on the closest of the vertical planes passing through the legs.

SHOT ACCURACY

Together with each shot, the DistoX2, DistoXBLE and Cavway record and transmit also the values of the intensity of the magnetic field (arbitrary units), that of the gravitational field (arbitrary units), and the dip angle, that is the angle the magnetic field makes with the horizontal plane.

By comparing these values with their average values TopoDroid can detect whether a shot might be affected by some anomaly, and should therefore be redone. These unreliable shots are displayed with a red background.

The amount these values can differ from their averages before the shot is declared unreliable are specified by the accuracy settings.

Note *When surveying in an area with magnetic anomaly, the anomaly analysis based on the magnetic field is not valid.*

Accuracy settings

- *Acceleration: percent difference of the acceleration from its mean value, above which a shot is signalled anomalous (DistoX2, DistoXBLE, Cavway)*
- *Magnetic: percent difference of the magnetic field from its mean value, above which a shot is signalled anomalous (DistoX2, DistoXBLE, Cavway)*

- *Dip angle: difference of the magnetic dip from its mean value, above which a shot is signalled anomalous (DistoX2, DistoXBLE, Cavway) [units degrees]*
- *Sibling threshold: percent difference of leg values to identify two legs as "bad" siblings*

Sibling

Siblings are legs with same stations.

In principle the leg between two stations can be measured two or more times, even backwards, in particular if one measure foresight and backsight.

SURVEY DATA DISPLAY MODE

This dialog is opened from the  display mode button in the [Survey Data window](#).

Through this dialog the user can hide/show certain classes of shots:

- **shot id**
- **splay**: shots with only one station
- **scan**: scan sets
- **latest splays**: only available if Secondary settings > Survey data > Recent shots setting is enabled. Time limit defined by Secondary settings > Survey data > Recent data timeout setting (in seconds).
- **blank**: shots with no stations
- **repeated leg**: shots close to the preceding leg

The shot ids are unique increasing indices assigned by TopoDroid to the shots of the survey.

If the splay shots are hidden it is still possible to show the splay shots for individual stations. Tapping on the station name in a leg shot opens the Saved stations dialog through which the display of the splays of the specified station can be toggled on and off.

SHOT EDIT

This dialog is opened by tapping on a shot data entry in the Survey Data window.

It displays

- **length, azimuth, inclination.** Editable for manually entered shots; not editable for downloaded shots. DistoX backshots are marked with '*'
- differences of acceleration, magnetic field, and magnetic dip from the respective average values. Only for DistoX2 shots
- date and time of the shot
- **stations** and
 - **left arrow:** load previous leg
 - **reverse** button: swap stations
 - **right arrow:** load next leg
- shot **comment**
- profile **extend:** "left", "vertical", "right". The "extend" is "unset" if none is checked
- If the shot has a **mark**, this is displayed on the right of the extend: (x) feature, (||) ridge, (v) backsight, (o) generic.

Properties buttons:

-  "duplicate" flag [A]
-  "surface" flag [A]
-  "commented" flag [A] Commented shots are only displayed very dimmed in the sketches and are excluded from the cave length calculation.
-  "backshot" flag.
-  the shot belongs to the immediately preceding leg.
-  the shot belongs to the immediately following leg.
-  the shot is a backsight leg, a second leg being taken in the reverse direction (ie, marked as "backleg" type and excluded from the cave length calculation).
-  reassign shot stations from this leg onward following the current station naming policy. Only for legs: both FROM and TO stations must have been specified

-  apply the station change to all splays in the group of the splay shot that has the same station as the splay. Only for splays.
-  hide the splay in either the plan, or the profile view, or both (splay only) [A]
-  clear (or set) splay class for this splay, and possibly the previous splays [A]

Backsight-check splay

If the secondary setting Backsight splay is enabled, the first splay after a leg is taken as a backsight check of that leg: it is compared with the reversed leg and, if the two disagree, the station is highlighted in violet.

Changing the station of such a splay clears the backsight-check flag, as does

the  button among the additional actions.

Action buttons:

- **Cancel:** close the dialog without saving
- **More/Less:** show/hide additional actions [A]
- **Save:** save the changes without closing the dialog
- **OK:** save the changes and close the dialog

Station names

Station names cannot contain special characters ('/', '\', '?', '<', '>', '|', ':', '"'), and must end with an alphanumeric character or one of '-', '.', '_', '@'.

For BRIC shots the date and time are the values downloaded with the shot data from the device. For other devices and for manually entered shots, they are the time of download/insertion.

The shot flags, "duplicate", "surface", "commented" and "backshot", are mutually exclusive. The survey data info takes into account these flags. The export of the shot is also affected by the shot flag, either with an export flag, if the format supports it, or as comment.

There is a warning if, upon editing a leg shot, it differs from another leg between the same stations more than one fourth in length, or 15°.

Station cut-n-paste

Station names can be 'cut', 'copied', and 'pasted' across the app, in any dialog where there is a station entry field.

Long-tapping the station entry field opens the drop-down menu with the three options

- cut
- copy
- paste

Station reassignment

Station reassignment can be initiated only from a leg. By default the reassignment affects all the shots following the leg up to the end. If the secondary setting splay station is disabled, the stations of splay shots are not reassigned.

Normal reassignment should be used only for very last few legs, in case a leg has not been detected (eg. only two data), or erroneously detected (very close splays taken as a leg).

More control on the shots that are reassigned is achieved with the "multi-shot" selection, where the user can pick the shots to reassign individually.

Splay group

A group of splays consists of all the splays between two legs. Splay group station change applies only to the contiguous set of splays with same station as the edit shot. Only FROM station must have been specified.

If the secondary setting "Splay group" is set, only the splays after the editing shot are modified.

Splay classes

Splay classes are enabled with a secondary setting. There are four classes:

- normal splays: Blue
- x-section splays: Green
- horizontal, plan-view, splays: Dark blue
- vertical, profile-view, splays: Violet

The Splay class button cycles through these four classes. The splay shot is

set to the chosen class when the data is saved. If the  "all splays" button is checked, the class is set also to all previous contiguous "normal" splays.

Splay classes can also be set with the "multi-shot" selection. In multi-shot you can select splays that are not contiguous and assign them all a splay class.

If you take splays shots in a consistent way, splay classes can be set editing the leg shot and checking the splay group button. You must take first the x-section splays starting with +90 and going around in circle. Next you take the H-splays. Finally, you take the splays in the vertical plane starting again with +90.

Additional actions

At the usage profiles "Basic" or "Normal" the additional actions are only a button to delete the shot.

At higher usage profiles it has more buttons:

-  [photo](#)
-  [audio](#) comment
-  [Sensor measurements](#)
-  clear the backsight-check flag (only for a backsight-check splay, and if the Backsight splay setting is enabled)
-  [split the survey](#) at this shot (creating a new survey or append to another survey)
- **Color**, only for splays
-  mark the shot as calibration-check (only for legs)
-  [recompute](#) the direction of the survey shots, starting from this shot (only for Cavway) [T]
-  delete the shot

For legs there are also fields to add intermediate LRUD.

The **delete** button does not remove the shot from the database. The shot gets marked as "deleted" and can be later recovered with the [Recovery](#) dialog. If the shot is the first shot of a leg you can also delete all the shots of the leg, by checking the checkbox besides the buttons.

Leg shots have also a **calibration** button. This button marks the shot as "calibration-check". These shots are not used in the survey, and their data should be taken at different roll angles. They are recorded in the database, and are a record of the reliability of the device calibration when the survey was done. They can be displayed with the [Recovery](#) dialog.

Survey split

This button opens a dialog for the new survey info. Next the survey data are assigned to the new survey, beginning with this shot. Afterwards the program continues with the new survey. You may need to split surveys when you have data from two (separate) survey pieces stored in the device: these

are downloaded to a single survey which must be then split accordingly. At tester level you get also the option to **move** the data to another survey. Select the target survey from the pull-down option set, and tap the "Move" button.

Warning. It is advisable to make a zip archive before splitting/moving the survey and back it up to a safe place.

Splay color

For splays there is also a button to assign a color to the shot. The shot list is not immediately updated with colors. To display the splay with color press the "Refresh" button.

SHOT BUFFER

This dialog is opened from the Recover | Paste menu of the Survey Data window if the shot-buffer is not empty.

Shot buffer [T]

The shot buffer is filled with a multishot selection. Either the shots at the end of the survey are "copied" into the buffer, or selected shots are deleted from the survey and go into the buffer.

The data in the shot-buffer can be copied from the buffer and appended to any survey. With the "move" button the shot-buffer is cleared after the copy. The shots are appended in the order they are added to the buffer. Alternatively, they can be appended sorted by the index, if the appropriate checkbox is marked. The shot stations are replaced with new names different from those already in the survey. The graph structure of the shots in the buffer is preserved.

MULTI-SHOT EDIT

This dialog is opened by the "light bulb" button from the Survey Data window in multi-shot mode.

It has the following actions

- renumbering the shots stations. The first selected shot must be a leg. The stations of the first shot are used to seed the renumbering. They can be changed in the dialog. If both stations are entered, the first

selected shot is a leg and the following selected shots are renumbered consistently. If only the FROM station is entered (and the TO station is left empty), all the shots are renumbered as splay.

- *swapping stations*
- *adding a length **offset** (positive or negative) to the shots (in the configured unit). An offset larger than 0.1 m marks the shots as "tampered", ie, displayed in red*
- *coloring the splays [T]*
Setting the color of splays is enabled if only splays have been selected, otherwise it is disabled.
- *marking the splays as either regular (-), x-section (X), horizontal (H) or vertical (V). [T]*
- *hiding the splays in either the plan, or the profile view, or both. [T]*
- *computing the plane that fits the shots (only if the Plane interpolation option is enabled) [T]*
The computation of strike and dip of the bedding plane requires that all the selected shots have the same FROM station. Otherwise, it is disabled. The result is shown in the dialog and added to (replaced if already present) the comment of the first selected shot.

Splay coloring, marking, and hiding are available only if all selected shots are splays and the "splay coloring" setting is enabled.

Bedding computing is available only if the "plane fitting" setting is enabled.

SCAN SHOTS

The Scan shot dialog allows to change the FROM station for the shots in a group of scan shots.

Type the new station name in the edit text-field, and press the button OK to confirm.

The button Cancel closes the dialog without changing the station name. The button Delete deletes the scan set. There is a confirmation dialog. The scan set is only marked as "deleted" and can be retrieved through the [recover dialog](#)

REDUCTION ERRORS

This dialog is presented when the 3 dots menu with red exclamation mark button in the Drawing window receives a long click.

It displays the anomalies encountered during the survey data reduction:

- 1. **bad siblings**: legs repeated with considerably different values*
- 2. **bad shots**: legs with shots that differ too much*
- 3. **missing extend**: legs with no extend specified*
- 4. **disconnected survey**: pieces of survey are not connected in a single component*

ITEM RECOVERY and SHOT BUFFER

This dialog is opened from the Recover menu of the Survey Data window.

Item recovery

The dialog can be used to recover the survey's deleted items, shots or sketches, as well as "overshoot" data, and "calibration-check" data.

The "Status" button selects which items are displayed. Tapping it the display cycles through "deleted" shots, "overshoot" shots, "calibration-check" shots, and "deleted" sketches.

Select the items to recover by tapping their checkboxes. To finish and recover the selected items tap the "OK" button.

Only the selected items of the type on display are recovered.

For scan set only the first shot of the set is shown and the recover restores the whole set.

A sketch recovery restore both the plan and the profile view at the same time.

The Shot buffer can be accessed by long-pressing the RECOVERY menu item [T]

The button "Reset IDs" replaces the ID numbers of the survey data with incremental values [T]

PHOTO LIST

This dialog is opened from the Photo button in the Survey Info window, or from the Photo menu in the Survey Data window.

It display the list of the photos of the survey, both those taken for shots and those taken for sketches.

Each entry has

- *the number of the photo*
- ***shot** to which photo was assigned, if the photo was taken for a shot*
- *photo **comment***

Tapping an entry in the list opens the Photo edit dialog.

SHOT/SKETCH PHOTO

This dialog is opened from the Photo button in the secondary Shot edit dialog, or when a "photo" point or a photo x-section are inserted in the Drawing window.

*There is a text field for the **comment** to the photo you are going to take.*

Buttons:

- ***OK** closes the dialog and starts to take a photo.*
- ***Cancel** closes the dialog.*

TopoDroid contains a simple implementation of a camera. Its unique feature is that it stores the azimuth and inclination of the device in exif tags in the image file. The azimuth is stored as GPS_LONGITUDE and the inclination as GPS_LATITUDE (with sign).

The following exif tags are set by TopoDroid:

- *Software: "TopoDroid" and its version, separated by a space [Android 7 and up]*
- *Orientation*
- *Datetime [yyyy.MM.dd]*
- *GPS latitude: inclination [1/100°]*
- *GPS latitude ref: "N" or "S"*
- *GPS longitude: azimuth [1/100°]*
- *GPS longitude ref: "E"*
- *Image description: azimuth and inclination in 1/100°, separated by a space*

The direction of the photo is the direction of view of the camera.

For "photo" point or photo x-section the display of the TopoDroid camera has a centered overlaid cross-hair.

TopoDroid uses Android sensors to get the direction (azimuth and clino). A warning is displayed if the sensor accuracy is low or there is a sensor failure.

The TopoDroid camera has buttons on the lower right.

- **camera**. Red: ready, tap to take a picture. Blue: on hold, tap to turn camera active.
- **edit** saves the photo and allows to draw over it (only with new camera) [T]
- **floppy** saves the photo (JPG) or the drawing (PNG)
- **back** closes the window without saving the image.

A warning is displayed, if the JPEG image is not available. You should wait a second and retry.

With the screen up the picture is portrait. When the screen is turned to the left or to the right the picture is landscape.

Zoom buttons are brought up by tapping the screen in the mid-bottom. To go through large zoom steps tap the zoom buttons repeatedly fast.

Camera implementation

TopoDroid has two camera implementations: the first uses the old Android camera API, the second uses the new Android camera2 API.

From Android-5 onward TopoDroid uses the new camera implementation. On old device it uses the old implementation.

With the new camera2 API there is a limit of five attempts of picture capture, after which the camera button is no longer available.

PHOTO EDIT

This dialog is opened tapping an entry in the Photo list dialog.

- **thumbnail** of the photo
- **orientation** (azimuth and clino) of the photo, for photos taken with TopoDroid
- **date and time** of the photo
- **comment** (editable)

Buttons:

- **Save** saves the changes to the comment

- **Delete** deletes the photo (if the photo is for a shot)

The image is displayed when you tap on the thumbnail. The full image is not reoriented to portrait.

The Android BACK button closes the dialog.

Notes

[1] Photo taken for sketch points can be viewed also through the item edit dialog, and are deleted when the photo point is deleted.

[2] When a shot-photo is deleted it is dropped from the photo list dialog. However, in the shot listing the shot is still marked as if there was the photo, until the shot-list is refreshed.

PHOTO DISPLAY

This dialog displays a photo.

The "Close" button closes the dialog.

Geomorphology code [T]

This dialog lets you define a geomorphology code to associate to a photo, or a station, or change it.

The list of geomorphology codes are read from the text file "geocodes" which must be placed in the app private folder.

Each line of the "geocodes" file must contain

- an integer, for the class of the geomorphology
- the string of the code (no spaces)
- a description of the code (which can have spaces)

The geomorphology classes are shown with different colors. Therefore it is advisable to list together the geomorphologies of the same class.

The code is not shown in the interface, but it is stored in the database and the sketch "tdr" file.

DATA AUDIO COMMENT

This dialog is opened from the Audio button of the secondary shot edit dialog, or when you add a point audio to a sketch.

There are three icon-buttons:

-  *plays the audio comment*
-  *records the audio comment (if there is already one this is overwritten)*
-  *deletes the audio comment (if any) and closes the dialog*

When you delete the audio for a sketch point, the point is also deleted.

Audio files are saved in "wav" format in the audio directory.

The Close button closes the dialog.

AUDIO LISTING

This dialog is opened from the Audio menu Survey Info window.

It contains a list of the survey audios.

A short tap an entry plays the audio.

A long tap brings up the audio dialog, where you can play or change the recording.

SENSOR MEASUREMENTS LIST

This dialog is opened from the Sensor button of the Survey Info window or from the Sensor menu option in the Shot list window.

These options are only available if the Secondary settings > Survey data > Sensors setting is enabled.

It displays the list of the sensor measurements.

A tap on an item in the list opens the sensor data [edit dialog](#).

SENSOR MEASUREMENTS

This dialog is opened from the Sensor button in the secondary Shot edit dialog.

It has list a checkboxes to select a builtin sensor, or an external sensor (in this case you must enter the "sensor" name).

There are three text fields

- **name** of sensor. Either one of the listed builtin sensors, or it must be typed in if you select an "external" sensor
- **values** Automatically updated for builtin sensors. Manually entered for "external" sensors.
- **comment**

Save stores the measurement in the database.

The Android BACK key closes the dialog without saving.

SENSOR MEASUREMENT EDIT

This dialog is opened tapping a sensor measurement entry in the Sensor list dialog.

*It displays some info about the measurement (title, type, shot, and value) and a field to change the sensor **comment**.*

Buttons

- **OK** saves the changes and closes the dialog
- **Delete** deletes the measurement

Delete does not remove the sensor measurement from the database, but marks it as "deleted". It can be recovered with a SQLite app.

TRILATERATION

Trilateration is a special "loop compensation" applied to triangles. It can be used in short surveys in non-uniform magnetic environment as the azimuths are computed by measuring the sides of triangles, relying on the fact that the angles of a triangle are determined by its sides. The values measured by the device are used for the inclinations.

Trilateration is enabled by setting the Loop compensation to "Triangles".

The survey should be made of a net (or a ladder) of triangles. Therefore you measure 0-1, 1-2, and 0-2; then 2-3 and 1-3, and so on.

The trilateration algorithm carries out a global estimate for the angle azimuths, and is rather slow. Therefore it is advisable not to use it for large surveys.

NEW SKETCH

This dialog is opened from the New sketch button in the Sketch list dialog.

It has two text field and a confirmation button.

- **name** is the name assigned to the sketch.
- **reference station**
- **projected** profile checkbox (by default the profile is "extended") [E]
- **dangling** sketch (origin not attached to midline)

*You can use either the **name** suggested by TopoDroid (an increasing integer number) or enter a mnemonic name. Each sketch is saved with two files, one for the plan, the other for the profile. The filenames are obtained composing the survey name, the sketch name, and a suffix ("p" for plan, "s" for profile). For example the sketch "2" of the survey "cave" is saved in the two files "cave-2p.tdr" and "cave-2s.tdr".*

TopoDroid will give a warning if the name coincides with the name of an existing sketch. In particular it is not possible to have two sketches with names differing only by the characters case since the two sketches would have the same file because the filesystem is case insensitive.

WARNING. *The name of the sketch cannot contain special characters: '/', ':', '<', '>', '*', '\\', '?', '|', '\"'.*

The name of the sketch cannot contain spaces. Spaces are replaced with '_' (underscore).

Reference station

The reference station (origin) defines how the sketch is translated with respect to the survey midline. The survey midline defines the sketch scale (the drawing is scaled at 40 canvas units per meter) and rotation, but leaves undefined how the drawing is translated with respect to the midline. This uncertainty is resolved by specifying the reference station to be at (0,0) in the reference frame of the sketch canvas.

The reference station also determines the survey midline **data reduction** for the sketch. From the sketch point of view the midline is built starting from the reference station and continuing as long as data can be attached. The resulting midline forms a tree, rooted at the reference station. In order to make the canvas less cluttered, is possible to hide the display of portion of the midline, either after a station, or before it (i.e., between the reference station and the station). This is explained in the "Sketch station" dialog.

If the survey has "fixed station origin" [setting], TopoDroid prefills the "reference station" with the first survey station. Otherwise it uses the last station or the active station if this is set. You can change it if you want another station as sketch reference.

The app signals an error if the reference station does not exist in the survey data. It can be forced to use an non-existing station by checking the "dangling" checkbox.

In this case the sketch will not have a midline reference, neither splays nor stations, until the reference station is assigned to a leg.

If the **projected** checkbox is checked the sketch profile is projected, The direction of view, ie, the projection azimuth, is chosen with a graphical dialog that allows to rotate the survey midline around a vertical axis.

Otherwise the sketch profile is extended and the **projection azimuth** is not shown.

DANGLING SHOTS

This dialog lists the unattached shots.

It is opened by a tap on the line with the number of unattached shots in the statistics of a sketch.

For each shot it displays the index, the stations (FROM and TO), and the data (distance, azimuth, clino).

SURVEY SKETCHES

This dialog is opened from the Sketch button in the Survey Data window.

It has a button to create a new sketch and the list of the existing sketches of the survey.

New sketch creates a new sketch. It takes you to another dialog, where you can enter the sketch information, before opening the Drawing window.

The list of the existing sketches has two columns:

- **plan** views on the left
- **profile** (either extended or projected) views on the right

Each sketch consists of a plan and a profile view, and is listed in both columns.

Tap on one sketch entry to open it. If you tap the left-hand-side entry the sketch is opened in the plan view. If you tap the right-hand-side entry the sketch is opened in the profile view.

SCAN SET DIALOG

The Scan-set dialog for the Sketch Window allows to change the station and the type of the scan set.

Buttons:

OK: confirm the new station

Cancel: close the dialog without changing station.

PROJECTED PROFILE AZIMUTH

This dialog is opened when you create a new sketch with a projected profile.

The central canvas display the survey midline in the projected view. The dialog opens with the projected midline centered on the canvas. It can be translated and zoomed.

The direction of view (projection azimuth) of the projected profile is set by dragging the seekbar at the top, or by entering the azimuth value in the text field. The "Plus" and "Minus" button change the azimuth by one degree at a time (short tap) or by 10 degrees (long tap).

The midline display rotates about the reference station of the midline.

If oblique projection are enabled there is an additional seekbar at the bottom, to set the oblique angle with respect to the projection azimuth.

The OK button creates the sketch and opens it in the Drawing window.

The BACK button closes the dialog without creating the sketch.

SKETCH REFERENCES

This dialog is opened from the Reference button of the Drawing window.

The column contains the sketch references,

- **legs**, survey midline (white)
- **splays** (grey)
- **latest** splays [G]
- **stations** names (violet)
- **sketch grid**, metric or yard
- **scalebar**, reference scalebar and magnetic North/Upward arrow
- **fix refs**, allow to shift the sketch independently from the shots (be careful!!)
- **current scrap**, whether to shift only the current scrap

The "latest" splays can be highlighted with an adjustable color (default light blue) [G]. With on-demand download mode, these are the splays that have been downloaded most recently. With continuous mode, these are all the splays downloaded during the last highlighting interval.

There are three sketch grids,

1. a grid at 1 unit spacing (dark grey)
2. a grid at 10 units spacing (grey)
3. a grid at 100 units spacing (light grey)

The sketch grid units can be 1m, 1y, 2ft, 0.1m. The default is 1 m. TopoDroid shows only the sketch grids that are appropriate for the zoom of the sketch display.

The scalebar is adapted to the sketch grid units (m, y, ft, dm) and has a small arrow indicating the magnetic North/Upward direction. If the survey declination is set the true North is indicated as a light blue line, at the declination angle to the magnetic North. If the magnetic declination is small

the North blue line is very close to the magnetic North arrow. The legend on the scalebar has the same size as the station names.

Sketch outlines

If there are other sketches, the [outline](#) of another sketch can also be used as reference for sketching.

The outline of an x-section can be displayed at the relative section point. The display of all the outlines is cleared when the sketch is closed.

Sketch shift and scale

A further checkbox [E/G] specifies whether the reference should be held fixed when shifting the drawing with a two-finger touch. This is used to adjust a sketch imported from PocketTopo to the reference in the case it gets imported with a translation. The sketch can be zoomed with a three-finger touch.

Canvas views (layers)

When canvas views are enabled [G], there is another column, on the right, to switch the views on and off.

A drawing item is displayed if it belongs to at least a view that is ON.

*The **Base** (B) view is always ON. The other views are*

- **Floor** (F)
- **Decoration** (D)
- **Ceiling** (C)
- **Artifacts** (A)

The views are denoted by an uppercase letter, B, F, D, C, and A, as indicated above.

The mnemonic names carry no semantics.

SKETCH OUTLINE

This dialog is opened checking the sketch Outline button of the Sketch Rename Etc. dialog

Tapping a sketch on the list, displays its outline in the drawing canvas.

The Clear button removes the sketch outline from the canvas

The Merge button of the Sketch Rename Etc. dialog includes the outlined sketch into the current sketch.

DRAWING TOOLS PALETTE

The drawing tools bundled in the program include about 9 areas, 16 lines and 46 points.

When the app starts the first time it installs the drawing tools files in the symbol directories point, line, and area. There are files for points (iconic symbols, like "entrance", or "stalagmite"), lines (like "wall", or "pit"), and areas (regions shown with a transparent color).

New versions of the drawing tools are not automatically installed, but there is a button to update them. The coming dialog tells you the installed and the current version and asks whether to update the drawing tools.

It is likely you do not need many of them, or maybe you need different ones. You can customize the drawing tools and the way you select them in four ways:

- *with the choice of main drawing tools (dialog)*
- *at the level of the survey/sketch*
- *at the level of the project (directory)*
- *on the file system*

The drawing tools are loaded from files in the app private folder. When the drawing tools files are updated, TopoDroid does not install the new tools automatically.

Global palette

Except for the "system" tools, TopoDroid loads the drawing tool from files stored in the symbol directories of the app private folder.

The system tools are the "user", "label" and "section" points, "user", "wall" and "section" lines, "user" and "water" areas. These are always enabled.

The tools button of the Main window opens the dialog to enable the drawing tools.

You have to do this independently for points, lines, and areas

When a new sketch is created its palette is initialized with the enabled tools of the global palette.

Survey drawing tools

The drawing tools enabled at survey level are the global palette, by default. If a sketch is opened that has tools that are not enabled by default, these are added to the survey palette.

If the sketch has tools that are not in the palette, these are replaced by

"user" symbols with the name of the tool written in the symbols' option string.

Sketch drawing tools dialog

The tools bottom bar contains only six drawing tools (for each type). If you need to use another tool, you must go to the [drawing tools dialog](#) which display the list of the drawing tools, subdivided under three tabs: Point, Line and Area.

Each tool shows the symbol icon, the tool name, and the tool-group if any. When you start a new sketch it gets the tools configured in the survey palette.

Sketch palette

Each sketch has its own palette. When you close the sketch it is saved with a header listing its palette. When you reopen it, the palette is merged with the survey palette. If the sketch contains items which are not enabled in the survey palette they are added to the survey palette. If an "item" tool is not among the main drawing tools, it is replaced with the "user" tool. The "user" tools (point, line and area) are always in the palette.

The sketch palette is specified through the PALETTE menu of the Drawing window. This opens a dialog that lets you decide which tool enable in the sketch palette. You switch on and off the tools to your needs then close the dialog (with the BACK button) to save the changes. You have to do this independently for points, lines and areas.

Custom drawing tools

At a even higher level you can customize the tool files, and write your own tool files.

The tool files are plain text files, stored in the app private folder under "/sdcard/Android/data". The syntax of the tool files is described in the pages for [Points](#), [Lines](#), and [Areas](#).

Even if TopoDroid comes with many drawing tools preinstalled, it can happen that it does not contains the tools you need (or you do not like the provided ones). You can modify them, delete those you do not need, and write and add new ones (for tools not included in the program). After a cold restart, TopoDroid will see and use them.

If you modify a tool file or write a new tool file that may be useful to other users, consider making a pull request on the TopoDroid github repository.

The "system" drawing tools cannot be modified, nor deleted.

Additional symbol sets

TopoDroid comes with these additional symbol sets:

- *Extra speleo symbols*
- *Mining symbols*
- *Geology symbols*
- *Archeology symbols*
- *Anthropic symbols*
- *Paleology symbols*
- *Biology symbols*
- *Karst symbols*

In order to install additional symbol sets, the secondary setting "Palette" must be enabled.

DRAWING TOOL PICKER

This dialog is used to select the current drawing tool. It is opened with a tap on the >> button in the tools bar of Drawing window in "drawing" mode.

*There are three tabs at the top, **Point**, **Line**, **Area** to switch among the three types of tools.*

*The dialog displays the tools of the selected type, as a **list**, with a selection checkbox, tool icon and name.*

To select a tool check its box in the list, and close the dialog (Back key). A shortcut is a double-tap on the check-box of the tool.

*The **Slider** sets tool default orientation (orientable point tools, and area tools).*

The +/- button changes the (global) default icon size cycling through XS, S, M, L, XL (point tools).

When canvas views [G] are enabled and the selected tool is associated only to disabled views, it is replaced by the corresponding "user" tool.

POINT TOOLS

The "point" tools are defined in text files.

Three point tools: "user", "section", and "label" (text), are built in the programs. The others are loaded from the TopoDroid private folder point. This directory is populated with the point tools packaged in the distribution apk. You can delete the tools you do not need, edit the ones you want to change, and add new ones that are not provided in the apk.

The point tool file contains lines specifying the property of the drawing tool.

- "symbol point" line [mandatory]
- "name NAME", where NAME is the point default name [mandatory]
- translated names, eg, "name-es ..." [translated names are optional]
- "th_name NAME", where NAME is the Therion name. The filename must be the same as the Therion name, without the possible prefix "u": [mandatory]
- "color rgb" (rgb in hex format, for example red is '0xff0000') [mandatory]
- "orientation FLAG" [optional, false if missing, true if FLAG is '1' or 'yes']
- "has_text FLAG" [optional, false if missing, true if FLAG is '1' or 'yes']
- "roundtrip VALUE" [optional, for SVG-roundtrip, VALUE can be 1 (walls_shp), 2 (walls_sym), 3 (detail_shp), 4 (detail_sym) - default 4]
- "level" view_flag [optional, viewing level of the point; if missing level is set to 1]
- "path" begins the point symbol path. This is specified with the following commands
 - "moveTo X Y"
 - "lineTo X Y"
 - "cubicTo X1 Y1 X2 Y2 X Y"
 - "addCircle X Y R": (X,Y) center, R radius
 - "arcTo X0 Y0 X1 Y1 FROM_ANGLE SPAN_ANGLE": (X0,Y0 - X1,Y1) bounding rectangle, angles in degrees, 0 along X-axis, direction is CW
- "endpath" marks the end of the tool path
- "endsymbol"

Name

The English point "name" is mandatory.

Translation of the name in other languages are optional.

Names must not contain spaces; use underscore '_' in their place.

The Therion name must coincide with the filename. This may cause a problem when the Therion name contains a dash '-' and this is an illegal character in filenames. In this case use the equal character '=' in the filename, the script that creates the zip archive of the symbols files replaces '=' with '-' in the filenames.

Therion names and filename must not contain spaces.

Orientation

If the attribute "orientation" is set, the point symbol can be drawn at an angle, and it can be rotated.

Canvas views (layers)

The "level" attribute takes value the flag for the views enabled on this point items.

The flag is a number sum of 1 (base), 2 (floor), 4 (fills), 8 (ceiling), and 16 (artifacts).

LINE_TOOLS

The "line" tools are defined in text files.

Three line tools: "user", "section", and "wall", are built in the programs. The others are loaded from the TopoDroid private folder line. This directory is populated with the line tools packaged in the distribution apk. You can delete the tools you do not need, edit the ones you want to change, and add new ones that are not provided in the apk.

The line tool file contains lines specifying the property of the drawing tool.

- "symbol line" line [mandatory]
- "name NAME", where NAME is the line default name [mandatory]
- translated names, eg, "name-es ..." [optional]
- "th_name NAME", where NAME is the Therion name. The filename must be the same as the Therion name, without the possible prefix "u": [mandatory]
- "group GROUP", specifies the group this line belongs to [optional, default no group]
- "color rgb alpha" (rgb and alpha in hex format, for example '0x00ff00' is green, and '0xff' is fully opaque) [rgb mandatory, alpha is optional, if not specified 0xff is taken]
- "width SIZE", line width in units of the line width setting [optional, default 1]
- "dash PATTERN", example "15 5" [optional dash pattern]
- "style STYLE" where STYLE can be 'straight' or 'xN' [optional]
- "effect" begins the line path-effect and is followed by path commands, and closed by
- "endeffect"
- "roundtrip VALUE" [optional, for SVG-roundtrip, VALUE can be 1 (walls_shp), 2 (walls_sym), 3 (detail_shp), 4 (detail_sym) - default 3]
- "level" view_flag [optional, 1 if missing]
- "endsymbol"

Name

Refer to the [Point Tool](#) for how to specify the names.

Group

This attribute is used to replace a missing symbol with the group symbol.

Dash

The "dash" directive specifies that the line is drawn with a dashed scheme. For example, "dash 15 5" means that the line is drawn with segments of 15 units, separated by 5 units spaces.

Style

The "style" directive is used to draw lines using sparse points.

A "style x3" means that the line is drawn using one every three points. The effect is that the line is less "curvy".

If you want straight segments use "style straight".

Effect path

This option is used to define lines with complex style.

Canvas views (layers)

The "level" attribute has value the flag for the views enabled on this line items.

The flag is a number sum of 1 (base), 2 (floor), 4 (fills), 8 (ceiling), and 16 (artifacts).

AREA TOOLS

The "area" tools are defined in text files.

Two area tools, "user" and "water", are built in the programs. The others are loaded from the TopoDroid private folder area. This directory is populated with the area tools packaged in the distribution apk. You can delete the tools you do not need, edit the ones you want to change, and add new ones that are not provided in the apk.

The area tool file contains lines specifying the property of the drawing tool.

- "symbol area" line [mandatory]
- "name NAME", where NAME is the area default name [mandatory]
- translated names, eg, "name-es ..." [optional]
- "th_name NAME", where NAME is the Therion name. The filename must be the same as the Therion name, without the possible prefix "u": [mandatory]
- "color rgb alpha" (rgb and alpha in hex format, for example '0x0000ff' is blue, and '0x99' is semi-transparent) [mandatory]
- "close-horizontal" [optional, false if missing]
- "orientable" [optional, false if missing]
- "level" view_flag [optional, 1 if missing]

- "bitmap WIDTH HEIGHT X_TILE Y_TILE" starts of the bitmap pattern. The tile modes can be M (mirror) or R (repeat). This line must be followed by HEIGHT lines of length WIDTH with the bitmap pixels (1: foreground, 0: background).
- "endbitmap" marks the end of a bitmap
- "roundtrip VALUE" [optional, for SVG-roundtrip, VALUE can be 1 (walls_shp), 2 (walls_sym), 3 (detail_shp), 4 (detail_sym) - default 3]
- "endsymbol"

Name

Refer to the [Point Tool](#) for how to specify the names.

Close-horizontal

If this attribute is set the area border is closed horizontally in the profile view and in the x-sections.

If the Y coordinate of the last point of the drawn line is close to that of the first point, the area border is interrupted at the first point that has the Y coordinate close to that of the first point. By this means you can draw areas with a flat horizontal top (or bottom) profile.

If the Y coordinates of the drawn line endpoints differ too much, the border is closed with a slant segment.

Currently, this attribute it is only set for the built-in "water" tool.

Bitmap pattern

With this option you can specify a background pattern, eg, for differentiating rock formations.

Area patterns are not used in the default symbol set. Shading (as defined through the "color" attribute) is preferred for performance instead of tiling (as defined through the "bitmap" attribute).

This attribute is not used in any tool packaged in the apk.

Orientation

If the attribute "orientable" is set, the area pattern can be rotated through the area edit dialog.

Canvas views (layers)

The "level" attribute has value the flag for the views enabled on this area items.

The flag is a number sum of 1 (base), 2 (floor), 4 (fills), 8 (ceiling), and 16 (artifacts).

DRAWING TOOL SETS

The drawing tools are divided in three types: points, lines, or areas (regions).

The point tools are stored in the point subdirectory of the private app files. The line tools are stored in the line subdirectory, and the area tools in the area subdirectory of the private app files.

TopoDroid comes with one default set of drawing tools, and other eight additional sets [T]. Overall that makes more than 200 points, 60 lines, and 30 areas. The basic speleology set has about 40 points, 15 lines, and less than 10 areas.

The drawing tool files are stored in the subfolders of com.topodroid.TDX/ files. These files can be edited and modified, or even deleted. Furthermore, if other custom tool files are placed in the proper subfolder, TopoDroid will load and use them.

Eight system tools are always present: the three "user" symbols, points "label" and "section", lines "wall" and "section", and area "water". These tools are not stored as files, and cannot be changed. A custom tool file with the same name as one of these tools is skipped.

By default only the speleology tools are installed. The installation of additional tools is enabled through a setting [G]. A few tools are included in more than one set. The additional symbol sets are provided mostly as examples of custom symbols.

To install additional tool sets tap the the PALETTE menu [G] of the Main window. The coming dialog lists the sets of tools. Check the tool sets you want to add and press the button Install.

If you press the button Replace the currently installed tools are removed before installing the tool sets you selected.

After installing new tool sets you must enable the tools you need, using the Palette button of the Main window. The choice of enabled tools is stored in the database and is not lost when tool files are removed, therefore you will find them enabled if you re-install the set later.

Warning

Raplacing the palette of symbols may invalidate existing sketches.

When a sketch is opened and an item type is not found, TopoDroid tries to substitute it for another type in the same group. If there is none, the "user" tool is used.

Tool groups *The tools of each type are further divided in groups (classes). A tool can belong to a group or have no group.*

The tools groups are displayed in the tool enabling dialog. Group names are in English.

SKETCH RELOAD

This window is opened by the Reload menu of the Drawing window.

TopoDroid maintains backup copies of sketch files, from which the sketch can be recovered.

TopoDroid keeps a set of backup copies for each sketch file. These are named with suffixes "bck", "bck0", "bck1", and so on. They are stored in the tdr subdirectory (binary sketch files). Whenever a sketch is saved the backup copies are rotated (eg, "bck1" is renamed "bck2", and so on; "bck" is renamed "bck0"), the old "tdr" file is renamed with suffix "bck", and the sketch is saved to a new "tdr" file.

A sketch is saved whenever it is closed, or you toggle between plan and profile view, or open a x-section, or switch to another sketch.

It is also saved whenever it is modified (eg, a new item is added, or part of the sketch is erased) while it is open. Upon a modification a timer is started and when it expires the sketch is saved. The wait-time is a setting (default 60 seconds).

The number of backups is another setting.

The Sketch Backup window displays a preview of the sketch backups. One backup at a time is displayed. Some info about the backup are shown in the title:

- how long ago it was written, with suffixes for minutes (m), hours (h), days (d), months (M) and years (Y)
- size of the file, in square brackets, in bytes
- backup number

There are two buttons to move back and forth in the backup list



and one  to return to reload the backed-up sketch and close the Sketch Backup window.

The menus are

- Close the window without reloading the sketch from a backup
- Help

THE UNDO STACK

TopoDroid implements undo/redo with a stack of draw/erase/edit actions. When you draw something a draw action is put at the top of the stack. Undoing amounts to moving the stack pointer down the stack (beside affecting the sketch drawing). Redoing moves the stack pointer up (and add the drawing to the sketch).

The effect of the "continuation" drawing on the undo stack is that a new line is not created, and the line that has been continued is moved on top of the undo stack.

Erase and edit actions can also be undone/redone. However their effect on the stack is not as straightforward as for drawing.

Erasing is a raster operation while the drawing is made of vector items. The points of the vector items that come under the erased region are removed. This affects the undo stack in two way. If an item is completely erased (eg, a point item), it is dropped from the stack. Undoing the erase action puts the item back, but at the top of the stack. Erasing the middle of a line remove the line object from the stack and replaces it with two lines that are added on the top of the stack. A following "undo" removes the two pieces and puts the line back, however at the top of the stack.

SKETCH ITEM PROPERTIES EDITOR

The sketch item properties editor dialog is opened by selecting a point (or the point) of the sketch item and from the Note button.

Point items properties:

- **size** (one of XS, S, M, L, XL)
- **orientation**, if the point item is orientable
- **layers**, the canvas views on which the point is displayed, available when the Layers option is set to "by item" [T]
- **text**, for "label" points
- generic Therion **options**

Line items properties:

- **line type**
- generic Therion **options**
- **outline** (either "out" or "in")
- **layers**, the canvas views on which the line is displayed, available when the Layers option is not set to "none" [T]
- **direction**, i.e. whether the line is directed or reversed
- a polycubic line can be made **polyline**, i.e., composed of segments instead of cubics
- the line can be **subsampling**, by removing intermediate points
- the line can be **closed**, by appending a last point at the position of the first

Area items properties:

- **area type**
- **visibility** of the area border
- the area border line can be **subsampling**, by removing intermediate points
- **orientation**, if the area is orientable
- **layer**, the canvas views on which the line is displayed, available when the Layers option is set to "by item" [T]

Survey stations properties (see also [sketch station editing](#)):

- make the station the **active** one, i.e. to continue the survey from it
- show the hidden midline (only if applicable)
- toggle the display of **splays** at the station (overriding the global choice of splay display)
- adding the station to the Therion export (if "therion auto-station" option is disabled)
- at-station **x-section** buttons

Survey shots properties:

- **stations**
- shot **comment**
- the **extend** value
- the **flags**: "duplicate", "surface". ("commented" shots are not displayed in the sketches)
- midline hiding
- **auto-walls** generation [E]
- custom **color** (for splays) [T]

The splay custom color requires an active splay color setting, either "picker" or "palette". It is associated with the survey splay in the list and is used to identify the splay in the sketches. The color is saved in the database. It is exported in the ZIP and imported from a ZIP only when the splay color setting is enabled.

The custom color of a splay can be cleared from the Shot edit dialog.

The splay custom colors can be all cleared by a menu in the Survey Info window.

SKETCH STATION

This dialog is presented when you tap the Edit properties button on the Sketch drawing window. It displays the station position, East, (magnetic) North, and Z (upwards) relative to the sketch origin, and the station comment (editable field) [T].

Actions:

- enter or edit the station comment and save it [T]
- make the station "active", ie, continue the survey from this station.
- show the hidden midline (if applicable)
- force showing/hiding the display of the splay shots at the station
- at-station x-section buttons
- go to the [saved-station](#) dialog [A]

Active station

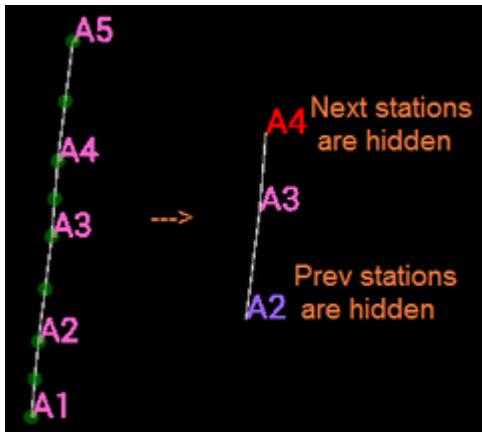
The "active" station is where the next downloaded shots will be attached.

When a leg is downloaded (or inserted) the new station becomes the "active" station. By changing the "active" station you can continue surveying from a station in the middle of the survey. The active station is shown green in the shot data list.

Midline hiding

Hiding parts of the midline is useful when the survey is complicated or goes over itself. In this case it is useful to see only the relevant portion of the midline, by hiding what lies beyond a station. If the midline beyond a station is hidden, the station is highlighted: violet if the hidden midline lies between them and the sketch origin station, red otherwise. Stations in the part of midline behind a "red" station, are not automatically saved as Therion station points. Therefore, by choosing the sketch origin station inside the scrap and marking stations as barrier you can avoid spurious stations in the scrap file.

The display of the midline can be limited through the property edit dialog of leg shots. When the midline beyond the station is hidden, there is a button to revert and show it in the station edit dialog.



Station splays

The display of the station splays can be forced to override the global setting of the sketch. If showing is forced, the splays are shown even when they are hidden globally. If hiding is forced, the splays are not shown even when they are displayed globally.

This flag does NOT affect the edit-selection of splays; the selection green dots are always displayed according to the global setting.

At-station x-section

X-sections can be created at stations. In plan view they are in the vertical plane (their plane is determined by the azimuth). In profile view, x-section can have an inclination. However when selected through the "stations" checkboxes they are vertical if the legs inclination is below a specified value (a setting), otherwise they are horizontal.

If the station has only two legs attached, the plane of the x-section can be inferred by TopoDroid as the bisecant of the angle between the two legs. The plane is perpendicular to the leg of there is only one leg. In order to disambiguate the direction of view of the x-section, there are two buttons that specify the direction as an arrow between the two nearby stations (or the station itself and the nearby station).

With the last button the user can pick the x-section direction with the Android compass.

The name of an at-station x-section in the plan view is composed of the "survey_name", "xs", the "station_name" and, if the x-sections are private, the "sketch_name". For example, "survey-xs-0-1p". The name of an at-station x-section in the profile view has "xh" instead of "xs".

The at-station x-sections can have a comment, just like the section-line x-sections.

Deleting the section-point does not remove the station x-section. The station x-section is deleted only through the station dialog.

Therion STATION POINTS

Therion scraps must have a scale defined or at least two stations so that Therion can compute their scale.

TopoDroid has an option to automatically write into the scrap file all the stations that fall in the convex hull of the sketch. This is enabled by default, and for simple cave maps it is ok.

However this may not be convenient if you work with xtherion and stations from other portions of the survey fall inside the scrap and there are many "spurious" station points. In this case you might want to disable this option and decide by yourself which station to add to the scrap. When the "Therion station points" option is disabled, the sketch edit dialog of stations has an option to mark the station as "Therion station point".

Alternatively, you can let the "Therion station points" option enabled, and mark "barrier" stations to limit the automatic station points. In fact TopoDroid does not add to the Therion file stations that are beyond a "barrier" station.

SKETCH SHOT

This dialog is opened when you select a survey shot and tap the Item button in the Drawing window (in "edit" mode).

Shot properties:

- *length, azimuth, and inclination. If the shot has a **mark**, this is shown with symbols as in the [shot edit dialog](#): (*) feature, (||) ridge, (v) backsight, and (o) generic mark.*
- **stations**
- **shot type** (D forward, B backward, M manual).
- **comment**
- **profile extend**: "left" (-1), "vertical" (0), or "right" (+1)
- **profile fractional extend** slider (extended profile view legs only) [T]
-  "duplicate" flag
-  "surface" flag
-  "commented" flag
-  "backshot" flag

- checkboxes to **hide** the midline (legs only)
- color, if enabled (splay only)

The **shot type** is orange if the leg is inconsistent with the siblings, and red if there is a possible magnetic problems.

Save confirms the changes and closes the dialog.

There is a warning if, upon editing a leg shot, it differs from another leg between the same stations more than one fourth in length, or 15°.

Cancel and the Back key close the dialog without saving the changes.

Midline hiding

Leg shots have checkboxes to hide the display of the midline on the side of either stations. The hiding stations are displayed either red or violet in the drawing. The hiding is reverted through the edit dialog of the station.

Leg extend [E]

In the extended profile legs are usually drawn either to the left, or vertical, or to the right (or not drawn at all). The choice among these options is the "extend" value, that specifies whether the leg is extended left (-1), vertically (0), or right (+1). When a leg is selected in the extended profile view a horizontal bar is drawn through its middle point. by swiping from the midpoint along the bar the user can quickly change the "extend" of the leg.

Shot fractional extend [T]

It is also possible to set a "fractional extend": a value between -1.5 and 1.5, instead of only the integer values -1 (left), 0 (vertical), and 1 (right). By default, the "extend" has an integer value and the fractional part is zero. The slider below the "extend" checkboxes changes the integer part and the fractional part of the shot "extend".

The fractional part of the shot "extend" is set to zero when

- the shot extend is set with a multi-shot action (except for the "flip" action)
- the shot extend is changed in the Drawing window using the yellow extend line
- the shot extend is changed in the shot edit dialog (of the Survey Data window)
- the shot extend is assigned by TopoDroid

To set a fractional extend, you must enable the relative setting. As most map-drawing programs do not support fractional "extend", this setting is disabled by default.

Splay custom color [T]

It is possible to assign a custom color to each splay shot. The color is stored in the database and is thus shared among the sketches.

Splay coloring is determined by a setting [G]: either disabled, or selectable with a color picker, or among a color palette.

The color dialog has also a button to clear the splay custom color.

SKETCH POINT

This dialog is opened when you select a point symbol and tap the Edit button in the Drawing window (in "edit" mode).

Point properties:

- **size** (XS, S, M, L, XL)
- **orientation**, if the point is orientable
- **layers**, the canvas views on which the point is displayed [T]
- **text**, if the point has text associated
- additional Therion **options**

The canvas views are presented as checkboxes with the view letter-code:

- B: base layer (default)
- F: floor layer
- D: decoration layer
- C: ceiling layer
- A: artifacts layer

You can change the views the point belongs to. The point icon is not displayed if all its views are OFF. The selectable point will nevertheless be shown in edit mode, so that the point edit dialog remains accessible, and the point can be assigned to a canvas view.

Cancel closes the dialog.

Save confirms the changes and closes the dialog.

With corner-drag enabled, the point orientation can be changed also by sliding down from the upper corners of the canvas (which turns the point orientation anti-clockwise and clockwise, respectively).

The BACK key closes the dialog without saving the changes.

[Section](#) point dialog

SKETCH LABELS EDIT

This dialog is opened when you edit a label, in drawing mode of the Drawing window by selecting it and then clicking in the Note button.

Label properties:

- **size:** one of XS, S, M, L, XL
- **orientation**
- **text**
- **layers**, canvas views, available when the "Layers" setting is set to "by item" [T]
- additional Therion **options**

Canvas views (layers)

The canvas views are presented as checkboxes with the view letter-code. By default, labels belong to the "base" and "artifact" views. It is possible to switch off some views.

Beware that, if the label is not assigned to a visible view, it will not appear in the canvas even if it is actually added to the sketch. In this case, however, the label point will be visible in edit mode, so that the label edit dialog can be opened and the label assigned to a canvas view.

PHOTO-ITEM EDIT

This dialog displays the properties of sketch points of type "photo".

- **thumbnail** of the photo
- **orientation** (azimuth and clino) of the photo, for photos taken with TopoDroid
- **date and time** of the photo
- **comment** (editable)

Buttons:

- **Save** saves the changes to the comment

The image is displayed when you tap on the thumbnail. The full image is not reoriented to portrait.

The Android BACK button closes the dialog.

SKETCH LINE

This dialog is opened when you select a point of the a line, in editing mode of the Drawing window, and tap the Edit button.

You can change the line type by selecting it from the scroll-down list. The type cannot be set to "section" line.

Line properties:

- line **type**
- **outline**: "in" or "out", or "unset"
- **layers**, canvas views
- additional Therion **options**

With this dialog you can also

-  reverse the line direction
-  make the line more straight
-  reduce the number of line points (decimation)
-  make the line with sharp-corner
-  make the line into a segment
-  mark the line closed or open

The second, third, and fourth are mutually exclusive.

Canvas views (layers)

You can change the canvas views to line belongs to. The views are presented

as checkboxes with the view letter-code. The line is not displayed if all its views are OFF. The line points will nevertheless be shown in edit mode, so that the line edit dialog remains accessible, and the line can be assigned to a canvas view.

Point decimation

The line points can be reduced either keeping one point every two (single check), or one every four (double check).

Section line dialog

Section lines are special as they are associated with x-sections (both sketch and photo). Therefore they have different fields:

- "section comment" (text field)
- the photo thumbnail and info (orientation and date), if the x-section has a photo.
- four action buttons: take a photo, open the sketch, delete the x-section, save the comment.

Tapping the thumbnail displays the photo.

The **OK** button saves the changes and closes the dialog.
The **BACK** key closes the dialog.

SKETCH AREA

This dialog is opened when you select a point of the border of an area (in edit mode of the Drawing window), and tap the Edit button.

You can change the type of area, by selecting it from the scroll down list.

Area properties:

- **visibility** of the area border
- **layers**, canvas views the area belongs to

The area border visibility is written in the Therion export, so that it can be used when the survey map is generated.

TopoDroid displays the area shaded region without a border if the visibility is off. Otherwise a light grey border is shown.

The default value of the area border visibility is a setting.

Point decimation

The number of border points can be reduced by either keeping one point every two (single check) or one every four (double check).



Canvas views (layers)

Finally you can change the canvas views to area belongs to. The views are presented as checkboxes with the view letter-code.

The area is not displayed if all its views are OFF. In this case the border points will nevertheless be shown in edit mode, so that the area edit dialog remains accessible, and the area can be re-assigned to a canvas view.

The BACK key closes the dialog.

SKETCH STATISTICS

This dialog is opened from the Stats menu of the Drawing window (plan or profile view). It displays both data-reduction statistics and sketch related statistics.

The data-reduction statistics are

- *Legs*: number of (regular) legs, sketch-leg total length, profile and plan view cave-leg total lengths.
- *Duplicate shots*: number of duplicate legs and their total length.
- *Surface shots*: number of surface leg and their total length.
- *Splays*: number of splay shots, and sketch splay shots.
- *Stations*: number of stations, and sketch stations.
- *Leg sigma*: average position error between repeated shots.
- *Survey length*: length of legs not including flagged or repeated legs.
- *Dangling shots*: number of shots not attached to the sketch (if any) and their total length. Tapping this line opens the dialog with the list of unattached shots.
- *Cycles*: number of cycles (including the backshot-cycles) if any. This item is only shown if there is at least one loop defined.
- *Components*: number of disconnected survey components if greater than 1. This is 1 if the legs are all attached together in one survey.
- *Sketch origin station*: sketch origin (zero station).
- *Sketch length*: the midline length.
- *3D length*: profile-view (3D) midline length.
- *Plan length*: plan-view midline length.
- *W E N S*: the sketch extension (West, East, North and South).

- *Sketch denivelation: the sketch denivelations, both positive and negative, relative to the sketch center (the sketch origin unless re-centered).*
- *Cave denivelation: the cave denivelation relative to the sketch origin.*
- *Angle errors: average angle error between shots of the same leg [degrees].*
- *the number of loops, those used in closure compensation and those not used*

When there are two values separated by a number sign (#), the first value is the total survey value, and the second is the data reduction value, i.e., the value for the connected component that contains the reference station of sketch. These two values are different when the survey midline data contains more than one connected component.

For each loop:

- *closing shot*
- *number of loop shots*
- *closing error: 3d length over total loop length*
- *horizontal and vertical error lengths*
- *percent error: ratio between error and loop length*
- *estimated shot angle error [degrees]. This is the error times the square root of the number of shots and divided by the length. The result, in radians, is converted to degrees.*

Tapping on the loop closure entry switches the text between the closure error, and the list of loop stations.

Length are in the units set in the app settings.

Denivelations

The sketch center is the sketch origin, unless the sketch is re-centered.

The cave denivelation includes the survey data that are not "surface", "duplicate", or "commented". It includes the splays, therefore if you have taken "surface" splays you need to mark them as "surface".

SKETCH FLIP AND SHIFT

Profile sketch flip [E]

When viewing the profile, a long tap on the Plan/Profile button opens a confirmation dialog to flip the profile sketch horizontally.

The dialog has a checkbox to reverse the "extend" value of the shots as well.

This is enabled by default. Only the shots that are displayed in the profile, are reversed, ie, their "extend" is changed. Therefore shot hidden behind a station are not affected. By this means you can reverse only a portion of the survey data.

Shots "extend" can also be reversed with a multi-selection in the Survey Data window.

Sketch transform [E]

Shifting, scaling and arbitrary affine transformations are enabled by secondary settings [G].

The drawing can be shifted as a whole by selecting the Fix ref checkbox in the sketch reference dialog. Then in move/zoom mode the drawing can be shifted by dragging it with two fingers.

Scaling and affine transformation are accomplished by touching the screen with three fingers. The motion of the fingers defines the parameters of the affine transformation. In this way the sketch can be arbitrarily scaled, rotated and skewed.

If the general affine transform is not enabled, the sketch is only scaled.

WARNING

Sketch transformation is CPU intensive, especially for large sketches, because it changes the coordinates of every drawing point, and the app may appear unresponsive if the sketch is big.

SKETCH OPERATIONS

With this dialog you can do maintenance operations on the sketch:

- delete the sketch*
- change the name of the sketch*
- change the origin station of the sketch [T]*
- split the sketch and create a new sketch moving or copying drawing items to it*
- move or copy drawing items to a new scrap of the sketch*
- move or copy drawing items in a buffer*
- clear the item buffer or paste them in a scrap*
- select another sketch and add it to the outlines of the current sketch*
- merge the items of the outline sketches in the current sketch*

The dialog is opened from the Operations menu of the Drawing window.

It has a field for the new name of the sketch (or the name of the new sketch in case of "split") and a text box with the sketch reference station (sketch origin).

Type the new sketch name in the edit box and tap the Rename button to confirm.

WARNING. The name of the sketch cannot contain special characters: '/', ':', '<', '>', '*', '\\', '?', '|', '\"'.

The name of the sketch cannot contain spaces. Spaces are replaced with '_' (underscore).

The Delete button marks the sketch as "deleted". It can be recovered with the Survey Data window recover menu.

With the sketch button [T] you can cut a portion of the sketch and save it as a new sketch. You must write the name of the new sketch in the edit box. The base station of the new sketch is the same as that of the current sketch. With the checkbox you choose whether to copy or to move the selected items to the new sketch.

After you press the sketch button, you must select the items of the sketch to move/copy by tracing a line around them.

With the scrap button [T] the selected items are copied or moved to a new scrap.

With the buffer button they are put in a buffer.

The buffered items can be pasted to a scrap with the Paste button.

The Clear button empties the buffer.

Outlines

The Select button opens a dialog to pick another sketch and add its outline as a reference. The outlines are not saved with the sketch, and are lost when the sketch is closed.

The Merge button copies the items of the sketches shown in outline in the current sketch.

Close the dialog with the Cancel button to cancel.

SKETCH SCRAPS

With this dialog you can add a scrap to a sketch or switch among the scraps of a sketch.

It is opened from the Scraps menu of the Drawing window.

A sketch can be divided in more scraps.

Only one scrap is active at a time, the others are shown only in outline (the walls).

The number of the active scrap and the total number of scraps are shown in the dialog message.

The New button add a scrap to the sketch. The newly added scrap becomes the active scrap.

With the Next and Prev button you can switch to another scrap, to work on that.

These buttons are greyed out when the active scrap is the last or the first, respectively.

If there is more than one scrap, there is a Delete button to remove the currently active scrap.

The Back button closes the dialog without doing anything.

SKETCH MERGE AND SPLIT

In the Sketch rename/delete dialog you can either rename or delete the opened sketch.

If split/merge [G] is enabled there are also buttons to split/copy the sketch (ie, extract part of the drawings creating a new sketch), include the display of the outlines of other sketches, and merge the drawings of the sketch shown as outline in the current sketch.

Sketch split

The split button creates a new sketch populated with items selected from the current sketch. To select these items trace a border surrounding them in the Drawing window.

You can specify the name of the new sketch. However, its origin is the same as that of the current sketch. Likewise it has the same type of profile view as the current sketch.

If the Drawing window is displaying the plan view, the new sketch plan view is populated with the selected items, and the profile view is empty. The converse, if the Drawing window is displaying the profile view. Only visible items are selected for the split.

You can choose whether to move the selected items to the new sketch or only to copy them to it.

After this operation the Drawing window displays the new sketch.

Sketch outline

With this button a new dialog is opened to select another sketch that is shown in outline (only the walls) on the display.

Sketch merge

If the outline of another sketch is being displayed you can merge it within the current sketch.

The items of the outlined sketch are copied in the current sketch.

SKETCH ZOOM-FIT

This dialog is opened by the menu "Zoom fit".

With this dialog you can

- *Choose the orientation of the presentation, and fit the drawing into the display*
- *Set a graph-paper or a metric sketch grid*
- *Center the drawing at a given station*

Orientation

This orientation refers to the presentation of the drawing by TopoDroid and is not related to the Portrait/Landscape orientation managed by Android.

The orientation applies only to the plan/profile views. You can choose to display the drawing with the magnetic North/Upward direction to the top of the screen or to the left.

Graph-paper sketch grid

Instead of the metric sketch grid, with cells corresponding to the selected sketch grid units, it is possible to have a graph-paper sketch grid with cells of 1 mm size, as for technical drawing. With graph-paper the sketch units is 1 m.

The sketch scale must be chosen among 1:100, 1:200, 1:300, 1:400, and 1:500. Choose "---" to go back to metric sketch grid. Tap the "Graph-paper" button to confirm the choice.

The size of the graph-paper is approximate, as it is not possible to get an accurate value for all Android devices. The secondary setting [graph-paper scale](#) allows to calibrate the graph-paper by adjusting a 5 cm long bar (using a real ruler).

Center at a station

Enter the center station and tap the button "Center at" to confirm. The station is highlighted white. The "clear" button cancels the highlight.

Close the dialog with the Cancel button to cancel any action.

Graph-paper scale

This dialog is opened tapping the value of the Graph-paper scale sketch secondary setting.

It allows to set a factor in the computation of the scale of the graph-paper.

It displays a segment of (world) length 5 m at scale 1:100. Therefore the segment should be 5 cm long on the screen.

You can adjust the length of the segment with the "plus" and "minus" buttons.

Tap the "OK" button to confirm your choice, "cancel" to discard it.

SKETCH EXPORTS

Sketches are stored in TopoDroid binary format, and can be exported as

- Therion ".th2" file
- cSurvey ".csx" file
- Tunnel ".xml" file
- 2D DXF (LibreCAD/AutoCAD)
- SVG (Inkscape/Adobe Illustrator)
- PDF
- shapefile (QGIS)
- xtherion image ".xvi"

In addition to the export, there is an option to automatically save a sketch in one of the supported export formats whenever it is closed.

*The **cSurvey** export file contains both the plan and the profile sketches. If the "section" points are defined, it has also the x-sections. X-sections are not exported as cSurvey files.*

*Sketches can be exported as 2D **DXF** files. Supported DXF versions are 9, 12, and 14.*

*The **shapefile** export includes stations, shots, and drawing. All the drawing objects are converted to linestring. The attribute tables contain the object data (eg, point type, orientation, etc.). The files comprising a shapefile export are compressed in a zip archive.*

A few export settings, specific to the selected format, can be overridden in the export dialog.

Overview export

Sketches can be exported also from the "Overview Window" in the following formats

- Therion ".th2" file
- 2D DXF (LibreCAD/AutoCAD)
- SVG (Inkscape/Adobe Illustrator)
- shapefile (QGIS)
- PDF
- ".xvi" file (xtherion)

The export of the overview window includes all the sketches of the current view (either plan or profile).

If there are "section" points, the x-sections are also included.

The exported file can be further shared by checking the Share checkbox.

OVERVIEW REFERENCES

This dialog is opened by the Reference button of the Overview window.

- **legs**, survey midline (white)
- **splays** (grey)
- **stations** names (violet)
- **sketch grid**, metric or yard
- **scalebar** and North arrow
- **x-sections**
- **Outline** only

There are three sketch grids,

1. a grid at 1 m spacing (dark grey)
2. a grid at 10 m spacing (grey)
3. a grid at 100 m spacing (light grey)

TopoDroid shows only the sketch grids that are appropriate for the scale of the sketch view.

THE FINAL MAP

This section describes the use of Therion or cSurvey to draft the final map.

Therion

If you want to compile the Therion project you must supply a thconfig file which can be as simple as

source my_survey.th

export map -proj plan -o my_cave_p.pdf

export map -proj extended -o my_cave_s.pdf

You need to have exported the survey data as Therion ".th" file, and the survey sketches as Therion ".th2" files.

The data file contains commands to input sketch files. However, they are commented and you need uncomment them.

The data file contains commands to define a map for each sketch (scrap). For simple surveys you can leave these commented.

Alternatively you can export the sketches as Therion ".xvi" files and retrace them using xtherion.

cSurvey

Export the sketch as cSurvey ".csx" file.

Open the exported file with cSurvey: you will have the data sheet filled with survey data and the sketches properly aligned in the drawing canvas of cSurvey.

PROJECT MANAGER - Project window

The cave Project window is opened tapping a project name in the listing of the Project Manager window.

It displays the list of surveys that comprise the project. The list is empty if the project has no surveys.

Each survey can be selected for further actions by tapping its checkbox.

Menus

- Close the window
- Export the cave project (Therion or Survex)
- Delete the cave project
- Help

Projects can be exported as Therion or Survex files. The export file is selected with a document provider.

Buttons

- insert surveys
- remove the selected surveys
- color: change the color of the selected surveys
- [2D display](#) of the selected surveys, in plan view
- display the survey geo-points
- list the project equates
- 3d view of the project

The insert button display the list of surveys in the TopoDroid database. Surveys are added to the project by tapping their checkboxes and confirming with the OK button.

Project 3D view

The project is displayed in the 3D viewer beginning with the first survey. The following surveys are shown in the 3D only if they are (directly or indirectly) connected to the first survey.

If several surveys are georeferenced, they can all be displayed in 3D, together with the connected surveys.

Surveys 2D plan view

The plan-view of the surveys allows defining "equate" graphically.

This window displays surveys in plan view, with midline and station names. Each survey has a different color. After a while the colors repeat; therefore, you should display only two or three surveys at a time. Equates are shown as dashed red lines, and equated stations have a round background. The size of the station names is equal to that in the [Drawing window](#).

The view can be panned and zoomed. A survey can be grabbed (tap on a station: it gets circled) and dragged around while the other surveys do not move. In this way you can overlap stations of different surveys that you want to equate. When they overlap, circle them both by tapping the point: you can then insert the equate between them with the "add equate" button.

Buttons

It has six buttons:

- Add equate: opens a dialog where the user can set which stations are to be equated. In order to manually add an equate between surveys of the cave project, tap the "OK" button".

- *Show possible equates, ie, stations with same name in different surveys.*
- *List equates between surveys of the cave project.*
- *Reduce the number of displayed stations*
- *Increase the number of displayed stations*
- *Select a single survey display of stations. Long-tap to deselect*

Equates

On the dialog presented when clicking on the List equates button, you can drop an equate by tapping its entry in the dialog listing.

There is also a button to delete all the equates.

PROJECT MANAGER - Project dialog

The cave Project dialog is opened tapping the Insert button of the Project Window.

It is used to create a new project.

It has only a field to enter the project name.

Buttons:

- ***Cancel*** - *closes the dialog without creating the project*
- ***OK*** - *creates a new project with the specified name*

PROJECT MANAGER - Surveys geo-points

This dialog is opened by the GNSS button of the cave Project window.

The dialog lists the geolocation info of the project surveys:

- *name*
- *station*
- *WGS84 coordinates*
- *additional coordinate system*
- *coordinates in the coordinate system*

Surveys without geolocation are not listed.

PROJECT MANAGER - Plan View

The Project Plan View window is shown when you tap the Plan button in the Project Manager window.

It displays the surveys of the cave project that have been selected, on the horizontal plane (thus in plan view), with midline and station names.

*The midlines are displayed with the color of the respective survey.
The display can be panned and zoomed in/out.*

By selecting a station of one survey (tapping on its point) the station is highlighted with a circle. Next the station survey can be dragged around while keeping the other survey fixed.

By this means the station can be brought close to a station of another survey.

When they are close, the two stations can be selected at the same time with a tap (both become circled), and "equated" in the cave project, ie, the two surveys are connected at the "equated" stations.

Equates are shown as dashed red lines, and equated stations have a round background.

The selection circle is displayed when you touch down the screen, but the selection becomes active only after you lift the touch without moving from it the station.

The selected survey can be moved by a drag touch on the display. By this means you can see where the selected station go.

A drag that is too fast loses the selected station.

When no station is selected all the surveys are moved around together.

The size of the station names is equal to that in the Drawing window.

Equates can be also added manually through the [Equate Dialog](#).

Menus

- Close the window
- Help

Buttons

- [Add equate](#): opens a dialog where the user can type the stations to be equated. Tap the "OK" button to confirm. If close stations are not selected and no survey is disconnected it opens a dialog to enter the survey stations to equate.
- Display possible equates
- List equates of the cave project
- Display fewer station names
- Display more station names
- Select a single survey and display only its stations. A long tap deselect the single survey

Possible equates

With the "possible equates" button the set of stations in different surveys, with coincident name are displayed with yellow dashed lines joining the stations with same name.

Tapping on a station of a yellow line, the corresponding possible equates are added to the project.

The display of the possible equates is cleared by tapping on the "possible equates" button a second time.

Warning There is no check that by adding equates the project graph acquires loops.

Equates list

The [Equates](#) dialog is presented when clicking on the List equates button. You can drop an equate by tapping its entry in the listing.

PROJECT MANAGER - New equate dialog

This dialog is opened by the New equate button on the Project Display window

You can manually add an equate between stations of the project surveys by entering the station names of the displayed surveys.

The "-" (dash) sign excludes the given survey from the relationship.

There is an error message if a name is entered that does not correspond to a station in the survey.

If a survey is not connected, and is splitted with subsurveys, the first station is used as hint for the subsurveys.

Buttons:

- *Cancel: closes the dialog*
- *Search: finds the next points with the same name in the measurements*
- *One: creates a loopless connection of all selected surveys based on stations with the same name*
- *All: creates a equate between all points with the same name*
- *OK: confirms the changes and closes the dialog.*

PROJECT MANAGER - Equates dialog

This dialog is opened by the Equates button of the cave Project window or of the Project Display window.

The dialog lists the project equates.

An equate can be removed by tapping it. There is a confirmation dialog.

Buttons:

- *Close: closes the dialog*
- *Clear all: remove all equates. There is a confirmation dialog.*

PROJECT MANAGER - Surveys dialog

This dialog is opened by the Insert button of the cave Project window

The dialog lists the surveys in the TopoDroid database.

To add one or more surveys to the project, check the box to their left and click the OK button.

PROJECT MANAGER - Export dialog

This dialog is opened by the Export button of the cave Project window

Cave projects can be exported as Therion ".th" files, or as Survex ".svx" files.

The exported file contains the list of surveys that make the project and the list of stations identifications ("equate" commands).

3D VIEWER - Viewpoint

This dialog is opened from the menu Viewpoint of the 3D Viewer window.

It sets the 3D viewpoint of the model to one among East, North, West, South, and Top.

There is also a button to reset the zoom of the display.

3D VIEWER - Model projection

The 3D projection dialog is opened with a long-tap on the Projection button in the 3D Viewer window, when the model view mode is perspective projection.

With this dialog it is possible to adjust the focal length of the perspective projection.

3D VIEWER - Model info

The 3D info dialog is opened by the menu Info in the 3D Viewer window.

It presents some global information

- *Size of the grid cells*
- *Current direction of view*
- *Number of surveys*
- *Number of legs and number of splays*

- *Number of stations*
- *Total length*
- *Total denivelation*
- *Estimated volume (if the wall model has been computed)*
- *Longitude extent*
- *Latitude extent*
- *Altitude range*

This is followed by the list of surveys that compose the model. Tapping a survey pops up a dialog with the number of legs and splays of the survey and their lengths.

3D VIEWER - Legs visibility

The 3D legs visibility dialog is opened by a long-tap on the Frame button in the 3D Viewer window.

It is used to hide or show special legs, namely

- *Surface legs*
- *Duplicate legs*
- *Commented legs*
- *Backshot legs*

3D VIEWER - Stations

The display of stations and the actions on them are controlled by the third

button .

If the button has a red dot on the top-right the stations are selectable. The button switches between the two states with a long press.

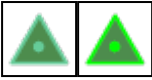
The stations can have three display modes

- *invisible, ie, hidden*
- *visible as points*
- *visible as names*

When visible and the state is "selectable" (red dot), the stations are selected with a touch on the dot or the name. The selected station is highlighted with a red dot or a boxed name. Its data (name and coordinates) are shown in a bottom blue bar (short selection tap) or in the station dialog (long tap).

Station bar

The station bar is opened when a station is selected with a short tap
It contains the station name, its coordinates (in the model), and four buttons:

- Close deselects the station. The station can be also deselected with a touch on it on the screen.
-  center makes the selected station as the center of the rotations of the model
-  ruler toggles the measure mode. In this mode the program measures the distance between the selected station and any another station, chosen with a touch on the screen. It computes the 3D distance and the displacements in the North, East and vertical directions. If the two stations are path connected it computes also the shortest path length, and highlights the path in green on the 3D.
- Switch to the station dialog.

Station dialog

The station dialog is opened when a station is selected with a long tap
It contains the station name, with possible equated stations, and its coordinates (in the model).

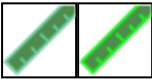
It has two buttons:

- Close closes the dialog and deselect the station, thus possibly closing the station-bar
- Cancel closes the dialog

3D VIEWER - Measure dialog

When the fixed station is selected, its coordinates appear in a blue bar at the bottom of the screen.

This bar has four buttons:

- Close the station bar
-  the first sets the station as center of rotations
-  the second switches the program in "measure mode": tapping on another station brings up the Measure dialog with the result of the measurement.
- Open the Station dialog

The Measure dialog show the distance between the fixed station and another one.

- *the full names of the two stations*
- *the East, North, and vertical displacement between the two stations*
- *the 3D distance*
- *the distance in the horizontal plane*
- *the azimuth and inclination of the straight line between the two stations*
- *if it exists, the length of the shortest path along the cave midline between the two stations, and the positive and negative path denivelation*

Denivelation

The positive and negative path denivelation is computed by summing up the vertical components of upwards and downwards legs, respectively. Only legs with inclination over 10° are considered. Above 30° the leg vertical component is fully taken into account. Between 10 and 30° the vertical component is linearly interpolated, ie, multiplied by (clino-10)/20.

3D VIEWER - Surveys list

When you open the 3D of a cave project, a long press on the pencil button



*opens a dialog with the list of the surveys in the project.
For each survey it shows*

- *color box*
- *visibility box*
- *name of the survey*

The color box allows to change the color assigned to the survey through a color picker dialog.

The visibility box toggles the visibility of the survey in the 3D.

Buttons

- *OK*
- *Cancel*

3D VIEWER - Survey

This page needs writing.

3D VIEWER - Surface DEM

A surface DEM (Digital Elevation Model) can be added to TopoDroid models or can be shown if it is enclosed in a loaded model file (eg, in a lox file)

The Surface button of the 3D Viewer window toggles the DEM display on and off.

This dialog is opened from the Surface menu of the 3D Viewer window.

It is used to load a DEM, a surface texture to overlay on the DEM, and adjust its opacity.

It contains

- a button to select a DEM file*
- a button to select a texture file, after a DEM has been loaded (or if the model already has a DEM)*
- a slider to adjust the opacity*
- a checkbox to activate the display of the projection of the model on the DEM*
- a checkbox to activate the display of the surface texture (if there is one)*

DEM file

The 3D viewer supports DEM file in ascii format, or Therion grid format (see Therion manual for details on the file formats and coordinate reference systems).

DEM ascii format is a text file with six header lines ("cols", "rows", "xllcorner", "yllcorner", "cellsize", and "NODATA_value") followed by data lines with values arranged left-right (west-east), The lines are arranged top-down (north-south).

Texture file

The 3D viewer supports texture GeoTIFF image files, and OSM map files.

The DEM and texture X-Y coordinates must be in the Coordinate Reference System of the model (which can be changed in the fixed station dialog), because the 3D viewer does not perform coordinate conversions.

OSM files are in WGS84 geographical coordinates. The values can be used with models in UTM cartographic, but the conversion is approximate.

3D VIEWER - Model walls

If the model has a sufficient number of splays, the walls of the model can be estimated. There are multiple algorithms to compute the walls:

- *Hull*
- *Convex hull builds the walls as a quasi-convex local envelope of the splay endpoints*
- *Powercrust works with a huge number of splays, and interpolate the walls through the splay endpoints [debug only]*
- *Tube walls are suitable for old-style LRUD-surveys [debug only]*
- *Bubble walls [debug only]*

Once computed the display of the walls is toggled on and off with the Walls button of the 3D Viewer window.

The dialog has a slider to adjust the walls transparency.

3D VIEWER - 2D rose diagram

The dialog is opened from the menu Rose diagram of the 3D Viewer window.

It displays the 2D view of the directions, in the horizontal plane, of the survey legs.

The North is vertical upward, and the East is horizontal to the right.

3D VIEWER - 3D rose diagram

The dialog is opened from the menu 3D rose diagram of the 3D Viewer window.

It displays the 3D view of the directions of the survey legs.

The reference is made of a green line pointing East, a blue line pointing North, and a red line pointing upwards.

The direction of view is displayed on the top of the screen.

The inclination is restricted to negative values (ie, -90 means looking down and 0 means looking from a side).

Station Highlight

This page needs writing.

3D VIEWER - Export

The dialog is opened from the menu Export of the 3D Viewer window.

The 3D model can be exported as

- *glTF (GL transfer format)*
- *CGAL*
- *STL, either binary or ascii*
- *LAS, binary*
- *DXF*
- *KML (if the model is georeferenced)*
- *Shapefile (if the model is georeferenced)*

When supported by the format, the exported file can include

- *Stations*
- *Splays*
- *Walls (if generated)*
- *Surface DEM (if loaded)*

The exported file che be further shared by checking the Share checkbox.

Index

- 3D: [project settings survey](#)
- A3 (DistoX): [info](#)
- Accuracy (shot): [calibration check-shots settings settings](#)
- Active (station): [saved sketch sketch station station](#)
- AI: [API key dialog](#)
- Altitude: [coordinate](#)
- Android: [11+ hardware permissions](#)
- Anomaly (magnetic): [station-naming](#)
- Area: [properties settings sketch tools tools](#)
- Audio: [dialog folder list menu permission shot](#)
- Azimuth: [button projection reference](#)
- Backsight: [leg memory-data station-naming](#)
- Backshot: [DistoX setting](#)
- Backup: [sketch](#)
- Bezier: [line-style settings](#)
- Bluetooth: [actions aliases bluetooth button DistoXBLE permission reset](#)

- Blunder: [setting station-naming](#)
- BRIC4: [calibration connection-mode data-download guide index info memory selection settings](#)
- Calibration: [check-shot check-shot computation data database data-colors data-distribution data-memory data-search dialog check-shots coefficients groups guide settings validation window](#)
- Camera: [photo permission x-section](#)
- Cavway: [guide selection](#)
- Class: [splay](#)
- Clipboard: [point](#)
- Close: [TopoDroid area-tool line](#)
- Color: [accuracy calibration-data clear colors multi-shot packets PocketTopo shot sketch-legs sketch-splays splay splay](#)
- Commented: [leg](#)
- Compass: [import/export import settings](#)
- Continuous mode: [connection-mode download](#)
- Convex hull: [3D walls](#)
- Coordinates: [conversion DEM](#)
- cSurvey: [export map sketch-export](#)
- CSV: [data-export](#)
- CWD (current work folder): [folder menu](#)
- Database: [databases files](#)
- Data: [connection-mode download mode mode reduction sketch](#)
- Dangling: [sketch](#)
- Decimation: [area-border line](#)
- Declination: [sketch geopoint survey survey](#)
- Delete: [calibration dialog saved-station sensor shot sketch](#)
- DEM: [3D](#)
- Denivelation: [cave 3D-path sketch](#)
- Device: [database forget nickname nickname selection settings settings troubleshooting](#)
- Dip (magnetic): [shot](#)
- DistoX: [firmware guide memory packets selection troubleshooting](#)
- DistoXBLE: [selection troubleshooting](#)
- Diving: [data-export data-mode manual-shot survey](#)
- Drawing: [line-tracing point-icon stylus-only tools](#)
- Dump: [memory](#)
- Duplicate: [leg search shot-dialog statistics survey-info](#)
- DXF: [export settings sketch-export](#)
- Equate: [listing new project-stations](#)
- Erasing: [eraser mode undo](#)
- Export: [3D calibration data export folder overview overview project settings sketch surveys](#)
- Extend: [button button fractional leg leg reduction reference setting sketch-item splay](#)
- Extended (view): [flip new profile shot-dialog sketch](#)
- File: [x-section](#)
- Flip: [sketch](#)
- Focal length: [perspective](#)
- Folders: [private survey](#)
- Geocoding: [dialog](#)

- Geolocation: [button dialog GNSS import listing permission point settings](#)
- GNSS: [apps input advice](#)
- GPSTest: [geo-point selection](#)
- Graph paper: [sketch sketch](#)
- Grid: [3D overview sketch](#)
- Groups: [calibration calibration data line-tool splay tools](#)
- Help: [AI-help TopoDroid user-manual](#)
- Horizontal: [area-tool x-section](#)
- Import: [calibration calibration geo-point settings settings import old-surveys survey](#)
- Internet: [permission](#)
- JedEye: [guide selection](#)
- Keypad: [setting TopoDroid](#)
- KML: [data-export](#)
- Label (point): [dialog settings sketch-item](#)
- Landscape: [sketch sketch presentation](#)
- Language: [TopoDroid translation](#)
- Last time: [BRIC4](#)
- Layers: [area area-tool label line line-tool point point-tool section-point sketch sketch-items](#)
- Leg: [duplicate overshoot overview repeated search sketch-color sketch-midline statistics surface](#)
- Libraries: [native](#)
- Line: [dialog line-tool properties reduction retracing section settings setting style tools](#)
- Loop: [compensation setting statistics survey-info trilateration](#)
- LRUD: [leg manual sketch-color](#)
- Measure: [3D overview sensor](#)
- Memory: [BRIC4 DistoX reset](#)
- Merge: [sketch sketch sketch](#)
- Midline: [hide hide statistics](#)
- Mobile Topographer: [geo-point selection](#)
- Model (3D): [info](#)
- Move: [shot-buffer survey-data](#)
- Multi: [connection-mode](#)
- Multi-path: [multiselection multiselection setting](#)
- Multi-shot: [dialog selection splay-class](#)
- Name: [calibration survey](#)
- Notes: [button button dialog](#)
- On-demand mode: [connection-mode download](#)
- Options (Therion): [point sketch-item section-point](#)
- Orientation: [area-tool photo point point-tool sketch-items](#)
- Outline: [line overview scrap sketch sketch-lines](#)
- Packets: [log](#)
- Palette: [area-tools line-tools point-tools reload tools tools tool-picker](#)
- PDF: [sketch-export](#)
- Permissions: [granting TopoDroid](#)
- Perspective: [projection](#)
- Photo: [button dialog display edit-dialog edit-dialog folder geocoding listing menu permission shot-dialog](#)
- Plan (view): [project sketch x-sections](#)

- Plane interpolation: [interpolation](#)
- Point (sketch): [dialog photo point-tool properties settings tools](#)
- Portrait: [sketch sketch](#)
- Profile (view): [shot-dialog sketch x-sections](#)
- Proj4: [proj4 input](#)
- Project: [equates export folder info manager manager plan-view sources](#)
- Projected (view): [azimuth sketch sketch profile](#)
- Projection: [3D](#)
- Range: [selection](#)
- Recovery: [shot sketch](#)
- Reduction: [data line-points](#)
- Reference: [overview sketch station statistics](#)
- Reload: [palette sketch sketch](#)
- Remote: [bluetooth-commands command-settings](#)
- Rename: [dialog sketch survey](#)
- Renumber: [groups multi-shot](#)
- Retracing: [line/area](#)
- Reverse: [line shot-dialog splay](#)
- Roll: [calibration](#)
- Rose diagram: [2D 3D](#)
- SAP5: [connection-mode guide PIN selection settings](#)
- Scale: [sketch](#)
- Scalebar: [sketch](#)
- Scrap: [sketch](#)
- Screenshot: [screenshot](#)
- Search: [button calibration leg setting splay station](#)
- Sensor: [button dialog listing measure menu setting shot-dialog](#)
- Settings: [secondary main-window search settings TopoDroid](#)
- Shapefile: [export settings sketch-export](#)
- Shift: [sketch sketch](#)
- Shot: [blunder buffer color dangling display-mode edit listing manual marks marks multi-shot paste recompute reduction search settings settings sibling sketch sketch-item statistics](#)
- Sibling (leg): [color reduction threshold](#)
- Side-drag: [point](#)
- Size: [label point](#)
- Sketch: [cache secondary_settings dialog delete display-fit file-editor flip folder graph-paper listing merge new rename scale scraps settings shift shift shot-colors station-center statistics switch x-section zoom](#)
- Splay: [clear-color color color custom-color overview settings shot shot-dialog sketch sketch-color station statistics toggle](#)
- Split: [sketch survey survey](#)
- Station: [3D 3D active color naming naming naming-colors names project-equate overview reference remap remap-menu saved saved search sketch sketch-center sketch-color sketch-item survey-info](#)
- Storage: [folder permission](#)
- Stylus: [drawing](#)
- Surface: [3D DEM leg search shot-dialog statistics survey-info](#)
- Survex: [import/export import](#)
- Survey: [3D 3D 3D database data-colors data-move folder info new notes project-listing project-surveys reduction rename settings settings split split](#)

- SVG: [export settings sketch-export](#)
- Team: [entering new-survey survey](#)
- Text: [point label sketch-items](#)
- Therion: [file-editor import/export import map settings sketch-export station-point](#)
- TopoRobot: [station-naming](#)
- Transform: [sketch](#)
- Transparency: [3D surface 3D walls](#)
- Tripod: [station-naming](#)
- Units: [settings units](#)
- Usage Profile" [profiles setting](#)
- Viewpoint: [3D](#)
- Visibility: [area legs \(3D\) sketch-items](#)
- VisualTopo: [import/export import settings](#)
- V-threshold: [manual-shot](#)
- Walls: [3D 3D auto auto](#)
- Weeding: [line-style settings](#)
- WENS: [threshold manual_data](#)
- Window: [3D calibration calibration-data device main overview project shot sketch survey](#)
- Winkarst: [data-export](#)
- Work folder: [work folder](#)
- X310 (DistoX2): [info](#)
- XBLE (DistoXBLE): [info](#)
- X-section: [at-station at-station at-station cross-sections leg leg-sections line photo point point slant survey](#)
- Zip archive [import/export](#)
- Zoom: [3D](#)